

# Managing in the Digital World

An **information system** is the combination of people and information technology that creates, collects, processes, stores, and distributes useful data. That one sentence is doing a lot of work, and this module unpacks all of it: the world these systems now run, what the parts actually are, why the same technology can make one company and break another, what responsible use of it looks like, and how a manager turns any of that into a decision worth defending.

1

**OBJECTIVE 1.1**

Describe the digital world, the societal issues it raises, and how rising digital density is shaping what comes next

2

**OBJECTIVE 1.2**

Explain what an information system is, contrasting its data, technology, people, and organizational parts

3

**OBJECTIVE 1.3**

Describe the dual nature of information systems in the success and failure of modern organizations

4

**OBJECTIVE 1.4**

Describe how computer ethics affect the use of information systems, including privacy and intellectual property

5

**APPLICATION 1.5**

Use supplemental strategy frameworks to turn a business problem into a defensible technology recommendation

6

**APPLICATION 1.6**

Decide where artificial intelligence belongs in a business workflow, and where a person must stay in the decision

7

**APPLICATION 1.7**

Write four of this module's definitions as working functions, and let the tests say whether you understood them

## How to use this page

- 1. **Read the short setup first.** Every term is explained in ordinary language before any activity uses it, so nothing here assumes a background you have not been given yet.
- 2. **Answer as you go.** The activities sit inside the reading, not at the end. When you answer a question, **every option explains itself** — including the three you did not pick, because knowing why a wrong answer is wrong is the part that transfers.
- 3. **Use the diagrams.** The tabs redraw one idea from several angles. On a phone, a wide diagram scrolls sideways rather than shrinking to nothing.
- 4. **Retry anything.** Restart one activity, one section, or the whole page. Your completion record stays in this browser until you clear it; the answers themselves reset on reload so you always get a fresh attempt.
- 5. **Finish with the final challenge.** It unlocks once you have answered all thirty situations, then reports the four chapter objectives and the two application supplements separately so a weak area shows up instead of being averaged away.

## Where this module sits in the course objectives

The three modules are organized around the textbook’s own learning objectives, but the course is graded against a different list. This table says plainly which of the eleven course objectives this module serves, and which ones it does not, so you always know why you are reading something.

| COURSE OBJECTIVE                                                                                     | WHAT THIS MODULE DOES WITH IT                                                                                                                                                                                                                                                                                                                                                                                                  |
|------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 2. Discuss how companies use information systems to manage internal operations and to make decisions | <b>This module’s main job.</b> Systems inside an organization works through the whole catalogue, from transaction processing up to decision support and enterprise-wide systems, and says which level of the business each one serves. What an information system is supplies the data, information and knowledge foundation underneath it, and the artificial-intelligence supplement adds the newest tool in that catalogue. |
| 1. Describe what a company’s information systems department does                                     | <b>Also central.</b> The people in information systems covers the roles that group holds, the three competencies it hires for, and where an ordinary user fits. Systems inside an organization covers how the group is organized, who its staff report to, and why it now behaves more like an internal consultancy than a gatekeeper.                                                                                         |
|                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                |

| COURSE OBJECTIVE                                                                                                                              | WHAT THIS MODULE DOES WITH IT                                                                                                                                                                                                                                                                                                                        |
|-----------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 8. Describe the individual and organizational consequences of the use of information technology                                               | <b>Also central.</b> The digital world and digital density take the societal side; when systems succeed and fail takes the organizational one; ethics, privacy and property takes the personal duties of privacy, accuracy, property and access; and the artificial-intelligence supplement asks what the newest tools change about the work itself. |
| 3. Discuss how companies use information technology for competitive advantage by changing their marketplace                                   | Substantial. When systems succeed and fail shows the same class of investment carrying one firm and sinking another, and the strategy supplement adds competitive forces, the value chain, and the four-part shape of a recommendation somebody can actually fund. Module 2 is where the same material becomes the whole subject.                    |
| 7. Recognize a business problem which might be solved using information technology, then work with the information systems team on a solution | The first half. Matching a complaint to the class of system that answers it, and carrying a finding through evidence to a named initiative and a measure, are both practiced here, and the working relationship with the information systems group is described. Developing and rolling out the system is not covered.                               |
| 9. Recognize security breaches and computer crimes                                                                                            | One kind of attack. Ransomware — what it does to data, why paying does not guarantee getting it back, and the backup, recovery and awareness practices that answer it — plus the duty to protect data against unauthorized access, disclosure or destruction. That is one attack and one duty, not a survey of computer crime.                       |
| 4. Apply a working knowledge of computer hardware, software, database technology and computer networks                                        | Introduced only. What an information system is defines the technology half — hardware, software and telecommunications networks — and separates it from the people and organizational halves. The working knowledge itself is built in Module 3.                                                                                                     |
| 5. Delineate how businesses use internet technology for electronic commerce, communicating with employees and working with business partners  | Named, not developed. The catalogue defines the electronic-commerce, collaboration, customer-relationship and supply-chain systems that carry that work, one entry at a time. How the technology underneath them actually works is Module 3's job.                                                                                                   |
|                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                      |

**COURSE OBJECTIVE****WHAT THIS MODULE DOES WITH IT**

**6.** Outline the life cycle of an information system

Not taught here. The phrase appears once, in the list of what a systems competency covers, and nothing in this module explains or assesses it.

**10.** Solve a business problem by creating a database application using SQLite · **11.** Use advanced spreadsheet functions to create a decision support system

Served by the course's separate database and spreadsheet work rather than by any reading module. This module defines what a decision support system is; building one is that separate work.

**WORLD****The setting**

Ubiquitous computing, knowledge work, globalization, and the widening reach of connected devices.

**SYSTEM****The parts**

Data becomes information becomes knowledge — carried by hardware, software, networks, and people.

**STAKES****The double edge**

The same investment that carries one company past its rivals can sink another. Both happen for reasons.

**JUDGMENT****The responsibility**

Privacy, accuracy, property, access — and a method for recommending technology you can defend.

**ONE IDEA CONNECTS THE WHOLE MODULE**

Technology by itself is not the story. An information system is technology *plus* the people who use it and the organization it sits inside, and it is those last two parts that decide whether the first one is worth anything. That is why a company can buy exactly what its competitor bought and get a completely different result.

**SECTION 1-1A**

# The digital world we already live in

You already know computers are everywhere. What you have not done is look at that the way a manager must: as forces deciding who gets paid well, who gets locked out, and where work happens.

## Information systems are ubiquitous

An **information system** combines people and information technology to create, collect, process, store, and distribute useful data. Something **ubiquitous** is present everywhere at once, so present you stop noticing it. Information systems are now both.

**Shipping** — FedEx and UPS route trucks and track packages, so the delivery estimate on your phone came out of a system.

**Public infrastructure** — cities run adaptive traffic control and variable speed limits, so a light that changes with traffic is deciding something.

Some companies do not merely use one; they are one. A **business model** summarizes a firm's strategic direction and how its objectives will be met. Delete Uber's matching software and only strangers with cars remain.

## The post-PC era, and the cycle it feeds

Computing once meant walking to a machine and sitting down at it. In the **post-PC era**, connected devices, smartphones, and tablets replace traditional desktops and laptops. Tablets arrived as consumer goods and are now standard in warehouses, cockpits, and hospitals.

**Place** — people once tied to a stationary PC are tied to none, because what matters is the service and the data, not the device.

**Time** — the workday once ran from switching the computer on to switching it off, and email now fills small gaps at any hour.

Mobile broadband makes computing **instant-on**, and **cloud computing** keeps files reachable from whichever device you hold. Society pushes back, and the loop earns two names.

**Virtuous cycle** — social networking and online investing almost require mobility, so people grew used to checking stock quotes on the go.

**Vicious cycle** — work creeps into leisure and the fixation on being permanently on call grows, until employees bring their own devices to work.

## Knowledge workers and the knowledge society

In 1959 **Peter Drucker** coined **knowledge worker**: a professional, relatively well educated, who creates, modifies, or synthesizes knowledge as a fundamental part of the job. They are paid better than agricultural and industrial counterparts, learn continually, hold far more bargaining power, and make up about a quarter of the workforce in developed nations.

Because education matters that much, Drucker predicted it would become the cornerstone of a **knowledge society**, where knowledge counts as much as land, labor, or capital. This book calls that world the **digital world**.

**What the earnings data show.** Median annual earnings for workers 25 and over, U.S. Census Bureau American Community Survey, 2023 data:

| HIGHEST LEVEL OF EDUCATION      | MEDIAN EARNINGS |
|---------------------------------|-----------------|
| Graduate or professional degree | US\$86,524      |
| Bachelor's degree               | US\$67,256      |
| High school diploma             | US\$39,428      |
| Without a high school diploma   | US\$31,660      |

That gap is widening rather than closing.

Three questions on the era, the cycle, and the society.

### CHECK YOURSELF

## The post-PC era, the cycle, and the knowledge society

Pick the option that matches how the chapter actually describes each idea, then read every explanation.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. The chapter says we are living in the **post-PC era**. What does that mean?

- **A.** The PC is finished as a work machine, so an organization can issue every employee one mobile device and cover every task with it. — This is the one-device version of the idea, and it is what people usually mean when they say the PC is dead. The chapter says devices with newer form factors work in tandem with older ones, providing different devices for different users and different tasks, which is why a warehouse runs on tablets while a design studio still runs on workstations. It would be right if it said mobile devices are becoming the first thing people reach for.

- **B.** Connected devices, smartphones, and tablets are replacing traditional desktops and laptops, and new and old form factors work in tandem. — **correct.** The chapter defines the post-PC era as connected devices, smartphones, and tablets replacing traditional desktop and laptop computers, with newer and older form factors complementing each other because not the device but the services and data provided are of primary importance.
- **C.** It is another name for the move to cloud computing, so what changed is where files and applications live rather than which devices people use. — Cloud computing is real, related, and named separately in the chapter (Gmail, Microsoft 365, Dropbox) as what lets you reach email, files, and notes from different devices. But it answers where the data and applications sit, while the post-PC era answers which devices people carry and use, so the chapter treats cloud computing as an enabler of the shift rather than the shift itself.
- **D.** Computing shifted toward social and casual activity, which is why tablets remain consumer devices rather than professional tools. — The first half is right: computing changed from an activity primarily focused on automating work to one that also carries social and casual activity. The second half contradicts the chapter, which says tablets began as consumer devices and are now commonplace in warehouses, showrooms, airplane cockpits, and hospitals.

**2. What does the chapter mean by calling the relationship between technology and society a virtuous cycle — or a vicious cycle?**

- **A.** Technology drives society in one direction: new devices change how people live, but how people live has no real effect on what gets built next. — This is the one-way version of the story, and it is the most common mistake here. The chapter makes the causation run in both directions: changes in technology enabled new ways of working and socializing, and then the blurring of work and leisure led employees to demand devices that support both and to bring their own devices into the workplace. It would be right only if what people wanted never changed what gets built next.
- **B.** The virtuous cycle is the helpful technology and the vicious cycle is the harmful technology, so they refer to two different groups of devices. — It is one cycle viewed two ways, not two sets of devices. Mobility and connectivity are simultaneously the convenience and the reason people feel permanently on call; the label depends on whether you are describing the benefit or the cost of the same loop.
- **C.** Technological change produces societal change and societal change reshapes technology; the same loop turns vicious when work invades leisure time. — **correct.** The chapter describes a loop in which technological change produces societal change, which in turn shapes further technological change, and it offers 'vicious' as the same loop judged by the creep of work into people's leisure time.
- **D.** Companies must keep upgrading their information systems as competitors upgrade theirs, so falling behind for even one cycle costs market share. — This describes competitive pressure to adopt technology, which is a real idea in this course, but it is about firms racing each other rather than technology and society reshaping one another. It would answer a question about competitive advantage instead.

3. Drucker argued that in a **knowledge society**, possessing knowledge would be as important as possessing land, labor, or capital. Which evidence does the chapter offer for that claim?

- **A.** Knowledge workers now make up roughly three quarters of the workforce in the United States and other developed nations. — The direction is right but the size is wrong: the chapter says knowledge workers make up about a quarter of the workforce in the United States and other developed nations, with their numbers rising quickly. The separate claim that almost every worker can now be considered a knowledge worker is about how the work has spread, not about that percentage.
- **B.** 2023 Census survey data show median earnings of US\$67,256 for bachelor's degree holders versus US\$39,428 for high school graduates. — **correct.** The chapter cites the U.S. Census Bureau's American Community Survey (2023 data): US\$86,524 for a graduate or professional degree, US\$67,256 for a bachelor's degree, US\$39,428 for a high school diploma, and US\$31,660 without a high school diploma, and it notes the gap is increasing.
- **C.** The UPS delivery person uses GPS to take the best route, and the farmer in Iowa flies a drone to monitor soil conditions and crop yield. — Every word of this is in the chapter, which is what makes it tempting, but it is evidence for a different claim. Those two examples show that traditional occupations now use information systems too, which is why the line between knowledge workers and manual workers is blurring. They say nothing about knowledge being worth as much as land, labor, or capital; the earnings comparison is what carries that argument.
- **D.** A college degree raises an individual worker's pay, which is exactly what a median earnings figure reports. — This misreads what a median is. A median is the midpoint of a whole group, so it compares two populations rather than predicting what any one person would earn. It would be right if it said degree holders as a group earn more at the midpoint than non-degree holders as a group.

## The digital divide

All of that assumes you are on the inside. The **digital divide** is the gap between those with access to information systems, who hold great advantages, and those without — a major ethical challenge, because computer literacy decides who can compete.

**Inside America** — the gap is shrinking fast, yet rural communities, the elderly, people with disabilities, and minorities lag national averages for internet access and computer literacy.

**Outside developed countries** — infrastructure and money are lacking, and most developing countries have neither affordable internet access nor efficient electronic payment methods.

Two people can both be called offline when the fix for one is a cheaper data plan and for the other a power grid.

Meanwhile the line blurs: the UPS driver follows the route GPS picks. Workers of the future must become **learning workers**, because what matters is knowing how to learn.

#### COMPLETE IT

### The vocabulary of the knowledge society

Choose the word or phrase the chapter uses in each sentence; the explanation appears once you pick.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

- Professionals who are relatively well educated and who create, modify, and/or synthesize knowledge as a fundamental part of their jobs are **knowledge workers**, a term Peter Drucker coined in 1959. Drucker coined knowledge worker; manual worker is the category the chapter contrasts it with, and learning worker is the future role the chapter says everyone must grow into.
- Drucker predicted that as those workers grew in number and influence, a **knowledge society** would emerge, with education as its cornerstone and knowledge as important as land, labor, or capital. Knowledge society is Drucker's own term; the chapter notes others call the same thing the knowledge economy, the new economy, the digital society, the network era, or the internet era, and that this book calls it the digital world.
- Knowledge workers currently make up about **a quarter** of the workforce in the United States and other developed nations, and their numbers are rising quickly. The chapter's figure is about a quarter, which is easy to overestimate because the chapter separately says the lines between knowledge workers and manual workers are blurring.
- The gap between those who have access to information systems and those who do not is the **digital divide**, which the chapter calls one of the major ethical challenges facing society today. It counts as an ethical challenge because of the strong linkage between computer literacy and a person's ability to compete in the digital world, not merely because it is inconvenient.
- Inside America that gap is rapidly shrinking, but people in rural communities, the elderly, people with disabilities, and minorities still lag behind national averages for internet access and **computer literacy**. The chapter pairs internet access with computer literacy because having a connection and knowing how to use it are two separate barriers, and closing only one leaves the divide open.
- In most developing countries the obstacles are much harder to overcome, because infrastructure and financial resources are lacking and modern informational resources such as affordable internet access or **efficient electronic payment methods** are missing. The chapter names affordable internet access and efficient electronic payment methods specifically, which is why the divide abroad is an economic and infrastructure problem rather than a device problem.

- Because what will matter most is not the knowledge itself but the knowledge of how to learn, the chapter says workers of the future need to become **learning workers**. Learning worker is the chapter's forward-looking term; systems analyst and data entry clerk are examples of how narrowly people once pictured computer-related occupations.

## Globalization and outsourcing

**Globalization** is the integration of economies worldwide, enabled by technological progress and visible in the movement of commodities, money, information and labor. Falling transport and telecommunication costs opened new middle-class markets and made the talent pool global.

That last one has a name. **Outsourcing** is moving business processes or tasks — accounting, manufacturing, security — to another company or country, and cost is only one of eight reasons for it. U.S. hospitals have X-rays read abroad while American doctors sleep. Then the bill arrives, in three parts.

**Governmental challenges** — differences in political systems, regulation, laws, standards, or individual freedoms, so what is routine at home may be forbidden there.

**Geoeconomic challenges** — differences in infrastructure, demographics, welfare, or the expertise of the workers a company can actually hire.

**Cultural challenges** — differences in language, beliefs, attitudes, religion, or life focus, including sharply differing views on intellectual property.

Location is therefore chosen carefully, and the map moves: established destinations decline as salaries rise there, pushing outsourcers toward cheaper ones. Sort the statements below into chapter fact and common myth.

**SORT**

### The digital divide and globalization: accurate or a common misconception?

Drop each statement into the bucket where it belongs, then read why the chapter puts it there.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Accurate** — matches what the chapter actually says

- The digital divide in America is rapidly shrinking, yet rural communities, the elderly, people with disabilities, and minorities still lag national averages. — The chapter reports both halves at once: real national progress, and specific groups still behind on internet access and computer literacy.

- Outside developed countries the gap widens, because many developing countries lack affordable internet access and efficient electronic payment methods. — The chapter attributes the wider gap to missing infrastructure and financial resources rather than to individual choices.
- Outsourcing means moving business processes or tasks to another company, which may or may not be in another country. — The chapter's definition covers both: moving processes such as accounting, manufacturing, or security to another company or another country.
- U.S. hospitals outsource the reading of X-rays to skilled radiologists abroad, often while doctors in the United States are sleeping. — The chapter uses this to show that even highly specialized professional services, not just routine tasks, can move across borders.
- Singapore, Canada, and Ireland are declining as outsourcing destinations because salaries there are rising, pushing outsourcers toward Bulgaria, Egypt, Ghana, Bangladesh, or Vietnam. — The chapter treats the outsourcing map as something that keeps moving, which is why an outsourcing decision has to be revisited rather than made once.

**Common misconception** — sounds reasonable, but the chapter says otherwise

- The digital divide is essentially about who can afford the newest smartphone. — It is about access to information systems and the computer literacy that lets a person compete at all; a new phone with no affordable connection, no payment system, and no skills does not close it.
- Companies outsource mainly to pay lower wages, so cost is the reason that really matters. — Reducing or controlling costs is one of eight listed reasons, alongside freeing internal resources, gaining world-class capabilities, increasing revenue potential, reducing time to market, increasing process efficiencies, focusing on core activities, and compensating for missing skills.
- Once a company has compared labor costs across countries, it has the information it needs to pick an outsourcing location. — Location choice also involves governmental challenges (political systems, regulation, laws, individual freedoms), geoeconomic challenges (infrastructure, demographics, welfare, expertise), and cultural challenges (language, beliefs, religion, views on intellectual property), plus English proficiency and geopolitical risk.
- Globalization pushes cultures apart, since every country now produces media for its own audience. — The chapter says the reverse: movies, television, and other media have moved cultures closer together, creating shared understanding about norms, desirable goods and services, and even forms of government.
- Spreading internet access reliably expands the free flow of communication in the countries it reaches. — The chapter notes the same technology has facilitated an increase in authoritarianism and restrictions on the free flow of communication and content, naming WeChat and TikTok.

## Societal issues

Consulting firms, governments and the World Economic Forum treat these as interrelated megatrends: none can be managed alone. Six recur.

- **Demographic changes** — populations aging in developed countries while other regions grow fast, moving where workers and customers will be.
- **Urbanization** — the move from rural areas into cities, which now hold roughly half the world's population and have to stay livable.
- **Resource scarcity** — finite fossil fuels and materials against population growth, trade, and consumption that keep pushing demand up.
- **Climate change** — long-term shifts in temperature and weather, bringing rising seas and severer storms to individuals, businesses, and whole societies.
- **Shifts in economic power** — established economies losing dominant positions over purchasing power and resources, which has to be worked out politically.
- **Sustainable development** — the standard applied to all of it: meeting present needs without wrecking future generations' ability to meet theirs.

Open each below and read all four panels.

### EXPLORE

## The societal issues of the digital world

Open each issue and read all four panels; the last panel is the part managers underestimate.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

### Demographic changes · Population structure

- **What it is:** Changes in the structure of populations related to factors such as age, birth rates, and migration.
- **Where you can see it:** Many countries in the developed world see rapidly aging populations, while developing regions such as Africa are expected to rise rapidly in population, fueling massive global population growth.
- **Why it matters now:** These differences shift the balance of demand and supply of labor, so the workers an organization needs and the customers it sells to are increasingly in different places.
- **What makes it hard:** Differences in welfare are likely to keep increasing, and many countries are already experiencing both positive and negative effects of mass migrations, making this a political problem as much as an economic one.

### Urbanization · Rural to urban

- **What it is:** The movement of rural populations to urban areas.

- **Where you can see it:** 50 percent of the world's population is now living in cities, and cities are where the chapter's own public-infrastructure examples sit, such as adaptive traffic control systems and variable speed limits.
- **Why it matters now:** Every additional city dweller needs housing, transport, water, and power inside the same square miles, so density turns ordinary services into engineering problems.
- **What makes it hard:** Sustaining that growth while providing livable environments for the inhabitants will pose major challenges, and the growth does not pause while the solutions are being built.

#### **Shifts in economic power** · Where the leverage sits

- **What it is:** Changes in countries' purchasing power and control over natural resources.
- **Where you can see it:** Established economies are losing their dominating positions in the world's economy, while a new middle class in developing countries becomes a market of millions of new customers.
- **Why it matters now:** Where purchasing power sits determines which markets a company must serve and which countries set the terms of trade it has to accept.
- **What makes it hard:** The chapter notes the shift results in the need to resolve political struggles, and political struggle is exactly the governmental challenge that makes operating across borders risky.

#### **Resource scarcity** · Finite supply

- **What it is:** Pressure created by the limited availability of fossil fuels and other natural resources.
- **Where you can see it:** Population growth, global trade, consumerism, and other factors contribute to increasing waste and pollution as well as a growing need for resources.
- **Why it matters now:** Humans already live beyond the finite natural resources the planet can provide, so demand and supply are moving in opposite directions rather than converging.
- **What makes it hard:** Scarcity is entangled with every other issue on this list: more people, in more cities, with more purchasing power, all need more of the same finite materials.

#### **Climate change** · Long-term shifts

- **What it is:** Large-scale, long-term shifts in regional and global temperatures and weather patterns.
- **Where you can see it:** Changing weather patterns, a rise in sea levels, and an increase in the severity of storms.
- **Why it matters now:** These pose many challenges for individuals, societies, and the world, including for any organization whose stores, factories, or supply routes sit in the path of a storm.

- **What makes it hard:** The chapter is careful about scope here: the consequences arrive regardless of what one concludes about the causes, so an organization still has to plan around them.

#### **Sustainable development** · Present versus future

- **What it is:** Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.
- **Where you can see it:** The wording is the World Commission on Environment and Development's, from 1987, and the chapter attaches it to every issue above it: waste, pollution, and the need for resources keep rising while the planet's supply is finite.
- **Why it matters now:** The chapter says it will become an increasingly important aspect, which turns a moral argument into a test a manager can apply to a proposal: does this option leave the next generation the same range of choices?
- **What makes it hard:** The needs of the present are concrete and loud while the needs of future generations are abstract and silent, which is why the trade-off usually gets decided in favor of the present.

Technology sits on both sides of these problems. **Healthcare IS** support everything from diagnosis and treatment to analyzing disease data and running hospitals.

#### **CHECK YOURSELF**

#### **Globalization, societal issues, and healthcare IS**

Answer each question, then read all four explanations before moving on.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A U.S. hospital sends overnight X-ray images to skilled radiologists in another country to be read while American doctors sleep. Which term does the chapter use for this specific arrangement?

- **A.** Globalization, which the chapter defines as the practice of hiring workers in lower-wage countries. — Globalization is the right family but the wrong definition. The chapter defines globalization as the integration of economies throughout the world, enabled by innovation and technological progress; outsourcing is one activity that integration makes possible. This would be right if it defined globalization properly and called the X-ray example an instance of it.
- **B.** Outsourcing, which shows that even highly specialized professional services can move to another company or country. — **correct.** Outsourcing is the moving of business processes or tasks to another company or another country, and the chapter uses radiology precisely to show that even highly specialized services are candidates.

- **C.** The digital divide, because the country receiving the work has less access to information systems. — The digital divide is the gap between those with and those without access to information systems. It is a separate idea, and it does not fit here: the radiologists reading these images clearly have access to advanced systems.
- **D.** A geoeconomic challenge, because the arrangement rests on differences in infrastructure, welfare, and workers' expertise between the two countries. — Geoeconomic challenges are genuinely differences in infrastructure, demographics, welfare, or workers' expertise, so the words fit the situation. But the chapter uses that term for the difficulties an organization has to manage once it operates across borders, not as a name for the practice of moving the work. It would be right if the question asked what makes choosing this location risky.

2. The chapter quotes a 1987 definition of **sustainable development**. Which statement matches it?

- **A.** Growth a company can maintain year after year without losing market share to competitors. — This is the everyday business sense of 'sustainable' meaning repeatable, and it is the most common mix-up. The chapter's definition is about generations rather than quarters, and this option would be right in a discussion of business strategy instead of societal issues.
- **B.** Reducing waste through recycling programs so that materials are reused rather than discarded. — Recycling is one practice that can serve sustainable development, but the definition is broader: it covers every kind of development, including energy, housing, and agriculture, and it states a constraint on the future rather than a technique in the present.
- **C.** Development that meets the needs of the present without compromising the ability of future generations to meet their own needs. — **correct.** That is the World Commission on Environment and Development's 1987 wording, quoted in the chapter, which presents it as an increasingly important standard as resource scarcity and climate change intensify.
- **D.** Economic activity that avoids the use of fossil fuels, on the grounds that fossil fuels are the resource the chapter says is running short. — Fossil fuels appear in the chapter under resource scarcity, and using fewer of them may serve sustainable development, but the definition never names any specific resource. It would be right if it were offered as one example of sustainable development rather than as its definition.

3. How does the chapter define **healthcare IS**?

- **A.** Information systems supporting healthcare processes, from diagnosis and treatment to analyzing patient data to running offices and hospitals. — **correct.** The chapter's definition deliberately spans clinical work, analysis, and administration, which is why healthcare IS shows information systems reaching all the way into a traditional profession rather than being a single product.
- **B.** Software that lets patients look up symptoms, view test results, and book appointments online. — This is the patient-facing slice only. Patient portals are part of healthcare IS, but the definition also covers diagnosis and treatment, the analysis of patient and disease data, and the running of doctors' offices and hospitals, so this option is too narrow.

- **C.** The medical equipment itself, such as cardiac monitors and imaging machines, once it has been connected to a network and is sending its readings somewhere to be stored. — Devices are hardware. An information system includes the technology, the data, and the people and processes that use it; a cardiac monitor generates data that a healthcare IS then stores, analyzes, and routes to a physician.
- **D.** A specialization within computer science that prepares programmers to build software for hospitals. — This describes a field of study and the people trained in it, not a system. Healthcare IS refers to the systems themselves, which are used by professionals in the medical industry rather than only by programmers.

## Digital density and the digital future

Twenty years ago, driving to work left no record. The same drive today throws off tire pressure, engine temperature, position, and how hard you braked, poolable with a million other drivers' readings. The activity did not change; the connected data it produces did.

Two conditions set that up.

**Information technologies became pervasive** — used throughout society rather than locked in a computer room, so the technology sits inside ordinary activity.

**Innovation keeps accelerating** — radical innovations displace whole industries, which is how drones reached farms and film sets within a few years.

The result is an exponential rise in **digital density**: the amount of connected data per unit of activity, where every unit generates more connected data and enables new value-added interactions and business models. The chapter builds it in three moves — connections, data, interactions.

**Read that as a ratio, not a total.** Not how much data exists, but how much one unit of activity produces. A cash sale of a hammer in 2004 recorded one thing: a hammer left.

### INTERACTIVE DIAGRAM

### One idea, four angles

Pick a driver to redraw the diagram; each model is the same digital-density story told from a different starting point.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Connections** — Anything physical can now be wired into the digital world.

Mobile devices (Within reach 24/7) → Sensors (Detect, record, report) → IoT and IIoT (Things share data on their own) → Wearables (The body reports in)

- The old picture connected people, organizations, or computers; the new one connects **things** too, from a tire with a pressure sensor to a cow with an injectable ID chip.
- Cheaper **sensors**, better chips, and wireless radios are what made connecting objects affordable instead of theoretical.

- Phones stay within reach around the clock, so customer contact arrives in **micro-moments** rather than long desktop sessions.
- On a factory floor the same idea is the **Industrial Internet of Things**, where information technology and operations technology converge.

**Data** — Every connection leaves a record behind, and records pile up fast.

Volume (Extremely large datasets) → Variety (Many different types) → Velocity (Collected and analyzed faster) → Data quality (Fit to decide on?)

- **Big Data** is the chapter's name for extremely large and complex datasets high in volume, variety, and velocity.
- Social media poured in **unstructured** data, such as opinions about a product posted on blogs and networks.
- Storage got cheap and moved to the **cloud**, so keeping the data is rarely the constraint; making sense of it is.
- More data is not better data: **data quality** means completeness, accuracy, timeliness, validity, and consistency.

**Interactions** — Connected data becomes a different way of doing business.

Network effect (More users, more value) → APIs (Use someone else's system) → Servitization (Sensors sell it by usage) → Value from data (Small headcount, large value)

- The **network effect** means a network gets more valuable as more people use it, which is how Uber and Airbnb disrupted established industries.
- **APIs** let a company borrow functionality it never built; Lyft takes payments through Stripe and maps through Google.
- Sensors monitoring performance, temperature, or mileage turn a product into a service, which is how Bridgestone and Michelin can sell tires by usage while staying responsible for how the tires perform.
- Value now comes from data rather than headcount: digital-world firms reached the top with 2,500 to 35,000 employees.

**Workforce** — The same shift, arriving in your job description.

BYOD (Your device, real work) → Consumerization of IT (Consumer first, work second) → Computer literacy (Use the tool you know) → Computer fluency (Learn the next tool)

- **BYOD** means employees use personal devices for work that now reaches into customer relationship management and other enterprise systems.
- In the **consumerization of IT**, innovations reach shoppers first and organizations second, so managers must keep evaluating consumer technology.
- **Computer literacy**, knowing how to use a computer and certain applications, is now assumed rather than impressive.

- **Computer fluency** is independently learning new technologies as they emerge and assessing their impact on your work and life.

## Connections: anything physical can join the digital world

Connections once ran between people, organizations, or computers. Today almost any element of the physical world, **things** included, joins the digital realm. The first enabler is the mobile phone, kept within reach around the clock and, in the developing world, leapfrogging the PC entirely.

**Mobile apps** — companies must build these programs, each performing one well-defined function, because the customer visit moved onto the phone.

**Micro-moments** — interaction happens less in long desktop sessions and more in the instants when someone instinctively picks up a device to buy, know, do, or go.

The second enabler is the **Internet of Things (IoT)**: a network of physical objects that automatically share data over the internet — a tire with a pressure sensor, a smart meter, a cow with an injectable ID chip.

Sensors worn on a person are **wearable technologies** — clothing or accessories with electronics built in, such as a smartwatch, recording movement and heart rate alongside light, orientation and altitude. Logging that about yourself to improve health or performance is the **quantified self**.

It runs on **sensors**, devices that detect, record, and report changes in the physical environment. Cheaper sensors and better radios are why, already in 2008, more devices were connected to the internet than there were people on earth.

The same technology arrives under other names: **smart home technologies**, or **home automation**, and the **Industrial Internet of Things (IIoT)** in manufacturing. Open each connecting technology below and read all four facets.

### EXPLORE

## The technologies doing the connecting

Open each card and read all four facets; the fourth one is the part students usually skip.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

### Internet of Things · Connections

- **What it is:** A network of a broad range of physical objects that can automatically share data over the internet. The word automatically is the point: no person types the reading in.
- **Where you already see it:** A tire that reports its own pressure, a smart meter your utility reads remotely, an injectable ID chip in a cow, a thermostat you adjust from a parking lot.

- **Why it matters:** It is the reason a unit of activity generates data at all. Already in 2008, more devices were connected to the internet than there were people living on earth.
- **The catch:** Connecting an object is now the easy part. Analyzing the enormous, often unstructured data that results poses tremendous challenges for organizations.

### Sensors · Connections

- **What it is:** Devices that can detect, record, and report changes in the physical environment, such as temperature, movement, light, moisture, or heart rate.
- **Where you already see it:** Road surfaces that trigger dynamic speed limits when ice is likely, sensors reporting open parking spaces or traffic flow, soil-moisture sensors on golf courses.
- **Why it matters:** Decreasing sensor costs, plus advances in chips and wireless radios, are what turned the Internet of Things from an idea into an affordable build.
- **The catch:** A sensor reports; it does not judge. A miscalibrated sensor sends confident wrong numbers downstream, which is the data-quality problem in physical form.

### Wearable technologies · Connections

- **What it is:** Clothing or accessories that incorporate electronic technologies, such as the Apple Watch, Samsung's Galaxy Watch, or the Google Pixel Watch.
- **Where you already see it:** A smartwatch extending your phone by showing notifications and giving quick access to phone functions, or a Fitbit worn passively all day.
- **Why it matters:** Their sensors record physiological data such as body movements or heart rate and environmental data such as ambient light, orientation, or altitude.
- **The catch:** They enable the quantified self, the logging of all aspects of daily life including moods and physiological states, which is the most personal data a system can hold.

### Artificial intelligence · Interactions

- **What it is:** Using information technologies to simulate human intelligence, typically working alongside machine learning algorithms.
- **Where you already see it:** The software layer inside connected products: continuous sensor input paired with machine learning is what the chapter credits for advances in robotics, from self-parking cars to autonomous mining trucks.
- **Why it matters:** Without it the streams from connected things are just volume. AI is what lets organizations anticipate changes, coordinate resources, and personalize offerings.
- **The catch:** Its output inherits the quality of its input. Garbage in, garbage out applies here at enormous scale and without any obvious warning sign.

### Robotics · Interactions

- **What it is:** The use of robots to perform manual tasks, which the chapter treats as a direct result of sensors and AI advancing together.

- **Where you already see it:** Autonomous Caterpillar mining trucks already in use, self-parking systems sold in many vehicles, self-driving cars and trucks in active testing.
- **Why it matters:** It shows how connections and data become physical action: continuous sensor input paired with machine learning is what enables the advances.
- **The catch:** Machines taking over manual tasks is part of why the chapter argues that computer fluency, not one fixed skill, is what protects a career.

### 3D printing · Digital to physical

- **What it is:** Creating physical three-dimensional objects from digital models by successively adding thin layers of material.
- **Where you already see it:** Prototypes and manufactured parts that traditionally were drilled, cut, or milled out of a solid piece of material.
- **Why it matters:** Traditional machining can leave up to 90 percent of a slab as scrap, while additive printing builds only the object itself.
- **The catch:** Better, faster, cheaper printers also make counterfeit goods quick and inexpensive to produce, deceiving buyers and causing intellectual property losses.

## Data: what the connections leave behind

All of that generates **Big Data**: extremely large and complex datasets, defined by three characteristics.

**Volume** — the datasets are extremely large, because connected things report constantly rather than when asked.

**Variety** — many types of data arrive at once, from a sensor reading to a paragraph of opinion.

**Velocity** — data are collected and analyzed at ever-increasing rates, so the window for acting keeps shrinking.

Social media added a flood of **unstructured** data, text nobody designed to fit a spreadsheet column. Storage keeps getting cheaper, so organizations hold it in the cloud and analyze it there.

Volume is not virtue: garbage in, garbage out. **Data quality** asks whether data are reliable enough to decide on, in five tests — completeness, accuracy, timeliness, validity, consistency.

A report built on stale readings does not fail loudly; it quietly recommends the wrong thing.

Three questions on connections and data.

**CHECK YOURSELF****Connections and data**

Choose the best answer, then read every explanation, including the ones for options you did not pick.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A ride-hailing trip today produces route data, driver and rider ratings, payment records, and map corrections. The same trip by taxi in 2004 produced a paper receipt. Which chapter term names that change?

- **A.** The digital divide, because the two trips rely on different levels of technology — The digital divide is the gap in which those with access to information systems hold great advantages over those without. It would be right for a question comparing this rider to someone with no computer access or skills at all.
- **B.** Digital density, because each unit of activity now generates far more connected data — **correct.** Digital density is the amount of connected data per unit of activity, and here the unit of activity is identical while the connected data around it multiplied.
- **C.** The network effect, because many more riders joined the service over those years — The network effect is the rise in a network's value as more users join. That is true of ride-hailing, but it describes value from many users, not how much data a single trip creates.
- **D.** Big Data, because a trip record is an extremely large and complex dataset — Big Data names the datasets that result, defined by high volume, variety, and velocity. One trip's records are not Big Data; the pooled records of millions of trips are.

**2.** A dairy farm gives each cow an injectable ID chip that automatically uploads the animal's location and health readings. In the chapter's vocabulary, this is an example of:

- **A.** The Internet of Things, because a physical object automatically shares data over the internet — **correct.** The chapter uses the cow with an injectable ID chip as one of its own examples of a thing on the Internet of Things: a physical object automatically sharing data over the internet.
- **B.** The Industrial Internet of Things, because the animals are part of a production operation — The Industrial Internet of Things is the chapter's term for IoT technologies used in manufacturing, where information technology converges with operations technology. Sensors on an assembly line would fit; livestock in a field does not.
- **C.** Wearable technology, because the chip travels on the animal all day — Wearable technologies are clothing or accessories that incorporate electronic technologies, such as an Apple Watch or a Fitbit. This option would be right if the device were worn as an accessory rather than implanted.
- **D.** The quantified self, because everything about the cow's day is being logged — The quantified self is a person logging all aspects of their own daily life to improve their health and performance. The defining feature is self-tracking by a person, not a farm tracking an animal.

**3.** Which set of characteristics does the chapter use to define Big Data?

- **A.** Completeness, accuracy, timeliness, validity, and consistency — Those five are the chapter's definition of data quality, which is how you judge whether data are reliable enough to decide on. They describe the fitness of data, not its scale.
- **B.** Volume, veracity, and value — Veracity and value appear in other popular lists of data attributes and veracity is a genuine concern, but the chapter's three characteristics are volume, variety, and velocity.
- **C.** High volume, variety, and velocity — **correct.** Big Data is described as extremely large and complex datasets of high volume, variety (many different types of data), and velocity (collected and analyzed at ever-increasing rates).
- **D.** Cloud storage, machine learning, and unstructured social media posts — Those three are things the chapter associates with Big Data — cloud computing stores it, machine learning helps make sense of it, social posts supply unstructured input — but they are sources and tools, not the characteristics that define it.

## Interactions: connected data becomes a business model

Connections and data together enable new value-added interactions, because value now comes from data rather than headcount: Airbnb and Nvidia create it with a fraction of the workforce GE or Ford needed.

Continuous sensor input paired with machine learning and **artificial intelligence (AI)**, using information technologies to simulate human intelligence, makes sense of those streams and drives **robotics**, the use of robots for manual tasks. Two patterns recur.

**The network effect** — a network is worth more as more people use it, which is how Uber and Airbnb disrupted established industries: riders come for drivers, drivers come for riders.

**Servitization** — a firm shifts from selling a product to providing it as a service, so sensors let Michelin sell tires by usage and stay responsible for them.

Data once improved efficiency; connected data now lets firms anticipate change and personalize offerings.

## The API economy

**APIs (application programming interfaces)** are intermediaries that let software components exchange data or functionality over common web protocols, so a provider opens part of itself without anyone needing to know its inner workings. Think of a power socket: a standard interface delivering electricity you use without knowing how it was made.

**Providers gain revenue and reach** — Stripe processes payments for companies from Target to Lyft, on infrastructure none of them built.

**Users build on functionality they never wrote** — Lyft pulls maps from Google Maps, and when Stripe changes inside, the interface holds still.

Match each term below with its definition.

**MATCH**

## **The vocabulary of digital density**

Pair each term with the chapter's definition, then read the note to see what separates it from its nearest neighbor.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

### **Digital density**

The amount of connected data per unit of activity It is a ratio, not a total: the same trip or sale now generates far more connected data than it once did.

### **Internet of Things (IoT)**

A network of a broad range of physical objects that can automatically share data over the internet Automatically is the key word; a person emailing a photo is not the Internet of Things.

### **Sensors**

Devices that can detect, record, and report changes in the physical environment Falling sensor costs plus better chips and wireless radios are what made the Internet of Things practical.

### **Quantified self**

The logging of all aspects of one's daily life to improve health and performance Activity trackers such as the Fitbit are designed to be worn passively so the logging never has to be started.

### **Micro-moments**

The instinctive moments when a person picks up a mobile device to buy, know, do, or go They replaced the long, well-defined desktop session as the typical shape of a customer interaction.

### **Network effect**

The value of a network increases with the number of other users It explains why Uber and Airbnb could disrupt established industries so quickly once they had critical mass.

### **APIs**

Intermediaries that let different software components exchange data or functionality over common web protocols Like a power socket, an API delivers a service in a standard format without exposing how it is produced.

### **Big Data**

Extremely large and complex datasets of high volume, variety, and velocity Volume alone is not the definition; variety means many different types of data and velocity means data collected and analyzed at ever-increasing rates.

## Digital density and the workforce

Employees increasingly use their own devices and familiar software for work, not just for email but for enterprise systems. **BYOD** (bring your own device) worries managers over security, compliance, and support costs, though it can raise productivity and retention.

Innovations now reach the consumer marketplace first and organizations second: the **consumerization of IT**. Knowing how to operate a computer and common applications is **computer literacy**, which can decide whether a person is employed at all.

**Computer fluency** goes further: independently learning new technologies as they emerge and assessing their impact. Literacy gets you in the door; fluency sets you apart later.

### CHECK YOURSELF

## Interactions, APIs, and the workforce

Choose the best answer, then read all four explanations so the near-miss options stop looking attractive.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. Airbnb becomes more useful to travelers as more hosts list on it, and more useful to hosts as more travelers search it. The chapter's name for this is:

- **A.** Economies of scale, because Airbnb's cost per booking falls as the company grows — Economies of scale is about a company's own costs per unit falling as volume rises. That may well be true here, but it says nothing about the value each user gets from other users.
- **B.** The consumerization of IT, because ordinary consumers supply the rooms — The consumerization of IT is the pattern of technologies appearing in the consumer marketplace before organizations adopt them. It describes where a technology shows up first, not why a platform gains value as it fills up.
- **C.** The network effect, because a network's value rises with the number of other users — **correct.** The network effect is the notion that the value of a network, or of a tool or application based on a network, increases with the number of other users, and the chapter names Uber and Airbnb as examples.
- **D.** Servitization, because Airbnb provides a service rather than selling a product — Servitization is the shift from selling physical products to providing them as services, as when Bridgestone and Michelin sell tires by usage. Airbnb never sold the room as a product to begin with.

2. Lyft handles payments by connecting to Stripe's API and shows maps by connecting to Google Maps' API. Using the chapter's power-socket analogy, what does an API give the company that uses it?

- **A.** A copy of the provider's software that Lyft can host and modify on its own servers — That describes licensing software to run yourself, which is exactly what an API lets a company avoid. If Stripe handed over its code, Lyft would inherit the job of running payments reliably and securely.
- **B.** A direct connection into the provider's database so Lyft can query the provider's tables — APIs are intermediaries that exchange data or functionality over common web communication protocols, and they deliberately hide the provider's internal structure. Direct table access would break that, since Stripe rewrites its internals without API users noticing.
- **C.** Ownership of the data the provider processes on Lyft's behalf — Using an API is consuming someone else's service, not acquiring their data. The chapter frames the value as two-sided: the provider gains revenue and reach, the user gains functionality it did not have to build.
- **D.** Access to the provider's functionality through a standard interface — **correct.** Like a power socket, the API supplies the service in a standardized format while the user stays ignorant of how it is produced, which is why Stripe can change its systems and Lyft notices nothing.

3. An employee is fast with the payroll application she has used for six years, but when the company replaces it she waits to be trained rather than exploring it on her own. In the chapter's terms:

- **A.** She is computer fluent but not computer literate — This reverses the two terms. Fluency is the broader ability and is defined as learning new technologies independently, so it cannot exist without the basic skills that literacy names.
- **B.** She is computer literate but not computer fluent — **correct.** Computer literacy is knowing how to use a computer and certain applications, which she clearly has; computer fluency is independently learning new technologies as they emerge and assessing their impact, which is exactly what she is not doing.
- **C.** She is an information have-not, because she cannot use the replacement system without being trained — Information have-nots are people with limited or no computer access or skills, contrasted with information haves who have almost unlimited access to information. Waiting for training on one new system is not the same as having no access or skills at all.
- **D.** Neither term applies, because literacy and fluency describe an organization's technology rather than one employee's skills — Both terms describe a person, not a system: literacy is knowing how to use a computer and certain applications, and fluency is that person independently learning new technologies as they emerge and judging their impact on her work and life.

## What an information system actually is

Ask most people what an information system is and they point at the computers. An **information system (IS)** is the combination of people and information technology that creates, collects, processes, stores, and distributes useful data. Every clause is load-bearing.

### The definition, clause by clause

**People and information technology.** The people half is not politeness. Technologies may operate autonomously, but they do not build themselves and do not exist for their own sake: they are created to serve a useful purpose for people. A server humming in a closet, with nobody deciding what it does, is not one.

**Creates, collects, processes, stores, distributes.** Those five verbs are the job description, and a working system does all five. Watch them run in order.

- 1 **Creates** — tap your card at a store and the register creates a record of the sale.
- 2 **Collects** — the network collects that record into a central system.
- 3 **Processes** — software processes those records into the day's totals.
- 4 **Stores** — a database stores the result, so it outlives the moment it was made.
- 5 **Distributes** — a report distributes it to a manager who decides what to reorder.

Drop one verb and the chain breaks: data created but never distributed help nobody, and data distributed but never processed are noise arriving faster.

**Useful data.** Any information system involves data useful for someone, somewhere, often for two someones at once: a status update is useful to your friends and to the platform, a game score to the player and the developers. So the first question about any system is, useful to whom, for what decision?

#### COMPLETE IT

### Rebuild the two definitions word for word

Choose the option that makes each sentence match the chapter's definition exactly.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

- An information system is the combination of people and **information technology** that creates, collects, processes, stores, and distributes useful data. The definition pairs people with information technology as a whole. Hardware is only one of IT's three parts, and business processes are what the system supports, not a component of it.
- Information technology includes hardware, software, and **telecommunications networks**. An operating system is itself software, and a database is something software manages and hardware stores, so neither is a separate component. The third named component of IT is the telecommunications networks that link systems together.
- A program or set of programs that tell the computer to perform certain tasks is **software**. Hardware is the physical equipment; the instructions telling that equipment which processing functions to perform are software.
- A group of two or more computer systems linked together with communications equipment is a **telecommunications network**. An information system would also need people and useful data, and systems integration is the work of connecting separate systems, not the linked equipment itself.
- Physical computer equipment such as a computer, tablet, or printer, plus components like a monitor or keyboard, is **hardware**. Hardware is the physical equipment, and the chapter extends it to sensors, cameras, and actuators as well as traditional computer components.
- An information system creates, collects, processes, stores, and distributes data that are **useful** for someone, somewhere. Accuracy and timeliness are two of the five parts of data quality, which is a separate idea; the word in the definition itself is useful, as in transactional data that are useful for businesses and game scores that are useful for the player and the developers alike.

## The technology half

**Information technology (IT)** includes hardware, software, and telecommunications networks. Three parts, no more and no fewer.

**Hardware** is physical computer equipment — computer, tablet, printer, monitor, keyboard — and now sensors, cameras, and actuators, which is why a thermostat counts.

**Software** is a program or set of programs telling the computer to perform certain tasks. It supplies the instructions, so two identical laptops running different software run two different companies.

**Telecommunications networks** are two or more computer systems linked with communications equipment. Without them every system is an island holding data nobody else can reach.

Hardware is easier to place once you know what it replaced. Before computers, information systems were file folders and filing cabinets, with an abacus or slide rule for the arithmetic. Those were real systems; only the machinery changed.

**IS or IT?** Traditionally IT meant the hardware, software, and networking components of an information system. That difference is shrinking and many now use the terms synonymously. When a question forces the contrast: IT is the technology, an IS is that technology plus people, aimed at useful data.

#### CHECK YOURSELF

#### The definition and its technology components

Pick the best answer, then read every explanation, including the ones for options you did not choose.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. Which statement matches the chapter's definition of an **information system**?

- **A.** A computer program that stores and retrieves an organization's records. — That describes a piece of software, which is one part of the information technology half of an information system. It leaves out the people, the hardware the program runs on, the network it uses, and the useful data that is the point of the whole thing. It would be right if the question had asked what software is.
- **B.** The hardware, software, and telecommunications networks an organization owns, together with the budget and the contracts under which they were bought. — That is the chapter's definition of information technology, not of an information system. Add the people and the useful data the system produces and you would have the IS definition.
- **C.** The combination of people and information technology that creates, collects, processes, stores, and distributes useful data. — **correct.** and it is the chapter's wording exactly: two halves (people and information technology), five verbs (creates, collects, processes, stores, distributes), and an ending that names the output as useful data.
- **D.** The department that maintains an organization's computers and user accounts. — The chapter does note that information systems is also used to name the field of people who develop, use, manage, and study these systems, so this is a real part of the IS world. But a support department is a group of people doing a job, not the system itself.

2. A hospital installs a small sensor that reads a patient's heart rate and passes each reading over the hospital's wireless link to a monitoring program running on a server in another building. Which part of that description is **hardware**?

- **A.** The monitoring program that reads each number and raises an alarm. — A program is software: a set of programs that tell the computer what processing functions to perform. It runs on hardware but is not hardware itself.

- **B.** The sensor on the patient and the server that receives the readings. — **correct.** Hardware is physical computer equipment, and the chapter explicitly extends it beyond traditional computer components to input and output devices such as sensors, cameras, and actuators. A server is physical equipment too.
- **C.** The wireless link connecting the patient's floor to the other building. — That is a telecommunications network: two or more computer systems linked together with communications equipment. The equipment carrying the link is physical, but what the option names is the connection between systems, which the chapter counts as the third IT component rather than as hardware.
- **D.** The heart-rate readings the sensor produces. — Readings are data, the raw symbols the system creates and collects. They are what the hardware produces, not the hardware.

**3.** The chapter says the difference between IS and IT is shrinking and that many people use the terms synonymously. What distinction does it still draw between them?

- **A.** Information technology is what an organization buys from vendors; an information system is what it develops itself. — Where a technology came from has nothing to do with the definitions. A purchased payroll package and an in-house one are both software, and both sit inside an information system that also has people and data.
- **B.** Information technology is technology used by consumers; an information system is technology used inside businesses. — The chapter draws no consumer-versus-business line here. Its own examples of useful data include a personal news feed and scores in a computer game right alongside business transactional data.
- **C.** Information technology is the newer term, and it has replaced information system in modern organizations. — This has the history backwards. Information technology is the older, narrower term for the hardware, software, and networking components; the two terms are now often used interchangeably, but neither replaced the other.
- **D.** Information technology is the hardware, software and networks; an information system adds people and useful data. — **correct.** Traditionally IT named the hardware, software, and networking components of an information system, while the IS definition wraps those components together with people and points them at producing useful data.

## Data, information, knowledge

Unformatted data, or simply **data**, are raw symbols such as characters and numbers, with no meaning in themselves and little value until processed. Take the string 465889727: asked what it means, you could not say, because there is nothing there to know.

Garbage in, garbage out applies here, which is why **data quality** — completeness, accuracy, timeliness, validity, consistency — decides whether data are reliable enough to decide on.

Data formatted or processed to make them useful become **information**: a representation of reality that helps answer who, what, where, and when. Watch it happen to those nine symbols.

- 1 Present them as 465-88-9727, and the grouping alone looks familiar.
- 2 Add that they sit in a certain database, in John Doe's record.
- 3 Add a field labeled SSN, and you rightly surmise this is John Doe's Social Security number.

Nothing about the digits changed; a label did the work. A raw list of every payment you made last month is fairly useless data, while the same payments split into sent and received is incredibly useful information you could budget from.

A screen still decides nothing. **Knowledge** is the ability to understand information, form opinions, and make decisions or predictions from it. It has two faces.

**An ability a person brings in** — knowing that only one Social Security number can uniquely identify an individual, which the database never told you.

**A body of governing procedures** — guidelines or rules used to organize or manipulate data for a task, such as the rule that two records sharing one number are an error.

|                   | DATA        | INFORMATION    | KNOWLEDGE              |
|-------------------|-------------|----------------|------------------------|
| <b>Example</b>    | 465889727   | 465-88-9727    | 465-88-9727 → John Doe |
| <b>What it is</b> | Raw symbols | Formatted data | Data relationships     |
| <b>Meaning</b>    | ???         | SSN            | SSN → unique person    |

**The trap.** A well-formatted, clearly labeled, freshly generated report is still information, however impressive it looks. It becomes knowledge only inside someone who can read it, judge what it means, and decide.

Follow the three stages through one working system.

#### INTERACTIVE DIAGRAM

### From raw symbols to a working system

Select each stage to see what it contains and what has to be added to reach the next one.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Data** — Raw symbols, carrying no meaning yet

465889727 (Nine characters) → No label (Nothing states what it is) → No answer (You cannot say what it stands for)

- **Data** are raw symbols, such as characters and numbers, and they have no meaning in and of themselves.
- Data are of little value until processed, which is why the chapter calls them the root rather than the result.
- **Data quality** decides whether they are reliable enough to make decisions on: completeness, accuracy, timeliness, validity, and consistency.
- Garbage in, garbage out — every later stage inherits whatever defects are in the symbols.

**Information** — Formatted data that answers who, what, where, and when

465-88-9727 (Formatted) → Field: SSN (A label supplies context) → Record: John Doe (It now points at someone)

- **Information** is a representation of reality that helps answer questions about who, what, where, and when.
- Data become information when they are formatted, organized, or processed — the digits themselves never changed.
- Contextual cues, such as a label, are what make data familiar and useful to a reader.
- The month of payments split into **sent** and **received** is the same move: useless list in, usable budget table out.

**Knowledge** — The ability to understand information and act on it

SSN to person (A relationship, not a value) → One number, one person (A governing rule) → Decide or predict (Approve, flag, or reject)

- **Knowledge** is the ability to understand information, form opinions, and make decisions or predictions from it.
- You need knowledge to be aware that only one Social Security number can uniquely identify each individual.
- Knowledge is also a body of governing procedures — guidelines or rules — used to organize or manipulate data for a given task.
- It is the only stage that lives in a person rather than in a file, which is why the people half of the IS definition is not optional.

**The whole system** — People plus information technology, producing useful data

People (Build, manage, and use it) → Hardware (Equipment, sensors, actuators) → Software (Instructions for the hardware) → Networks (Two or more systems linked)

- These four together make an **information system** that creates, collects, processes, stores, and distributes useful data.

- The three technology boxes are exactly what the chapter means by **information technology**: hardware, software, telecommunications networks.
- **Systems integration** connects separate, often modular systems and their data using technologies such as APIs, so the boxes of different systems can work as one.
- Useful means useful to somebody: transactional data serve the business, a news feed serves your friends and the platform, game scores serve the player and the developers.

Now place each item where it belongs at the moment described, not where it could end up.

**SORT**

## Data, information, or knowledge?

Place each item in the bucket that matches what it is at that moment, not what it could eventually become.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Data** — raw, unformatted symbols with no meaning attached

- The string 465889727, with no label and nothing around it — These are raw symbols, and the chapter's whole point is that you cannot say what they mean or stand for.
- A list of every payment you sent and received last month, in the order they happened — The chapter calls exactly this a fairly useless list until it is organized, because nothing about the order answers a question you care about.
- The value 98.6 arriving from a thermometer sensor with no label attached — A number produced by real equipment measuring a real body is still a raw symbol until something says what it is a measurement of.
- The characters a barcode scanner sends to the register when an item is scanned — The scanner creates raw symbols; software has to process them against a product file before anyone learns a price or a product name.

**Information** — formatted or organized so it answers who, what, where, when

- 465-88-9727 stored in a field named SSN inside John Doe's record — Formatting plus a label plus a record is exactly the context that turns the same digits into a representation of reality.
- That same month of payments shown in a table split into sent and received — Dividing the transactions into categories is the formatting step, and the chapter calls the result incredibly useful for managing a budget.
- A chart line reading Temperature 98.6 F, taken 3:14 p.m. — The label and the timestamp answer what and when, which is precisely the work information does.

**Knowledge** — understanding, rules, and the ability to decide

- Being aware that only one Social Security number can uniquely identify each individual — No database supplied that; it is the understanding a person brings, and it is what lets someone spot a duplicate as an error.
- Your conclusion, after reading that table, that you should stop paying for weeknight delivery — Forming an opinion and making a decision based on information is the chapter's definition of knowledge.
- The standing rule that any temperature above 100.4 F must be reported to a physician — The chapter describes knowledge as a body of governing procedures, such as guidelines or rules, used to make data suitable for a task.

## Systems integration

Components evolve rapidly, which makes tying everything together ever more important. Two terms name that work.

**Internetworking** is connecting host computers and their networks together to form even larger networks, as the internet does.

**Systems integration** is connecting separate and often modular information systems and data, using technologies such as APIs, to improve business processes and decision making.

Integration is also why “what kind of system is this?” has no tidy answer any more. Ten or fifteen years ago systems fell cleanly into one category; today enterprise systems span the whole organization, and few sit in only one.

### CHECK YOURSELF

### The distinction that gets tested most

Answer, then read all four explanations, because the wrong options here are the ones students actually pick.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A fitness band records the value **72** every ten seconds and stores each one with no label attached. At that moment, what is the value 72?

- **A.** Information, because the band captured the value and wrote it to storage, and being held in a system is what makes a number information. — Storing something changes where it lives, not what it is. Data become information when they are formatted, organized, or processed, and none of those has happened here; a stored value with no label is still data.

- **B.** Data, because it is a raw symbol that carries no meaning until something formats, organizes, or processes it. — **correct.** Unformatted data are raw symbols such as characters and numbers, with no meaning in and of themselves and little value until processed. That is precisely the state of an unlabeled 72.
- **C.** Knowledge, because the band acts on the value to decide when to buzz your wrist, and acting on a number is what knowledge means. — An automated threshold is a rule applied to data, not knowledge in the chapter's sense. Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it, so it needs a person who can judge what the reading means.
- **D.** Information, because 72 sits in the normal heart-rate range, so a reader can work out its meaning without a label. — This is the most tempting wrong answer, because you personally can guess what 72 means. But the chapter is explicit that contextual cues, such as a label, are needed to turn data into information. Attach the label heart rate, in beats per minute, at 2:15 p.m., and the same value becomes information.

2. The chapter contrasts a raw month of peer-to-peer payment transactions with a table that divides the same payments into sent and received. Which reading matches the chapter?

- **A.** The raw list is data and the table is information, because sorting them into categories makes them usable. — **correct.** The chapter calls the raw list fairly useless data and the categorized table incredibly useful information, and the only thing that changed between them is organization.
- **B.** Both are information, because both are records of payments that really happened. — Being true does not make something information. Truth is a data-quality question (accuracy); the data-versus-information question is about whether the values have been given form and context.
- **C.** The raw list is information and the table is knowledge, because the table can be used to make budgeting decisions. — Half right and half wrong, which is what makes it tempting. The raw list has no organization at all, so it is data, not information. And a table cannot be knowledge: knowledge is the ability to understand the table and decide, which lives in the person reading it.
- **D.** The raw list is data and the table is knowledge, because the app processed the transactions to build it. — Processing is how data become information, not how they become knowledge. The app can produce the table all by itself; it cannot form the opinion that your delivery spending is too high.

3. Which of these is **knowledge** in the chapter's sense?

- **A.** The nine digits 465889727, with nothing attached to them. — That is the chapter's example of data: raw symbols you cannot interpret. It would become information the moment a label and a record put it in context.
- **B.** The value 465-88-9727 sitting in a field labeled SSN in John Doe's record. — That is information: formatted data with contextual cues that answer who and what. It is one step short of knowledge, which is the step most students skip.

- **C.** Understanding that a Social Security number identifies exactly one person, so two records sharing one must be the same person or an error. — **correct.** Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it, and it also covers governing rules such as one number, one person, used to organize or manipulate data for a task.
- **D.** A printed report listing every customer's Social Security number beside their account balance, sorted so that the largest balances appear at the top of the first page. — That is a lot of information at once, cleanly formatted and labeled, but volume and polish are not knowledge. The report becomes knowledge only in a person who can read it, judge what it means, and act — flagging a duplicate, catching an error, approving a loan.

## The people who build, manage, and use information systems

Hardware, software, and networks do nothing on their own. The IS field is the people who develop, maintain, manage, and study information systems — plus one the chapter refuses to leave out: you, the user.

### Why IS jobs keep landing at the top of the rankings

Because data became so valuable for competitive advantage, every company is now a technology company. The **Occupational Outlook Handbook** (2025) predicts employment for computer and IS managers will grow **17 percent through 2033**, much faster than average and in nearly every industry.

| RANK | CAREER                    | JOB SCORE (OUT OF 5.0) | MEDIAN PAY (US\$) |
|------|---------------------------|------------------------|-------------------|
| 1    | Enterprise Architect      | 4.1                    | 144,997           |
| 2    | Full Stack Engineer       | 4.3                    | 101,794           |
| 3    | Data scientist            | 4.1                    | 120,000           |
| 4    | Dev Ops Engineer          | 4.2                    | 120,095           |
| 5    | Strategy Manager          | 4.2                    | 140,000           |
| 6    | Machine Learning engineer | 4.3                    | 130,489           |
| 7    | Data engineer             | 4.0                    | 113,960           |
| 8    | Software Engineer         | 3.9                    | 116,638           |
| 9    | Java Developer            | 4.1                    | 107,099           |
| 10   | Product Manager           | 4.0                    | 125,317           |

Table 1.1 · Best Jobs in America (2022), ranked by Glassdoor.

**Read that table honestly.** It is evidence, not proof.

- Rank is not pay: Java Developer ranks 9th on US\$107,099, ahead of Product Manager on US\$125,317.
- Rank is not the job score either: Enterprise Architect leads at 4.1 while two lower-ranked jobs score 4.3.
- It is one job site's 2022 snapshot, and rankings differ by source. What survives is the demand.

Pay follows demand, though the figure depends on who counts: the Bureau of Labor Statistics put median pay for these managers at US\$187,990 in 2024, Salary.com US\$149,316, and graduates start nearer US\$71,556.

## Careers in information systems

The field covers those who build systems, those who maintain them, and those who manage them. Titles are slippery: firms use one title differently, or different titles for one function.

| IS ACTIVITY | JOB TITLE                       | JOB DESCRIPTION                                                                                        | MEDIAN SALARY (US\$) |
|-------------|---------------------------------|--------------------------------------------------------------------------------------------------------|----------------------|
| Develop     | IS analyst                      | Analyze business requirements and select information systems that meet those needs                     | 73,581               |
| Develop     | Software architect IV           | Create customized software for large corporations                                                      | 145,613              |
| Maintain    | IT auditor I                    | Audit information systems and operating procedures for compliance with internal and external standards | 70,790               |
| Maintain    | Database administrator II       | Manage database and database management software use                                                   | 104,390              |
| Manage      | IS and cyber security manager   | Manage security measures and disaster recovery                                                         | 146,790              |
| Manage      | Chief information officer (CIO) | Highest-ranking IS manager; oversee strategic planning and IS use throughout the firm                  | 344,404              |

| IS ACTIVITY | JOB TITLE                   | JOB DESCRIPTION                                                          | MEDIAN SALARY (US\$) |
|-------------|-----------------------------|--------------------------------------------------------------------------|----------------------|
| Manage      | Chief digital officer (CDO) | Executive focused on converting traditional analog businesses to digital | 298,890              |

Table 1.2 · IS titles, grouped by develop, maintain, manage.

Read each description, then find the title it belongs to.

### MATCH

## Who actually does what

Pair each IS job title from Table 1.2 with the work that person really does; read the description first, then find the title it belongs to.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

### IS analyst

Analyze business requirements and select information systems that meet those needs This is a develop role, but the work starts with the business problem, not with code — which is why business competency matters even in an entry-level analyst job.

### Software architect IV

Create customized software for large corporations Median salary US\$145,613, well above Software developer I at US\$81,690, because architecture decisions shape everything built afterward.

### IT auditor I

Audit information systems and operating procedures for compliance with internal and external standards Auditing is filed under maintain, not manage: the auditor checks that systems and procedures match the rules the firm is held to.

### Database administrator II

Manage database and database management software use Do not confuse this with senior database engineer, who develops, modernizes, and streamlines databases; the administrator keeps the running database healthy and used properly.

### IS and cyber security manager

Manage security measures and disaster recovery Note that disaster recovery sits in the same job as security — the same pairing the chapter recommends against ransomware.

### Chief information officer (CIO)

Highest-ranking IS manager; oversee strategic planning and IS use throughout the firm At US\$344,404 median, the highest-paid title in Table 1.2, and the job is strategy across the whole firm rather than any one system.

### Chief digital officer (CDO)

Executive focused on converting traditional analog businesses to digital The CDO exists because of transformation work — taking a business built on paper and processes into mobile

apps, social media, and other fast-changing digital sectors.

## What makes IS personnel valuable

Good IS personnel hold **integrated** knowledge in three areas — **technical**, **business**, and **systems**. It is the business and systems areas that separate the IS professional from someone with only technical skills.

| DOMAIN           | WHAT IT COVERS                                                                                                                                                  |
|------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Technical</b> | Hardware, infrastructure, cloud; operating systems, application software, AI, APIs; network administration, wireless, 5G, cybersecurity                         |
| <b>Business</b>  | Business processes and industry; managing people and projects; social (interpersonal, group dynamics, political); communication (verbal, written, presentation) |
| <b>Systems</b>   | Systems integration; development methodologies; critical thinking; problem solving                                                                              |

Table 1.3 · Core competencies — note that social and communication skills sit under **business**.

Each domain carries one idea students get backwards.

**Technical: just enough, not expert** — enough to understand how the technology works and what it can do for an organization, then direct the specialists who know more.

**Business: it may save your job** — understanding the business those systems serve is what propels IS professionals into project management and upper-level positions.

**Systems: four named parts** — integration, development methodologies, critical thinking, and problem solving: identify, compare, choose.

People skills are filed under business, not under extras: interpersonal skill, group dynamics, politics, and presentation are capability that earns money.

**CHECK YOURSELF****Rankings, titles, and the three competencies**

Pick the best answer, then read every explanation — the wrong options are the misreadings this section is designed to prevent.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** In the Best Jobs in America table, Enterprise Architect ranks 1st with a job score of 4.1 while Full Stack Engineer ranks 2nd with a **higher** score of 4.3; Java Developer ranks 9th on a median of US\$107,099, ahead of Product Manager, who earns US\$125,317. What does that pattern tell you?

- **A.** Rank blends several factors at once, so a career can rank higher without having the top score or the top pay — **correct**. The two columns disagree with the ordering in both directions, so the rank must combine factors the table does not print. That is why the chapter says the rankings differ between sources while the overall demand story stays clear.
- **B.** Rank is a straight ordering by median pay, so the pay column should fall steadily from row 1 to row 10 — Ordering by pay is the most natural reading of a jobs list, but the pay column does not fall steadily: rank 2 pays US\$101,794 while rank 9 pays US\$107,099. This would be right only if rank and pay moved together, and here they cross.
- **C.** Job score measures how demanding the job is, so a lower score earns a higher rank — Job score is a rating of the career, not a difficulty index — the chapter reports it as a score out of 5.0 alongside pay. If it measured difficulty, the highest-scoring careers would sit at the bottom, and they do not.
- **D.** Rank counts openings in each career, so the score and pay columns describe the job without affecting its position — Hiring volume genuinely is one ingredient such lists use, so this is a fair guess. But the table publishes only rank, job score, and median pay; it reports no opening counts, so this evidence cannot support the claim.

**2.** The chapter says the IS professional must know “just enough” about hardware, software, and networking. What does that phrase actually mean?

- **A.** Technical knowledge is optional once you reach management, since business skills are what get people promoted — Business skill does drive promotion, which is what makes this tempting. But the chapter’s whole argument is that the three areas are *integrated*; a manager who cannot evaluate the technology is the person vendors sell to. This would be right only if the chapter described two competencies instead of three.
- **B.** The technical area is the easiest of the three to keep current, because the fundamentals rarely change — The chapter says the opposite: technical is perhaps the most difficult area to maintain, because of the rapid pace of technological innovation. This would be right if the hot-skills list stayed the same each decade, and it does not.

- **C.** Enough to understand how these technologies work, what they can do for an organization, and how they should be applied — **correct.** Enough understanding to judge fit and application, not enough to out-build the specialists — the IS professional typically manages or directs those with deeper, more detailed technical knowledge.
- **D.** Expert-level depth in each technical area, because you cannot direct specialists unless you know more than they do — The instinct that authority requires out-knowing your specialists is common, and it is the one the chapter denies: the IS professional typically manages or directs those who have deeper, more detailed technical knowledge. This would be right for a role defined by a single technology, but it leaves no room for the business and systems competencies.

**3.** Table 1.3 lists **Social** (interpersonal, group dynamics, political) and **Communication** (verbal, written, technological communication and presentation). Which of the three competency domains are those two filed under?

- **A.** Technical knowledge and skills — Technical covers hardware, software, networking, and security — cloud computing, virtualization, prompt engineering, APIs, 5G. This answer would be right if the question had asked where non-relational databases or wireless networks sit.
- **B.** Business knowledge and skills — **correct.** Business knowledge and skills has four entries: business integration and industry, managing people and projects, social, and communication. Filing people skills here is deliberate — they are treated as business capability that earns money, not as decoration.
- **C.** Systems knowledge and skills — Systems covers systems integration, development methodologies, critical thinking, and problem solving — how the pieces fit and how problems get solved. This would be right for “connectivity, compatibility, integrating subsystems and systems.”
- **D.** A fourth domain of soft skills, kept separate from the three core competencies — Calling them soft skills is exactly the habit the table pushes back on. The chapter names three domains and no fourth; putting social and communication inside business is how it insists they are core, not supplementary.

## You — the user

In almost any business field — finance, accounting, operations, marketing — you will use information systems constantly and be pulled into decisions about them. Three habits set you apart.

**Understanding what systems can do** — and what they cannot, so a vendor’s promise can be measured against what technology delivers.

**Talking to technical staff** — describing a business need in terms the people who build the system can act on.

**Making educated decisions** — taking part in the choice rather than waiting to be handed someone else's pick.

The risk is sharpest in a small organization with no IS department: without a grasp of infrastructure, systems analysis, and security, you are at the mercy of vendors selling their technology of the week.

**EXPLORE**

### Three competencies, plus you

Open each card and read all four facets; the fourth card is the one that applies to you no matter what you major in.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

#### **Technical competency** · The nuts and bolts

- **What it covers:** Knowledge and skills in hardware, software, networking, and security. Hardware means platforms, infrastructure, cloud computing, virtualization, peripherals, and mobile devices; software means operating systems, application software, non-relational databases, AI and machine learning, prompt engineering, mobile apps, and APIs; networking means network administration, wireless, 5G, and cybersecurity.
- **Where it shows up in real work:** Knowing enough to tell whether a vendor's cloud proposal actually fits the firm, or why a wireless network keeps dropping in half the building.
- **What goes wrong without it:** You cannot judge what is being promised to you, and you cannot tell a hard technical problem from an excuse. But being *only* technical is exactly what the chapter says leaves a job exposed to outsourcing.
- **Why the chapter says it pays:** It is the hardest area to maintain, because innovation is so rapid. The IS professional usually manages or directs the deeper specialists rather than out-coding them.

#### **Business competency** · What the work is for

- **What it covers:** Business processes, the functional areas of a business and how they integrate, and industry characteristics. It also covers planning, organizing, leading, and controlling, managing people and projects, social skills (interpersonal, group dynamics, political), and communication (verbal, written, technological, and presentation).
- **Where it shows up in real work:** Running a project to a deadline, writing a proposal a non-technical manager will approve, reading the politics of a group, presenting a recommendation to the people who control the budget.
- **What goes wrong without it:** Someone with only technical skill builds exactly what was requested instead of what the business needed, and then cannot argue for the funding to fix it.

- **Why the chapter says it pays:** The chapter says this competency sets IS professionals apart and, in an era of increased outsourcing, may well save a person's job; it is what propels them into project management and upper-level management.

### **Systems competency** · Making the pieces fit

- **What it covers:** Systems integration (connectivity, compatibility, joining subsystems into systems) and development methodologies (steps in systems analysis and design, the systems development life cycle, alternative methodologies). It also covers critical thinking, meaning challenging your own and others' assumptions and ideas, and problem solving: gathering and synthesizing information, identifying the problem, formulating and comparing solutions, choosing.
- **Where it shows up in real work:** Getting a new checkout system to exchange data with an inventory system written years earlier by a different vendor, and deciding which of three candidate fixes is genuinely worth doing.
- **What goes wrong without it:** You end up with components that are each correct and still will not work together, or a system that gets built before anyone identified the real problem.
- **Why the chapter says it pays:** Those who understand how to build and integrate systems and how to solve problems ultimately manage large, complex systems projects — and manage the people who have only technical knowledge.

### **You — the user** · Every major, every job

- **What it covers:** Not a job title but a role you already hold: the person who uses the system and takes part in decisions about it. The chapter says an information system is of little use if it were not for you, the user.
- **Where it shows up in real work:** Choosing your next phone or operating system, deciding how you feel about sharing personal information online, securing the wireless network at home, keeping files in sync across devices — and at work, sitting in on a decision about what software to buy.
- **What goes wrong without it:** In a small firm with no IS department you still make the investment call. Without a basic grasp of IS infrastructure, systems analysis and design, and security, you are at the mercy of consultants or vendors selling their technology of the week.
- **Why the chapter says it pays:** Understanding what systems can and cannot do, and being able to communicate with the technical staff, is what the chapter says will set you apart from your competition in any business field.

**Security matters: ransomware.** Ransomware is a virus that encrypts the data it finds in place, leaving the victim unable to open it; the attacker then demands payment for the decryption keys. Paying guarantees nothing.

- It arrives as a spam attachment or a download disguised as a video, which is why the user is part of the security system.
- Demands grew from a 2018 average near US\$530 toward large organizations: Change Healthcare paid **US\$22 million** in 2024.
- The answer has two halves: awareness to avoid infection, and backup and recovery so infection is survivable.

Three questions on where the user sits.

#### CHECK YOURSELF

#### You are part of the system

Answer each question from the chapter's account of the user's role and the Security Matters box on ransomware.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** You take a marketing job at a 30-person company with no IS department, and the owner asks you to help choose a new customer database. Why does the chapter say your IS knowledge matters here?

- **A.** You will be part of the investment decision, and without basic understanding you are at others' mercy — **correct.** The chapter singles out smaller organizations without dedicated IS departments: you are likely involved in the investment decision, and lacking a grasp of IS infrastructure, systems analysis and design, or security leaves you exposed to people acting in their own interests.
- **B.** Small organizations face few IS decisions, so the stakes are low enough that any reasonable choice works out — The chapter argues the reverse — the absence of an IS department raises your exposure rather than lowering it, because there is no specialist to catch a bad decision. This would be right only if small firms did not depend on their systems, and they do.
- **C.** Hand the decision to an outside vendor, since their technical knowledge is deeper than anyone at the firm has — Vendors do have deeper technical knowledge; that is precisely the problem. The chapter warns that vendors are likely to act out of their own interests and sell their "technology of the week." Deferring to them would be reasonable only if their incentives matched yours.

- **D.** IS knowledge here is really product training, so what matters is learning whichever database the owner buys — Training on the tool you end up with does matter, but it comes after the decision this scenario is actually about. The chapter asks users to understand what systems can and cannot do so they can take part in the investment decision itself; this would be right only if someone qualified were reliably making that call for you.

## 2. What does ransomware actually do to a victim's data?

- **A.** It deletes the files permanently, which is why payment cannot restore them — Permanent deletion describes a destructive wiper, and it would make a ransom pointless — there would be nothing to sell back. The chapter's point is subtler: the data is still there, just unreadable, and paying still may not recover it.
- **B.** It encrypts the data in place in a format the victim cannot access, and the attacker demands payment for the decryption keys — **correct.** The chapter defines ransomware as a virus that, once it infects a system or network, encrypts the data it finds in place so the victim cannot access it, after which the attacker demands a ransom for the decryption keys.
- **C.** Its defining move is copying the data out and publishing it publicly — Data theft and public release is real and growing — the chapter says attackers increasingly threaten to release data if not paid, and for firms holding personally identifiable information the lawsuits and reporting duties can cost more than the lost access. But that is an added pressure tactic, not the definition.
- **D.** It locks the screen behind a warning message while the files themselves stay perfectly readable underneath, which is why a technician can recover them without paying anything — Screen-locking scareware exists and this is a common mental picture, which is why it is a trap. Ransomware attacks the files themselves through encryption, so removing the malware does not give the data back.

## 3. Given the chapter's account, what should individuals and organizations actually do about ransomware?

- **A.** Pay promptly, since paying is what gets the decryption keys released — Paying feels like the direct fix, and the ransom is designed to look cheaper than the outage. But the chapter states that many victims have painfully experienced that paying the ransom does not guarantee regaining access to the data.
- **B.** Budget for it as a routine cost, since the 2018 average demand was about US\$530 — That US\$530 figure is real, but it is the 2018 average demand from a period when attackers targeted individuals and small businesses. The chapter documents the shift to larger, higher-value targets — Change Healthcare paid US\$22 million in early 2024 — so budgeting from the old number understates the exposure badly.

- **C.** Stop storing personally identifiable information altogether, since that is the data ransomware goes after and an organization holding none of it has nothing an attacker could hold to ransom — Personally identifiable information does raise the stakes, because public release brings lawsuits and reporting requirements. But ransomware encrypts whatever it finds: family photos, personal records, customer records, financial data, intellectual property. Limiting what you store is not a defense against encryption.
- **D.** Raise security awareness and vigilance to avoid infection, and strengthen backup and disaster recovery so an infection is survivable — **correct.** The chapter closes the box with exactly this pair: increase security awareness and vigilance to better avoid infections, and improve backup and disaster recovery preparations to become more resilient to such attacks.

The same reasoning covers private life.

- Which phone to buy next, and which operating system you prefer.
- Whether to share personal information online, and how much.
- How to secure the wireless network you run at home.

They feel like preferences, but underneath sit real differences in privacy and security. Rate yourself below.

#### SELF-CHECK

### The competencies you are building

Rate each statement honestly; anything you cannot do yet has a pointer to the exact place in this section to reread.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

- I can use the chapter's own numbers to explain why IS careers rank so highly — and also say what the Best Jobs table does not prove. The 17 percent BLS growth forecast through 2033, and the honest-reading callout under Table 1.1.
- I can name at least six IS management job titles and say what each of those people actually does. Table 1.2, which groups titles by whether the work is to develop, maintain, or manage.
- I can name the three core competencies and give two sub-skills for each without looking. Table 1.3, then the three short sections that follow it.
- I can explain why the chapter files social and communication skills under business rather than treating them as extras. The Business knowledge and skills block in Table 1.3, and the closing lines of the systems competency section.
- I can explain what "just enough" technical knowledge means, and why the chapter calls the technical area the hardest to maintain. The Technical competency section, down to the hot skills for the next decade that sit in the business domain.

- I can describe what ransomware does to data and name the two defenses the chapter recommends. The Security Matters box: encryption in place, ransom for keys, then awareness plus backup and disaster recovery.
- I can explain what a non-IS major at a small company risks by not understanding information systems. The You — the user section, on consultants and vendors selling their technology of the week.

## Information systems inside an organization

Information systems do not exist in a vacuum. They are built and used inside a context, and that context is almost always an organization — the last piece of the definition.

Organizations put systems in place for four blunt reasons, and none is that the technology is new.

**To become more productive and profitable** — the system must pay for itself in work or money.

**To gain competitive advantage** — it does something rivals cannot, or does it faster.

**To reach more customers** — a website is open to anyone with a browser.

**To improve customer service** — answering a question faster is its own reason.

None of that is unique to business: the same four hold for religious, educational, and governmental organizations alike.

### One company, three questions

People at different heights in a company need different kinds of information.

**A cashier** — she needs the price of the item in her hand now, because a customer is waiting.

**Her store manager** — he needs to know which items ran out last week, to get next week's order right.

**A senior leader** — she needs to know whether a store in a new state will pay for itself.

Same data, three questions. Pick a layer of work below and see which system serves it.

#### INTERACTIVE DIAGRAM

### Which class of system serves which kind of work

Pick a layer of work to see what happens there, what the people there need to know, and which class of system serves them.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Operational level** — The people who run the day, one event at a time

An event happens (A customer checks out; a student registers) → A TPS records it (Scanner, register, registration screen) → The data are stored (Every layer above lives on this)

- A **transaction processing system** processes day-to-day business event data at the operational level of the organization — the grocery checkout register connected to a network is the classic case.
- The person using it is not making a strategic decision; they are recording something that just happened, accurately and fast.
- The by-product is the real prize: a TPS generates a tremendous amount of data the firm can use to learn about customers and changing product trends.
- Skip this layer and nothing above it can work, because there is no record of what actually occurred.

**Managerial level** — Managers running a department or a store week to week

TPS data pile up (Thousands of recorded events) → An MIS organizes them (Detailed reports on the unit) → A manager acts (Reorder, restaff, reschedule)

- A **management information system** produces detailed information to help manage a firm or part of a firm — an inventory management and planning system, or student enrollment management.
- TPS data are sorted and organized to support managerial decision making, and the MIS is the most common system doing that work.
- The question changes from “what happened just now?” to “what has been happening across my whole area, and what should I change?”
- Alongside it sits the **functional area information system**, which supports the activities within one specific functional area of the firm — a planning system for personnel training and work assignments, for instance.

**Analysis level** — The people modeling a choice before the company commits to it

Decision support system (Models a specific choice) → Business intelligence system (Analyzes Big Data) → Intelligent system (Emulates human capability)

- A **decision support system** provides analysis tools and access to databases to support quantitative decision making — product demand forecasting, or loan and investment analysis.
- A **business intelligence system** analyzes Big Data to better understand various aspects of a business, using online analytical processing (OLAP) and data visualization.
- An **intelligent system** emulates or enhances human capabilities: analyzing bank loan applications, self-driving cars, Siri, Alexa, ChatGPT, Gemini.
- Modern systems are built to present integrated data to busy decision makers along with the tools to manipulate and analyze those data.

### Systems that span every level — Enterprise systems that refuse to sit on one floor

ERP (Integrates all facets of the business) → CRM (Firm to customer) → SCM (Suppliers to production to delivery) → E-commerce (Selling from the firm's website)

- **Enterprise resource planning** supports and integrates all facets of the business, including planning, manufacturing, sales, and marketing, so it touches the clerk and the chief executive on the same day.
- **Customer relationship management** supports interaction between the firm and its customers, through things like sales force automation and lead generation.
- **Supply chain management** supports the coordination of suppliers, product or service production, and distribution — procurement planning is the everyday example.
- These three cannot be filed under one category, which is exactly the point: many organizations replaced stand-alone systems with enterprise systems that span the whole organization.

### The catalogue of system categories

Read the purpose column first: the names are only labels for the job beside them.

| CATEGORY OF SYSTEM                            | PURPOSE                                                                                        |
|-----------------------------------------------|------------------------------------------------------------------------------------------------|
| <b>Transaction processing system</b><br>(TPS) | Processes day-to-day business event data at the operational level of the organization          |
| <b>Management information system</b><br>(MIS) | Produces detailed information to help manage a firm                                            |
| <b>Decision support system</b> (DSS)          | Provides analysis tools and access to databases to support quantitative decision making        |
| <b>Intelligent system</b>                     | Emulates or enhances human capabilities                                                        |
| <b>Business intelligence system</b>           | Analyzes Big Data to understand aspects of a business                                          |
| <b>Office automation system</b>               | Supports a wide range of predefined day-to-day work activities of individuals and small groups |
| <b>Collaboration system</b>                   | Enables people to communicate, collaborate, and coordinate                                     |
| <b>Knowledge management system</b>            | Enables the generation, storage, sharing, and management of knowledge assets                   |
|                                               |                                                                                                |

| CATEGORY OF SYSTEM                            | PURPOSE                                                                                 |
|-----------------------------------------------|-----------------------------------------------------------------------------------------|
| <b>Geographic information system</b><br>(GIS) | Creates, stores, analyzes, and manages geographically referenced data                   |
| <b>Customer relationship management</b> (CRM) | Supports interaction between the firm and its customers                                 |
| <b>Enterprise resource planning</b><br>(ERP)  | Supports and integrates all facets of the business                                      |
| <b>Supply chain management</b><br>(SCM)       | Supports the coordination of suppliers, product or service production, and distribution |
| <b>Electronic commerce system</b>             | Enables customers to buy goods and services from a firm's website                       |

The transaction processing system is the foundation, and it also generates enormous amounts of data to learn from.

- Your grocery store scans bar codes, then prints coupons on the receipt for related products.
- Amazon processes thousands of transactions an hour, feeding warehouses that produce recommendations for later customers.

That data feeds the management information, decision support, business intelligence, and knowledge management systems. Pair each category with its purpose.

#### **MATCH**

### **System type to the job it does**

Pair each category of information system with the purpose it serves inside an organization.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

#### **Transaction processing system**

Processes day-to-day business event data at the operational level of the organization The checkout register that scans bar codes is the standard example, and the data it captures feed nearly every other system.

#### **Decision support system**

Provides analysis tools and access to databases to support quantitative decision making Product demand forecasting and loan or investment analysis are the sample applications: numbers in, modeled answer out.

#### **Knowledge management system**

Enables the generation, storage, sharing, and management of knowledge assets The everyday form is a knowledge portal where people find answers to common questions instead of asking a colleague again.

### **Geographic information system**

Creates, stores, analyzes, and manages geographically referenced data Data with a location attached behaves differently from ordinary data, which is why route planning gets its own category.

### **Customer relationship management**

Supports interaction between the firm and its customers Sales force automation and lead generation live here; the direction of travel is outward, from the firm to the buyer.

### **Supply chain management**

Supports the coordination of suppliers, product or service production, and distribution This one points the other way, upstream toward the people who supply the firm, with procurement planning as the sample application.

### **Office automation system**

Supports a wide range of predefined day-to-day work activities of individuals and small groups Word processing, spreadsheets, presentation software, and an email client are personal productivity tools, not enterprise systems.

### **Electronic commerce system**

Enables customers to buy goods and services from a firm's website Amazon, eBay, and Nordstrom.com are the named examples; the system exists to complete a purchase, not to analyze one.

Now name the class of system each situation calls for.

#### **CHECK YOURSELF**

### **Naming the right class of system**

Four options, one best answer; read every explanation, including the ones for answers you did not choose.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A grocery store's checkout lane scans bar codes as items are rung up, and the register is connected to the store network. Which category of information system is that register?

- **A.** A management information system, because the store uses the register data to manage the store — A management information system produces detailed information to help manage a firm or part of a firm, such as an inventory management and planning system. It consumes the register's data; it is not the register. Change the question to 'which system tells the manager what to reorder?' and this becomes correct.

- **B.** A transaction processing system, because it processes day-to-day business event data at the operational level — **correct.** The register captures a business event as it happens at the operational level, which is the definition of a transaction processing system. The coupons printed on the back of the receipt are a bonus built on that same captured data.
- **C.** A business intelligence system, because the scanned data are later analyzed for product trends — A business intelligence system analyzes Big Data to better understand aspects of the business, using tools like OLAP and data visualization. That analysis happens downstream, on data the TPS generated. This would be right if the question described the analytics team studying years of scanner data.
- **D.** An office automation system, because it is a predefined day-to-day work activity performed by one employee — An office automation system is personal productivity software — word processor, spreadsheet, presentation software, email client — supporting an individual's or small group's own work activities. A cashier ringing up a customer is processing a company transaction, not doing personal desk work.

**2.** A classmate says: "Categories like TPS and DSS are obsolete. Nobody uses those labels anymore." What is the accurate correction?

- **A.** They are right: organizations replaced categorized systems with enterprise systems, so the labels no longer describe anything real — Enterprise systems that span the organization did replace many stand-alone systems, which is why it is hard to say a system fits only one category. But the categories did not stop describing the work; they stopped being mutually exclusive. This would be right only if the categories had been abandoned, and they have not.
- **B.** The labels are exact if you go by a product's main feature, so a modern system still belongs in a single row of the table — This is the opposite error. Ten to 15 years ago it was typical to see systems that fell cleanly into one category, but internetworking and systems integration ended that: customer relationship management, supply chain management, and enterprise resource planning carry so many features and data types that no single row holds them. Naming a main feature hides the other jobs the system is doing.
- **C.** The labels still name the goals, features, and functions a system delivers, even though one modern system usually spans several of them — **correct.** Understanding the categories lets you see the myriad approaches, goals, features, and functions of modern systems, even though modern systems tend to span several categories at once.
- **D.** The labels described systems a company housed itself; cloud systems are described instead by the vendor and the subscription plan — Where a system runs and what job it does are separate questions. Many systems are no longer housed within organizations but sit in the cloud and are reached through a browser when needed, and they still process transactions, support decisions, or manage customer relationships. Hosting changes who maintains the servers, not the category of work.

**3.** An analytics team wants to study several years of purchase history to understand emerging product trends. Where do those data come from, and which class of system does the analysis?

- **A.** Managers type the monthly figures in by hand, and a management information system does the analysis — Managers do read MIS output, but they are not the data source. Transaction processing systems generate the raw event data automatically as business happens, which is why the volume is large enough to be worth analyzing at all.
- **B.** The data are generated by transaction processing systems and analyzed by a business intelligence system — **correct.** A TPS generates a tremendous amount of data as it processes transactions — Amazon feeds thousands of transactions per hour into large data warehouses — and a business intelligence system analyzes that Big Data to better understand the business.
- **C.** The data come from the company's office automation software, and a collaboration system does the analysis — Office automation software is word processing, spreadsheets, presentations, and email for individuals and small groups, and a collaboration system provides email, shared calendars, and threaded discussions. Neither one is where years of company purchase data live.
- **D.** The data come from the company's e-commerce website, and a customer relationship management system does the analysis — An e-commerce system is a real source of transaction data, but a firm with physical stores captures far more at the register, and CRM supports interaction with customers rather than large-scale trend analysis. This would be closer for a purely online retailer studying its own customer relationships.

## Why the boxes leak

Ten or fifteen years ago a system fell cleanly into one category. Enterprise systems, internetworking, and integration ended that, and many systems no longer sit in the building at all. A modern one spans several categories — CRM, SCM, ERP — yet the categories still name what a system is for.

**How to use the catalogue.** Do not ask what category a product is. Ask what job needs doing: recording an event, summarizing events for a manager, and modeling a decision are three different jobs. Name the job; the category follows.

### **DECIDE** Choosing the right system for the problem in front of you

You are the new operations manager at a 16-store regional grocery chain; pick the class of system that actually solves each problem.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

Leadership has money to spend and a list of complaints. Your job is not to name a vendor — it is to name the kind of system each complaint calls for.

**Decision 1.** Checkout is slow and every store rings sales on a standalone register that never talks to the network, so nobody outside the store knows what sold today.

- Buy a business intelligence system so leadership can analyze store performance across the chain — A business intelligence system analyzes Big Data to understand the business — but there is no Big Data yet. Nothing is capturing the transactions. You would be buying a telescope before there is anything in the sky.
- Give each store manager spreadsheet and email software to track that store's daily sales — That is office automation: personal productivity software supporting one person's day-to-day work. It would produce 22 hand-typed, inconsistent files instead of one automatic, reliable stream of transaction data.
- Install networked checkout registers — a transaction processing system — **the stronger call.** Right. Registers that scan bar codes and record each sale process day-to-day business event data at the operational level. They also fix the deeper problem: the chain finally has a record of what actually happened, which every other system will need.

**Decision 2.** Transactions are now captured, but the regional manager still cannot tell which stores are about to run out of which products until a shelf is already empty.

- Launch an e-commerce site so customers can order out-of-stock items online — An electronic commerce system enables customers to buy goods and services from the firm's website. Useful someday, but it does not tell your manager what is running low — it just moves the empty shelf online.
- Add inventory management and planning — a management information system — **the stronger call.** Right. A management information system produces detailed information to help manage a firm or part of a firm, and inventory management and planning is its textbook application. It turns the transaction data into detailed stock reports, so the raw events become something a manager can act on.
- Roll out a collaboration system so store managers can email each other about shortages — A collaboration system enables people to communicate, collaborate, and coordinate — email with a shared calendar. It helps them talk about the problem; it does not produce the detailed stock information they would be talking about.

**Decision 3.** Finance must decide whether to borrow money to build a 23rd store, and wants to model demand and repayment under three different economic scenarios before committing.

- Give finance analysis tools and database access — a decision support system — **the stronger call.** Right. A decision support system provides analysis tools and access to databases to support quantitative decision making, which is what finance needs to model the forecast; product demand forecasting and loan and investment analysis are exactly its sample applications.
- Pull the answer out of the transaction processing system, since it holds all the sales data — The TPS holds the raw record of what already happened, and it is the input you need — but it processes events, it does not model futures. Data alone is not a forecast.

- Search the company knowledge portal for what the firm did last time it expanded — A knowledge management system enables the generation, storage, sharing, and management of knowledge assets, so a portal of past answers is genuinely useful context. It just cannot run three quantitative scenarios against current demand data.

**Decision 4.** Suppliers keep shipping late, two trucks arrive at the same dock at once, and produce spoils while a store two towns over has none.

- Deploy customer relationship management so shoppers get told when produce is unavailable — CRM supports interaction between the firm and its customers — sales force automation, lead generation. It points downstream, toward the buyer. It would communicate the failure rather than prevent it.
- Deploy a geographic information system to plan better delivery routes — A geographic information system creates, stores, analyzes, and manages geographically referenced data, and route planning is its sample application, so this genuinely helps a truck get somewhere efficiently. It still does not coordinate suppliers, procurement, and distribution as a whole.
- Put in supply chain management, starting with procurement planning — **the stronger call.** Right. Supply chain management supports the coordination of suppliers, product or service production, and distribution, and procurement planning is where that coordination starts. The problem is upstream of the customer, so the system has to be upstream too.

## Organizing the IS function

The people who run technology inside a company are the **IS function**, and their old reputation was earned: they owned the computing resources, thought they knew better than users, and saw the job as saying no. Early departments shipped late, over budget, and hard to use.

Competition forced firms to treat IS as an enabler, and many realized the best ideas come from the employees using the system. IS units moved into a **consulting relationship** with users, built on four habits.

**Reach out first** — they seek user input instead of waiting for complaints.

**Change fast** — they modify systems at a moment's notice to meet a need.

**Welcome ideas** — they celebrate new ideas rather than explaining why they will not work.

**Hand over ownership** — the technology and information belong to the customer, supported by help desks and training centers.

A firm is unproductive when IS staff and everyone else are at odds, and remarkably productive when they work hand in hand.

## The pervasiveness of technology

Technology is entrenched inside the business units. Where firms use agile approaches such as **scrum**, a system's builders sit with its users, trained in information systems **and** in the area they support.

Hence the **dual-reporting relationships**: to the central IS group and to the business function. Decentralizing buys flexibility; centralizing buys coordination, economies of scale, and compatibility. Firms want both.

**When things go wrong: technology addiction.** Since **2018**, online gaming has been a diagnosable addiction. The mechanism is chemical: **dopamine**, released when we are stimulated, means a like can addict as cocaine does.

- Adults consume over **11 hours** of media a day, and half of 18- to 24-year-olds check the phone five minutes after waking.
- The average attention span fell from **12 seconds** in 2000 to **8 seconds**.
- The fixes are unglamorous: switch the gadgets off part of each day, eat and exercise properly, sleep enough.

Three questions on the IS function and what constant connection costs.

### CHECK YOURSELF

## The IS function, and what it costs us

Three questions on how technology staff are organized inside a firm and on the human side of being always connected.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** Which description matches the service-oriented mindset that replaced the old-school IS department?

- **A.** IS personnel own and control the computing resources and tell users what they may and may not do with them — This is the old-school mindset precisely: IS believed it owned the computing resources and knew better than users. It came bundled with huge backlogs and systems that were over budget, late, and hard to use. It is what the service orientation replaced.

- **B.** IS personnel act as consultants, seek user input before complaints arrive, and run help desks, hotlines, and training centers — **correct**. Personnel in many IS units took on a consulting relationship with users, reaching out proactively, modifying systems quickly, and building help desks, hotlines, information centers, and training centers to support them.
- **C.** IS personnel step back entirely, so each business unit buys and runs whatever technology it wants with no central coordination — Serving users is not the same as abandoning coordination. Even organizations that decentralize technology decisions still need centralized or centrally coordinated staff for economies of scale, systems integration, and enterprise networking.
- **D.** IS personnel wait for users to bring in systems complaints, then work through that queue in the order the complaints arrived — This is the reactive posture the service orientation replaced. Service-oriented units reach out to their internal customers and proactively seek their input and needs rather than waiting for customers to come in with complaints, and they modify systems at a moment's notice to meet a need. Working a complaint queue in arrival order is how the old project backlogs grew.

2. Firms increasingly place systems people inside business units with a desk, a phone, and a dual-reporting relationship. Why do they still keep centralized IS staff as well?

- **A.** Because business units are not permitted to make technology decisions of their own — Business units make plenty of technology decisions — that is the whole point of decentralizing and of embedding systems staff alongside users. The tension is about coordination, not permission.
- **B.** Because a central group costs less than embedded staff, so firms are gradually ending the embedded arrangement — Cost is not the driver described, and the embedded arrangement is expanding rather than ending: systems personnel are often permanently placed in the business unit, frequently with training in both IS and the functional area they support.
- **C.** Because centralization delivers the flexibility, adaptability, and responsiveness that decentralization cannot — This swaps the two lists. Flexibility, adaptability, and systems responsiveness are the benefits of decentralization. Centralization buys something different, which is why firms want both at once.
- **D.** Because centralization preserves coordination, economies of scale, compatibility, and connectivity — **correct**. Organizations want decentralization's flexibility, adaptability, and responsiveness, but they will not forgo centralization's coordination, economies of scale, compatibility, and connectivity — so they keep a centrally coordinated IS staff.

3. Which statement matches the evidence in the technology-addiction discussion above?

- **A.** Average adults consume more than 11 hours of media a day, and the average human attention span has fallen from 12 seconds to 8 — **correct**. Both figures are reported: more than 11 hours of media consumption a day for the average adult, and an attention span that dropped from 12 seconds at the turn of the century to 8 — less than a goldfish.

- **B.** Half of 18- to 24-year-olds open social media more than ten times a day, and a third of 25- to 34-year-olds check their phone on waking — The two age groups and the two behaviors are swapped. Fifty percent of 18- to 24-year-olds checked their phone within five minutes of waking up, and a third of 25- to 34-year-olds visited social media sites or used mobile apps more than ten times a day.
- **C.** The dopamine released by a like or a status update works through a different mechanism than addictive drugs do — The argument is the opposite: dopamine is released when we are stimulated, and the hit from a like, a status update, or a piece of gossip can have the same addicting effect on the body as drugs like cocaine or heroin.
- **D.** Younger generations are the group least able to adjust, precisely because they have been immersed in technology their whole lives — This inverts the one hopeful note. Because younger generations have been immersed in the technology their entire lives, they may be better able to adjust and adapt, developing a feel for their limits — even though a majority of adolescents already claim to be addicted.

## The dual nature of information systems

Information systems are now so important, and so expensive, that the chapter reaches for a blunt image: information technology is **like a sword**. You can wield it as a weapon, but those who live by the sword sometimes die by it.

A **competitive weapon** is anything that makes a customer pick you over the shop across the street — a live delivery map, say. On the night it fails, the customers who chose you for it feel most let down. That is the **dual nature of information systems**.

### A system gone awry: the Zoom outage

The 2020 pandemic pushed people to information systems for everything, and activities never online before moved there at once.

**Errands became orders** — delivery apps surged, because going out was no longer an option.

**Offices became clouds** — firms scrambled to move documents and workflows online, the building being shut.

**Social life became Zoom** — exercise classes and church services alike ran on one videoconferencing platform.

Videoconferencing was not new; the reach was. Daily meeting participants surged past **300 million**, and on **Sunday morning, May 17, 2020**, users suddenly could not get in. It was a Sunday, so few businesses were hit. Churches were, and services ended early.

**Notice what the chapter does not blame.** No hacker, no sabotage, no blunder. The growth that made Zoom indispensable made its bad Sunday everybody's, and relying on a **single provider** is the lesson.

Put the six stages of that failure back in order.

#### SEQUENCE

### How the Zoom outage actually unfolded

Drag the six moments into the order the chapter describes, from the pressure that built up to the lesson left behind.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

The chapter's failure case does not begin with a broken server. It begins with the world changing outside the company.

1. **Early 2020: stay-at-home orders, travel bans, and other mobility restrictions push people to use information systems for everything from working from home to the most mundane tasks.** — This is the trigger outside the company. Nothing about Zoom's software changed first - demand did.
2. **Activities that had never been online move online: delivery apps surge, companies move documents and workflows to the cloud, and entire social lives, from exercise classes to church service, shift onto Zoom.** — Videoconferencing was old news for business meetings. What was new was ordinary life depending on it, which is what set up the damage.
3. **Zoom's daily meeting participants surge to over 300 million, a tremendous and unforeseen increase.** — The chapter names the surge itself as the hazard: such a tremendous unforeseen increase in users can quickly cause problems.
4. **On Sunday morning, May 17, 2020, U.S. time, users suddenly start experiencing issues accessing the service.** — The failure lands after the growth, not before it, and the day of the week decides who it hurts.
5. **Comparatively few businesses and professionals are impacted; the largest impact falls on churches, whose worshippers cannot connect, so services across the United States are ended early.** — It was a Sunday morning, so the users with something scheduled at that exact hour and no fallback absorbed the loss.
6. **By the afternoon Zoom restores the systems to working order, leaving the lesson that relying on a single provider can quickly disrupt many activities in unforeseen ways.** — The recovery was fast, and the lesson still stands: what makes this a failure case is the dependence it exposed, not the length of the outage.

## A system that works: FedEx

**FedEx**, an US\$87.7 billion family of companies, is the world's largest express transporter, and it keeps updating its systems to sustain a competitive advantage.

| WHAT THE SYSTEMS COORDINATE           | THE CHAPTER'S FIGURE                                                       |
|---------------------------------------|----------------------------------------------------------------------------|
| FedEx.com traffic                     | Over 80 million visitors a month; over 500 million tracking requests a day |
| Time to divert a package after a scan | One to two seconds; decided in a few hundred milliseconds                  |
| Performance reengineered              | On average, twice a year                                                   |
| Delivered within one business day     | A quarter of all daily packages                                            |

Accuracy, not the web page, is the product, and no human decides a route in a few hundred milliseconds. Automation in the ground hubs is what the chapter calls an **enabler of competitive advantage**.

Open the three systems below.

#### EXPLORE

### One system that failed, two that worked

Open each card and read the failed system against the two successful ones facet by facet, rather than case by case.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

#### Zoom videoconferencing · The system gone awry

- **What the system is:** The videoconferencing software that, during the 2020 pandemic, absorbed not just business meetings but entire social lives, from exercise classes to church service.
- **What actually happened:** Daily meeting participants surged to over 300 million, and on Sunday morning, May 17, 2020, users suddenly could not access the service. Churches across the United States ended their services early. Zoom restored the systems by that afternoon.
- **Why it went that way:** The trouble came from a tremendous, unforeseen increase in users rather than from an obvious blunder. Everything that had moved onto one platform went down with that platform.
- **The lesson the chapter draws:** Relying on a single provider can quickly disrupt many activities in unforeseen ways – and the activities with no fallback at that exact hour take the hardest hit.

#### FedEx tracking · The system that works, part one

- **What the system is:** The customer-facing side of FedEx's extensive, interconnected systems: FedEx.com, which draws more than 80 million unique visitors per month and answers more than 500 million tracking requests per day.
- **What actually happened:** FedEx strives to provide the most accurate tracking information to each visitor, and continuously updates and fine-tunes the systems behind it to improve its services and sustain a competitive advantage.
- **Why it went that way:** Accuracy at that volume is not a by-product of shipping packages; it is something the company deliberately keeps investing in, which is what makes the choice strategic in intent.
- **The lesson the chapter draws:** A system can be the reason customers choose you, so it is developed to gain or sustain competitive advantage rather than merely to cut cost.

#### FedEx ground-hub automation · The system that works, part two

- **What the system is:** The sorting facilities each package typically travels through, where an extensive network of conveyor belts scans a package multiple times and reroutes it toward its intermediate and final destinations.
- **What actually happened:** Once a package passes an overhead scanner there is between 1 and 2 seconds to divert it, so decisions must be made in a few hundred milliseconds. On average FedEx reengineers and improves the performance twice a year and now manages to deliver a quarter of all daily packages handled within one business day.
- **Why it went that way:** No human can decide a route in a few hundred milliseconds, so the automation is not a convenience laid over manual sorting - it is the only way the service exists at all.
- **The lesson the chapter draws:** Automation in the ground hubs is another enabler of competitive advantage, and an advantage reengineered twice a year is one being renewed faster than rivals can copy it.

## What the two cases share

Both are typical of systems in large organizations, and share three traits.

**Large in scale and scope** — difficult to build, which is why the chapter insists on getting development right first time.

**Critical to the organization** — trouble inside the system is trouble in the business.

**Strategic in intent** — both were built, and are still updated, to gain advantage over rivals.

What separates them is which way the blade turned. The dependence that made Zoom valuable stopped everything at once; FedEx renews its edge twice a year rather than assuming it.

## What “strategic” means here

In everyday speech, strategic just means important. Here it is narrower: a choice is **strategic** when made to gain or sustain advantage over rivals, not merely to make a task cheaper. A cheaper email provider changes nobody's mind; tracking a hospital will trust does.

The advantage takes its name from Michael Porter: **competitive advantage** over rivals. It asks three things of technology.

**Efficiency** — the technology enables the same work with less time and expense than before.

**Return on investment** — a system that never pays back its cost fails the test, however clever.

**Competitive advantage** — technology use can **also** be strategic, and a powerful enabler of that advantage.

Three questions on the sword, the cases, and the word strategic.

**CHECK YOURSELF**

**The sword, the two cases, and the word strategic**

Pick the best answer, then read every explanation, because the wrong options are all positions students actually take.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. The chapter says information technology is **like a sword**. What is that image meant to convey?

- **A.** That the money sunk into information systems is the main danger, so the risk to manage is overspending on technology that underdelivers — Cost is real here - the chapter opens by noting how important and expensive information systems have become, and it insists that information systems must provide a return on investment. But that is one edge of the blade. The image is about the two-sided outcome: the same system that wins customers can be the one that takes an organization down.
- **B.** That information systems cut through the layers of an organization, letting any employee reach any other directly — This describes a genuine effect of information systems on communication, and it is a real thing that happens in organizations - but it is not the sword metaphor. To be right, the image would have to be about flattening hierarchy, and instead it is about a weapon that can wound the one holding it.
- **C.** That the same technology can be used effectively as a competitive weapon and can also be the thing that brings an organization down — **correct**. The chapter's framing is that you can use information technology effectively as a competitive weapon, but those who live by the sword sometimes die by the sword. Zoom and FedEx are the two edges of that same blade.
- **D.** That a system has to be kept sharp with constant updates, or it gradually stops doing its job — This is half true and it rests on a real chapter fact: FedEx reengineers and improves performance on average twice a year, and advantage from IS can be fleeting because competitors eventually do the same thing. But that is the maintenance point, not the metaphor. The sword is about which direction the blade turns, not about sharpening.

2. According to the chapter, who was hit hardest by the May 17, 2020, Zoom outage, and why?

- **A.** Churches, because their services had moved online and the outage came on a Sunday morning — **correct**. The chapter reports that the largest impact was for churches, who had moved their services online; with many worshippers unable to connect, churches across the United States were forced to pull the plug and end their Sunday services early.

- **B.** Businesses and professionals, because the outage struck in the middle of the workweek when most meetings were scheduled — This is the intuitive guess, and it is exactly backwards: the chapter says comparatively few businesses and professionals were impacted. The outage was on a Sunday morning, U.S. time, so the workweek was not running. This would be the right answer to an outage at 10 a.m. on a Tuesday.
- **C.** Schools, because students in remote classes could not attend their lessons for most of the day — Remote schooling is a real part of the pandemic shift the chapter describes, so this is a reasonable thing to expect – but the case in point does not name schools among those affected. It would be right only if the chapter had reported it, and reporting what the chapter actually says is the point of a case in point.
- **D.** Zoom itself, because the outage reversed the surge that had carried it past 300 million daily meeting participants — The 300 million figure is real, but it is the surge in daily meeting participants that preceded the outage, not a loss. Zoom had restored the systems by the afternoon, and the harm the chapter describes falls on the people depending on the service. This would be right if the chapter had reported a customer exodus, and it does not.

### 3. What do the Zoom and FedEx cases have in common in the chapter's telling?

- **A.** Both systems eventually failed under load that their builders had not anticipated — Failure is what separates the two cases, not what unites them: the FedEx case reports no outage at all. This would be right if the section were about two failures, but the chapter deliberately pairs one system gone wrong with one gone right to show the dual nature.
- **B.** Both companies keep their systems in-house rather than depending on outside providers, which is what made the systems dependable — The chapter never says who built which piece, and its competitive-advantage argument runs the other way: firms from Amazon to Zoom win by combining commoditized technologies with proprietary systems and business processes. Ownership of the parts is simply not the link the chapter draws between the two cases.
- **C.** Both were built to reduce operating costs, and any effect on competition was a welcome side benefit — This is the cost-only view of IT that the chapter reports as an argument and then rejects. It would be right if the chapter had said the systems were built for cost reduction and risk mitigation, but it says the opposite: the choices were strategic in their intent, aimed at competitive advantage.
- **D.** Both are large, hard-to-build systems, critical to their organizations, and both were developed with strategic intent — **correct.** The chapter says these systems are so large in scale and scope that they are difficult to build, that they were and continue to be critical to the success of the organizations that built them, and that the choices made in developing them were strategic in their intent.

### But hasn't IS become a commodity?

A **commodity** is so standardized that one supplier's version is interchangeable with another's, which is why nobody wins customers selling better electricity. Some therefore argued three things.

- Information systems are now a commodity, necessary for everyone.
- Firms should aim IT strictly at cost reduction and risk mitigation.
- Investing for differentiation is futile, since rivals can buy the same thing.

The chapter answers rather than dismisses that. IT keeps changing, and firms gained advantage from rising digital density.

**Amazon to Zoom** — they combined commoditized technologies with proprietary systems and business processes, so the combination is not for sale.

**Google and Facebook** — how customer-generated data are gathered and used can be a source of **sustained** advantage.

There is a hard limit: companies with bad business models tend to fail whether or not they use technology. A **business model** summarizes a firm's strategic direction and how its objectives will be met.

Technology carries out a business model; it does not supply one.

**The advantage does not stay bought.** Advantage from information systems is fleeting, because competitors eventually do the same thing.

#### CHECK YOURSELF

### Commodity, advantage, and business models

Choose the answer that matches the chapter's position, then read why each of the other three is a position the chapter does not take.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** Some argue that because information systems are standardized and ubiquitous, they are now a **commodity**, and that firms should therefore spend on IT strictly for cost reduction and risk mitigation. What is the chapter's answer?

- **A.** It agrees, which is why it warns that advantage from information systems can be fleeting — This runs two separate statements together. The chapter does say advantage from information systems can be fleeting because competitors eventually do the same thing, but fleeting means the advantage must be renewed, not that pursuing it is pointless. Its stated conclusion is that IS can be an enabler of competitive advantage.

- **B.** It holds for small firms, but companies the size of FedEx and Amazon can buy an advantage outright — Size is exactly what the chapter denies matters here: whether a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals. The chapter also describes advantage as built through combination, not purchased ready-made.
- **C.** It is right that bought technology cannot differentiate a firm, so a firm should rebuild the standardized parts itself — This gets the recipe backwards. The winning combination the chapter describes uses commoditized technologies for the standard parts and keeps the proprietary systems and business processes as the differentiator, so rebuilding standard parts spends scarce effort where it cannot set you apart.
- **D.** IT keeps changing rapidly, and firms from Amazon to Zoom pair commoditized technologies with proprietary systems — **correct.** The chapter grants that information systems are absolutely necessary for every company, then rebuts the rest: IT is changing rapidly, digital density is increasing, and companies from Amazon to Zoom created competitive advantages by combining certain commoditized technologies with proprietary systems and business processes.

**2.** The chapter warns that competitive advantage from the use of information systems can be **fleeting**. What reason does it give?

- **A.** Competitors can eventually do the same thing, so an edge that is not renewed gets matched — **correct.** The chapter's sentence is precise: the competitive advantage from the use of information systems can be fleeting, as competitors can eventually do the same thing. That is why FedEx reengineers and improves performance on average twice a year rather than treating its systems as finished.
- **B.** New devices come out every 6 to 18 months, so a system is outdated before it earns back what it cost — The 6-to-18-month device cycle is a real fact from this chapter, used where it explains that IS departments once planned in five-year time frames and must now adjust to a faster pace. But it explains why organizations have to keep up, not why an advantage fades. Advantage fades because of what rivals do.
- **C.** A technology that becomes standardized leaves the firms using it looking much alike to their customers — This is the commodity argument again, and the chapter rejects it: firms from Amazon to Zoom create advantage by combining commoditized technologies with proprietary systems and business processes, and Google and Facebook turn customer-generated data into sustained advantage. Standard parts do not make firms identical.
- **D.** Large information systems tend to cost more to run over time than the savings they produce — The chapter does say information systems must provide a return on investment, but it never claims large systems usually fail to deliver one - its position is that IS are necessary, can create efficiencies, and can enable advantage. This would be right only if the chapter had made a claim about typical project economics, and it does not.

**3.** A regional retailer buys the same analytics and logistics technology its most successful rival uses, but its stores stock merchandise its customers do not want at prices they will not pay. What does the chapter predict?

- **A.** It will close the gap, because in the chapter's view the technology is what creates the advantage — This reverses the chapter's causal order. Technology carries out a business model; it does not supply one. The chapter's success condition is a good business model plus information technology used successfully to carry that model out - both, in that order.
- **B.** It will likely fail anyway, because companies with bad business models tend to fail regardless of the technology — **correct.** The chapter states that companies with bad business models tend to fail regardless of whether they use information technology, while companies with good business models that use IT successfully to carry them out tend to be very successful. Merchandise and pricing are business-model problems, and buying software does not touch them.
- **C.** It will succeed once it layers proprietary systems on top of the technology it bought, the way Amazon and Zoom did — Layering proprietary systems on commoditized technology is genuinely the chapter's route to competitive advantage, and it is what companies from Amazon to Zoom did. But it is the route for a firm whose business model already works. Proprietary software on top of merchandise nobody wants still sells merchandise nobody wants.
- **D.** The chapter takes no position, because it treats the business model and the technology as separate subjects — The chapter links them explicitly: information systems conceived, designed, used, and managed effectively and strategically, together with a sound business model, are what let an organization gain or sustain competitive advantage - and a bad business model is its stated way to fail with good technology.

Now run the argument as a rollout decision.

#### DECIDE

### Rolling out a system that cuts both ways

Work through five decisions in order; after each one, read what the chapter says actually follows from your choice.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

You are on the team at a growing regional courier company that has just won a contract to carry deliveries for a large hospital network. Every choice below is the same sword the chapter describes: the system you are about to build can win the account or lose it.

**Decision 1.** Leadership asks what the new package-tracking system is actually for, so they know how to judge it later. What do you tell them?

- Its purpose is strategic in intent, and it still has to earn a return. — **the stronger call.** This matches the chapter. The purpose is to do something rivals cannot easily match - hospital-grade, package-level tracking - and the choices made in developing the systems at Zoom and FedEx were strategic in their intent: developed and continuously updated to gain or sustain competitive advantage over rivals (Porter, 1985; Porter and Millar, 1985). The chapter also insists information systems must provide a return on investment - technology use can be efficient and strategic at the same time.
- Its purpose is strictly cost reduction and risk mitigation, because information systems are a commodity now and chasing differentiation with IT is futile. — You have adopted the commodity argument the chapter reports and then rejects. IT keeps changing rapidly, and companies from Amazon to Zoom created advantage by combining commoditized technologies with proprietary systems and business processes. Judged on cost alone, the tracking accuracy the hospital actually cares about will be the first thing cut.
- Its purpose is strategic, so return on investment is not the right question to ask about it. — The chapter does not let you off that hook. It says plainly that information systems must provide a return on investment and can also be a powerful enabler of competitive advantage. Dropping the ROI test is how a strategic project becomes an unaccountable one.

**Decision 2.** The fastest route to launch is to run everything - scanning, routing, and customer notifications - on one outside provider's platform.

- Go all in on the single provider, since it is the fastest path and the provider is far larger and better staffed than you are. — Size did not protect anyone on May 17, 2020. Zoom was serving over 300 million daily meeting participants when the service became unreachable on a Sunday morning, and the churches with nothing to fall back on had to end their services early. Your hospital deliveries would have no fallback either.
- Build every piece in-house from scratch so that the company never depends on an outside provider. — This overcorrects past the chapter's point. Its advice is to combine commoditized technologies with proprietary systems and business processes, the way companies from Amazon to Zoom did. Rebuilding standard, interchangeable parts spends your scarce effort where it cannot differentiate you, and delays the part that can.
- Use the provider, but plan for the hours it is unavailable. — **the stronger call.** This is the exact lesson the chapter draws from Zoom: relying on a single provider can quickly disrupt many activities in unforeseen ways, because when one provider carries every function, all of them stop together. Note what you did not do - you did not refuse the provider. You refused to let one outage take down scanning, routing, and notifications at the same moment.

**Decision 3.** The contract assumes a steady daily volume. A flu surge across the hospital network could triple it with about a week of warning.

- Plan for the contracted volume; if demand triples, that is a good problem, and capacity can be added when it happens. — This is the assumption the Zoom case punctures. Demand arrived faster than anyone planned for, and the moment the system faltered was the moment its users needed it most. A week of warning is not enough time to rebuild something you did not design to stretch.
- Design and test for a spike you have not actually seen yet. — **the stronger call.** Right. The failure case in the chapter was caused by unforeseen growth rather than by a bug someone shipped: Zoom's outage followed a tremendous unforeseen increase in users, and the chapter presents that surge itself as the hazard. Systems this large are difficult to build, which is why the chapter insists on handling their development the right way the first time around.
- Cap how many customers can use live tracking at once, so that the system cannot be overloaded during a surge. — You have protected the servers by damaging the product. Accurate tracking is the reason the hospital chose you, and rationing it during the busiest week converts your competitive weapon into a complaint. FedEx answers more than 500 million tracking requests per day rather than limiting who may ask.

**Decision 4.** Year one goes well. Sorting is automated, the hospital renews, and the board asks whether IT spending can now drop to simple maintenance.

- No - keep reengineering the system on a schedule. — **the stronger call.** This is the chapter's warning and FedEx's practice. An advantage from information systems can be fleeting once competitors work out how to do the same thing, so FedEx reengineers and improves the performance on average twice a year and now manages to deliver a quarter of all daily packages handled within one business day. The edge is renewed, not banked.
- Yes - the system works and the contract is signed, so the advantage is locked in and further spending is waste. — The chapter says the competitive advantage from the use of information systems can be fleeting, as competitors can eventually do the same thing. A frozen system does not hold its lead; it simply waits to be matched.
- Yes, and redirect the savings into advertising the tracking feature, so customers remember who has it. — Advertising an advantage does not renew it. The claim in the ad stays the same while a rival's capability improves, and the chapter's mechanism for fading advantage is rivals doing the same thing, which no amount of promotion prevents.

**Decision 5.** Fourteen months later a rival launches tracking that matches yours. In the same month, a well-funded startup with slicker software than either of you shuts down.

- Conclude that the tracking investment was wasted, since a rival now offers the same thing. — Fleeting is not futile. The chapter's position is that information systems are a necessary part of doing business, can be used to create efficiencies, and can also enable competitive advantage - the fourteen months of exclusivity, the renewed hospital contract, and everything you learned building it were all real.

- Conclude that the startup failed because its technology was not good enough, and study its software for what to add next. — The chapter points somewhere else entirely: companies with bad business models tend to fail regardless of whether they use information technology. Copying features from a company that could not make its model work is how you inherit its problem.
- Read both events the chapter's way: advantage is fleeting, and a bad business model fails either way. — **the stronger call.** Exactly. Advantage from IS is fleeting because rivals eventually do the same thing, and companies with bad business models tend to fail regardless of whether they use information technology - the two facts are the two halves of the section. Good technology on a sound business model tends to succeed; good technology on a broken one tends to fail anyway; and any advantage you do build has to be renewed because it can be copied.

### Ethical dilemma: what the newest gadgets cost

An **ethical dilemma** arises when both options break a moral imperative — not right against wrong, but two harms. There is rarely a definite answer, so the chapter gives two steps.

- 1 Weigh the **consequences** of each option in benefits and harms, considering degree and time horizon.
- 2 Then judge the **actions** themselves, irrespective of consequences, for honesty, fairness, and respect.

Apple designs in California and assembles in China, controlling the designs but not how suppliers work. **Foxconn** drew scrutiny for repetitive strain injuries, 50,000 resignations a month, and an audit confirming excessive overtime. Against that sit shareholder obligations, a shortage of suppliers, and workers for whom the job beats farm work.

Each option trades one harm for another.

#### SECTION 1-4

### Computer ethics, privacy, and intellectual property

Every system here runs on data, much of it about people, including you. Three questions overlap without being identical: what is ethical, what an organization promised, and what the law permits.

Something can be legal and still be wrong. **Computer ethics** describes moral issues and standards of conduct as they pertain to the use of information systems. Whether a firm may sell personal data

depends on jurisdiction, consent, and its promises.

## Mason's four questions: PAPA

In 1986 **Richard O. Mason** reduced the debate to four issues, known since as **PAPA**. Almost every argument about technology and people is one of them.

**Information privacy** — what must an individual reveal, at work or in an ordinary transaction?

**Accuracy** — who answers for authenticity, when an edited photograph still looks like a record?

**Property** — who owns information about individuals, and how may it be sold on?

**Accessibility** — what may an organization obtain about others, and how may it be used?

Open each issue and read all four facets.

### EXPLORE

## The four PAPA issues, one card at a time

Open each of Mason's four issues and read all four facets before moving to the next.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

### Information privacy · What must I reveal?

- **What the question is:** What information an individual should have to reveal to others in the workplace or through other transactions, such as online shopping.
- **Where you already meet it:** An online store greets you by name and recommends the product you were about to search for, and the ads that follow you look aimed at you personally.
- **What goes wrong when nobody asks it:** Social Security numbers, credit card numbers, medical histories, and family histories end up online, where a coworker, a spouse, or a future employer can find them with a search.
- **The hard part:** You control only your half. A friend can post a photo of you, an abandoned profile keeps the address you typed years ago, and search engine caches keep pages reachable long after they come down.

### Accuracy · Is the record true?

- **What the question is:** Ensuring the authenticity and fidelity of information, and settling who answers for a record about a person that turns out to be wrong.

- **Where you already meet it:** The shopping habits and credit history a company compiles about you, and photographs, which anyone with ordinary software can now rearrange and otherwise change before publishing them.
- **What goes wrong when nobody asks it:** Decisions get made about you from a record you have never seen, and an altered picture circulates as though it were evidence, because nothing in the system distinguishes a faithful record from a convincing one.
- **The hard part:** The remedy exists on paper as Access/Participation, the right to see your data and request correction, but you have to learn the record exists before you can challenge it.

#### **Property** · Who owns the data?

- **What the question is:** Who owns information about individuals, and how that information can be sold and exchanged.
- **Where you already meet it:** The unwanted solicitations from credit card companies, department stores, and charities that follow one purchase, and the answer to the question of how you got on another mailing list.
- **What goes wrong when nobody asks it:** Separate pieces of you get joined. Demographic data, psychographic data, and card transactions combine into a profile precise enough to support targeted manipulation, discrimination, or a damaging breach.
- **The hard part:** The chapter frames database control as a property question, but control is not unlimited: promises, consent, the kind of data, and applicable law can all constrain sale or exchange.

#### **Accessibility** · Who may reach it?

- **What the question is:** What information a person or organization has the right to obtain about others, and how that information can be accessed and used.
- **Where you already meet it:** Which employees inside a company can open the customer database, and what a search engine hands a stranger who types your name, since almost anything posted by or about you can be found and sits in a long-term cache.
- **What goes wrong when nobody asks it:** Unsupervised staff browse records they have no business opening, and personal information such as Social Security numbers, credit card numbers, and medical histories becomes reachable by anyone who thinks to look for it.
- **The hard part:** It has to be answered twice: at a strategic and ethical level (should we be doing this?) and at a tactical level (hiring, training, supervision, and software and hardware safeguards).

Now drop each situation into the one issue it raises first.

**SORT****Which PAPA issue is this?**

Drop each situation into the one Mason issue it raises first and most directly.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Privacy** — what an individual should have to reveal to others

- A restaurant makes you type an email address to use its free Wi-Fi, and never says what happens to the address. — Giving an email address for Wi-Fi is one of the collection moments the chapter names, and the question is what you should have to reveal in an ordinary transaction.
- You give a birth date and home address to a social network, abandon the profile a year later, and the data stay out there. — You revealed personal information for one purpose and lost any say over how long it lives, which is the privacy question rather than an ownership dispute.

**Accuracy** — authenticity and fidelity of the information

- An insurer's file lists a surgery you never had, so your claim is denied. — Nothing was leaked or sold here; the record is simply not true, which is the authenticity and fidelity problem.
- Someone edits a photograph of a public figure to show an event that never happened, then posts it as news. — The chapter raises photo manipulation as an ethical problem precisely because the image no longer faithfully represents what happened.

**Property** — who owns the data and how it may be sold or exchanged

- A magazine shares its subscriber list with a department store, which starts mailing you catalogs. — Who controls a subscriber list and under what conditions it may be exchanged is exactly the property question; legality depends on the surrounding notice, consent, promises, and law.
- A company promised survey answers would be used strictly internally, then sold them to another firm years later. — The exchange raises the property question, and breaking the stated purpose is also an ethical failure that may trigger legal enforcement against deceptive promises.
- A retailer combines what you searched for, what you bought, and what you liked into one profile that predicts your next purchase. — Each piece was given separately, so the ethical question is ownership of the combined file, the aggregation the chapter calls the Database of Intentions.

**Accessibility** — who has the right to obtain it, and who can reach it at all

- A claims clerk with no reason to look opens a neighbor's file, because the system lets any employee open any record. — The data are accurate and legitimately held; the issue is who has the right to obtain them and how access is controlled.

- A hiring manager searches your name and reads a question you posted years ago on a page that has since been taken off the web. — Nothing here is inaccurate and nothing was sold; the question is what a person has the right to obtain about you and how they may reach it, and search engine caches keep pages available long after the original comes down.
- A bank ties access to its customer database to hiring, training, and supervision, backed by software safeguards. — This is the tactical half of the accessibility question, the controls deciding who may reach the data and how.

Those four ask about information that already exists about you. A fifth, the **digital divide**, asks who reaches the technology at all.

### Information privacy up close

Most personal data is handed over quietly: a card purchase records what you bought, and the address you type for free Wi-Fi is the price of the connection. Providing it enables that transaction, not every later use.

One survey is harmless; the combination is not, because two kinds of data work together.

**Demographic data** — who am I and where do I live, the part of you that appears on any form.

**Psychographic data** — what do I like, the part you give away by choosing rather than declaring.

Side by side they assemble an accurate profile, an aggregation called the **Database of Intentions**: what you want, buy, and like, and whom you know. Such profiles feed targeted advertising.

Providing data at separate moments never meant agreeing they be combined into one picture.

So who controls it? The chapter frames a company database as a property question, limited by what the company said. A firm that promised to keep marketing data in-house and then sold them breaks that promise, which regulators may enforce.

**Things you cannot control.** A friend's photo of you stays up whether you like it or not. In 2014 the European Court of Justice ruled that individuals have a **right to be forgotten**: search engines may have to remove links to inaccurate or excessive information.

Two newer rules go further.

**The General Data Protection Regulation (GDPR)** — applied across the European Union since 2018, establishing transparency, data minimization, and confidentiality.

**The California Consumer Privacy Act (CCPA)** — gives covered California consumers defined rights over personal information held by covered businesses.

Three questions on ethics, PAPA, and who owns your data.

**CHECK YOURSELF**

**Ethics, PAPA, and who owns your data**

Pick the best answer, then read every explanation, including the ones for options you did not choose.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. What does the term **computer ethics** refer to?

- **A.** The laws a government passes to punish computer crime and protect data — Those are statutes, such as the ECPA. Law and ethics overlap but differ: governments play catch-up, so much of what is legal is still an open ethical question. This would be right if the question asked what computer crime legislation does.
- **B.** The technical controls that keep unauthorized people out of a company database — That is Integrity/Security, one of the five fair information practices and the tactical half of the accessibility question. It is a way of acting ethically, not the field that decides what acting ethically means.
- **C.** The moral issues and standards of conduct that pertain to the use of information systems — **correct.** This is the chapter's definition, and it is deliberately broad: it covers conduct that may be legally permitted in a particular context, such as combining purchase data into a customer profile, and still worth arguing about.
- **D.** A company's written policy listing which software employees are allowed to install — That is an acceptable-use policy, one product of ethical thinking inside one organization. Ethics is the set of standards such a policy gets written from, which is why schools and businesses each write their own.

2. A hospital system lists the wrong blood type for a patient because a clerk mistyped it, and treatment decisions follow that record. Which of Mason's four issues does this raise first?

- **A.** Accuracy, because the authenticity and fidelity of the information have failed — **correct.** Accuracy is Mason's question about authenticity and fidelity, and it is the issue that can fail even when data remain private and access is correctly restricted.

- **B.** Information privacy, because medical data about an individual is exposed — Privacy concerns what an individual should have to reveal to others. Nothing was revealed to anyone who should not see it; the record is simply untrue. This would be the answer if a correct record had been leaked or sold.
- **C.** Property, because the hospital controls the record and decides how it is maintained — Property asks who controls information about individuals and how it may be exchanged. Control of the record is not what failed here, and nothing was transferred.
- **D.** Accessibility, because clinicians can open the record and act on what it says — Accessibility asks who has the right to obtain information and how it is accessed. The clinicians reading the file are exactly who should have access; the problem is what the file says.

**3.** You buy a lamp from an online retailer. Which statement best connects the chapter's property question to current privacy rules?

- **A.** Because the record is about you, the retailer needs affirmative written permission before repurposing it — Some laws require consent for particular uses, but written permission before every internal use is not a universal rule. This would be correct only where the applicable law or a prior promise specifically required that consent.
- **B.** Because the retailer maintains the record, its ownership interest controls unless it made a specific promise — The chapter uses database control to introduce the property question, not to erase privacy obligations. This would be correct only if no law, notice, consent choice, contract, or promise limited the proposed use.
- **C.** Because the card issuer processed payment, its rules control how the retailer may use the transaction record — A card issuer maintains its own payment record, while a retailer maintains its customer and order records. Processing one part of a transaction does not give the issuer exclusive control over every resulting record.
- **D.** The retailer controls its customer record, but notice, consent and applicable law can limit later use — **correct.** This preserves the chapter's property question while recognizing that current legal rights and duties vary by jurisdiction, data type, notice, consent, and the organization's promises.

## What a privacy policy is supposed to do

The accepted list of **fair information practices** comes from the Federal Trade Commission.

| PRACTICE         | WHAT IT MEANS                                                                                        |
|------------------|------------------------------------------------------------------------------------------------------|
| Notice/Awareness | What data are gathered, what for, and who gets access — the job of a <b>data privacy statement</b> . |
| Choice/Consent   | You <b>opt in</b> , agreeing to collection or further use, or <b>opt out</b> , refusing it.          |
|                  |                                                                                                      |

| PRACTICE             | WHAT IT MEANS                                                |
|----------------------|--------------------------------------------------------------|
| Access/Participation | Means to see the data about you and correct it.              |
| Integrity/Security   | Reputable sources, and controls against unauthorized access. |
| Enforcement/Redress  | Means to enforce the practices and give customers a remedy.  |

The catch: a privacy statement is notice, not a shield. It can be vague while a vendor records which pages you looked at, what you bought, and where it went.

## DECIDE

### One evening of ordinary privacy decisions

Work through the evening one decision at a time and read what each choice actually costs you.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

You are at a coffee shop on your laptop, planning to buy a pair of headphones. Nothing here is dramatic, and that is the point: this is where most personal data gets given away.

**Decision 1.** The shop's free Wi-Fi will not connect until you enter an email address on its login page.

- Enter the personal address you use for family, school, and banking. — The address can now be recorded in connection with a place and time, subject to the notice, choices, promises, and law that apply. It is also the same address your important password resets use, so unrelated marketing and security messages share one crowded channel.
- Enter a separate email account you keep only for sign-ups and online purchases. — **the stronger call.** This is the chapter's advice applied early. The address may still be collected and used as disclosed and permitted, but the marketing that follows lands where it cannot bury anything important and is not tied to your banking or work identity.
- Make up an address that does not exist, so nothing can be sent to you. — It hides you tonight, but you cannot receive the confirmation link many networks send, and the first real order needs a real address anyway. A separate real account solves it permanently instead of once.

**Decision 2.** The store you found is one you have never used, and you cannot find a privacy policy anywhere on the site.

- Order anyway; a missing policy just means a small company that has not written one yet. — A missing policy leaves you without clear notice of what is gathered, why, who receives it, or how it is protected. Applicable law may still impose duties, but you cannot evaluate the vendor's stated practices or choices before handing over the data.

- Buy from a competitor that publishes a clear policy. — **the stronger call**. This is the chapter's rule: review the privacy policy of every company you do business with, and refuse those that have no clear policy. Better still is a policy audited by an independent organization such as TRUSTe, because outside monitoring adds a check the site cannot perform on itself.
- Order, but pay with a card that has a low limit so the damage is capped. — That addresses payment exposure, which is a different risk. It changes nothing about the vendor's collection of the purchase, the browsing that led to it, or the shipping address, nor does it tell you how those records may be used.

**Decision 3.** At checkout a box is already checked: 'Yes, share my information with our trusted partners.' A link beside it says Privacy.

- Open the privacy statement, see what partners means and whether the data are required, then uncheck the box. — **the stronger call**. You used two practices in one minute: Notice/Awareness, which is what a data privacy statement is for, and Choice/Consent, by exercising the opt out. Reading alone would not have protected you; unchecking is what changed the outcome.
- Leave the box as it is, since it was checked by default and is probably required to complete the order. — It is not required; it is an opt-out arrangement, where sharing happens unless you act. Leaving it checked is how your details reach partners you never chose, and how the catalogs start.
- Uncheck the box and skip the policy, since the box is the only thing that matters. — Better than leaving it checked, but the statement is where you learn what else is collected, who has access, and how confidentiality is handled. The checkbox governs one use of your data, not all of them.

**Decision 4.** The order is placed. You keep browsing other stores on the same laptop, and those headphones start appearing in ads on unrelated sites.

- Log out of the store account, which should stop it from following you. — Logging out ends your session but leaves the cookies on your machine, and those are what let a site owner monitor where you go and what you do. The tracking continues while you are signed in to nothing.
- Rely on the store's privacy policy, since it promised not to misuse your information. — A policy states intentions, not capabilities, and the chapter is blunt that data privacy statements often do not protect consumers. The tracking here is being done by a file sitting on your own machine.
- Manage the browser's cookie settings and turn off ad personalization. — **the stronger call**. These are the chapter's suggested privacy steps, and the list runs further: a private browsing window and a search engine such as DuckDuckGo cut the same trail from the other end. Each limits part of the path between what you shop for and what follows you, but none should be mistaken for complete anonymity.

**Decision 5.** The headphones arrive late and damaged. You draft a furious post naming the clerk who answered the phone, with a photo of the receipt showing your address.

- Post it; it is all true, and you can delete it tomorrow if you regret it. — Truth is not the constraint. The internet never forgets: posts are stored, cached by search engines, and reshared, and content stays somewhere on the web after the original page is gone. The receipt also publishes your home address to strangers.
- Post it without the photo, since the text alone cannot identify anyone. — The photo was the worst part, so this is an improvement, but a named employee and an angry tone are exactly the regrettable material that can be devastating for a career later. The permanence problem is unchanged.
- Send a factual complaint through the company's support channel. — **the stronger call.** It solves the actual problem and leaves nothing a future employer or stranger can dig up, because the receipt and the employee's name stay out of public view. The chapter's warning is about permanence, and this is the version of the complaint that does not follow you.

## Email at work is not your mail

Many companies monitor the email accounts they supply. The **Electronic Communications Privacy Act (ECPA)** of 1986 regulates interception, but its exceptions guarantee nothing about messages on an employer's system. So treat work accounts as organizational resources and send private messages from a personal device.

## Steps that actually protect your privacy

None of these makes you invisible, but each closes one leak.

**Manage cookies** — sites leave **cookies** so the owner can monitor where you go; use browser settings or management software.

**Visit anonymously** — private windows, search engines such as DuckDuckGo, and ad personalization off keep marketers and identity thieves at bay.

**Beware what you post** — the internet never forgets, and one regrettable item can wreck a career.

Rate your habits honestly below.

### SELF-CHECK

## Your own privacy habits

Rate each statement honestly; anything you cannot claim points straight at what to re-read.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

- I can define information privacy in one sentence and name a moment this week when I traded some of it for convenience. Re-read the definition: what an individual should have to reveal to others in the workplace or through transactions such as online shopping.
- I know whether the last account I created asked me to opt in, or quietly relied on my not opting out. See Choice/Consent in the fair information practices table: opting in signals agreement, opting out signals refusal.
- I could find the data privacy statement on a site I use weekly, and I know the five things notice is supposed to tell me. Notice/Awareness covers what is gathered, what for, who gets access, whether it is required or voluntary, and how confidentiality is ensured.
- I have deliberately managed the cookie settings in my browser, and I can say what a cookie lets a site owner observe. See the steps for maintaining privacy online: cookies let the site owner monitor where you go and what you do on the site.
- I can name something I have posted that I would not want a future employer to read, and explain why deleting it may not remove it. Search engines keep long-term caches, and most content stays somewhere on the web after the original page is gone.
- I would be comfortable if every message I sent from a work account this month were read aloud by my employer. Re-read the email section: the ECPA protects voice mail far more than email, and the working rule is to send only what would not embarrass you if made public.
- I can list four behaviors the Computer Ethics Institute prohibits and the two habits it recommends, without looking them up. See the responsible computer use list, including the two positive recommendations that follow the prohibitions.

## Intellectual property in a world of perfect copies

**Intellectual property (IP)** means creations of the mind with commercial value: a song, a photograph, a program. Analog copies lost quality, protecting creators by accident; digital files removed that friction, and **3D printing** copies objects too.

The cases are ordinary: software your school licensed to you, kept after graduating. People defend it two ways.

**No real loss occurred** — the person would have used a free alternative, or bought nothing.

**Copying counts as praise** — in many societies using another's work honors the creator, so a famous song under a video reads as flattery.

Either way the act is the same: using intellectual property without permission, attribution, or compensation.

## Codes of conduct and responsible use

Statutes answer none of the daily questions, so most universities publish guidelines instead. The best known come from the **Computer Ethics Institute**, whose eight prohibitions cover harming others, interfering with their work, snooping, stealing, bearing false witness, copying software, using others' resources, and appropriating their work.

Beyond those it recommends two habits: think about the social consequences of what you build, and use a computer with consideration and respect for others.

### CHECK YOURSELF

### Email, consent, IP, and the code of conduct

Answer all four, then read every explanation before you move on.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. Your employer has never mentioned monitoring. What is the soundest approach to email you send from your work account?

- **A.** Treat it as private until the company publishes a policy, because notice normally defines when monitoring begins — Notice can be required by a policy or particular law, but it is not safe to infer privacy from silence. This would be correct only where the applicable rule made advance notice a condition of the specific monitoring at issue.
- **B.** Treat the ECPA as the controlling protection, comparable to the rule for a private phone conversation — The ECPA regulates electronic communications, but different communications, access methods, exceptions, and jurisdictions can produce different rules. This would be correct only if one protection applied identically to every channel, which it does not.
- **C.** Treat personal messages as exempt from monitoring, because their subject matter is not company business — Monitoring commonly turns on the account, device, network, policy, and applicable law rather than whether a message feels personal. This would be correct only if the governing policy or law created that subject-matter exemption.
- **D.** Read the policy and applicable rules, and assume an employer-provided account may be monitored — **correct.** This does not claim that every form of monitoring is lawful; it recognizes that rules vary and applies the chapter's practical lesson to keep sensitive personal communication off employer-provided accounts and systems.

2. A checkout page shows a box that is **already checked**: 'Yes, send me offers from our partners.' Which consent model is the page using?

- **A.** Opt in, because a checked box is a signal of agreement to further use of the data — Opting in is when you take the action that signals agreement, such as checking an empty box or answering a request for permission. Had the box started empty and you checked it, this would be right; pre-checking reverses who has to act.

- **B. Opt out**, because the data will be used unless you take an action to refuse — **correct**.  
Opting out means signaling that data cannot be collected or used in other ways, and here the sharing proceeds by default unless you uncheck the box.
- **C. Notice/Awareness**, because the wording on the box tells you what will happen to your data — Notice/Awareness is the disclosure of what data are gathered, what for, who gets access, and whether providing them is voluntary, normally in the data privacy statement. The box is the choice mechanism the notice describes.
- **D. Access/Participation**, because you can see the setting and change it yourself — Access/Participation means seeing the data a company has collected about you, checking it for accuracy, and requesting correction. Toggling a marketing preference is a consent setting, not access to your record.

**3.** You keep using software your school licensed to you after you graduate, and you pass the installer to a cousin. Why does the chapter treat this as an intellectual property problem?

- **A.** There is no real problem, since the publisher lost no sale: you would have used a free alternative if you had to pay — This is exactly the argument the chapter says people make, alongside the claim that students cannot afford expensive software. The chapter reports it seriously and still concludes the work is being used without permission and without compensating the creator.
- **B.** Because each copy degrades the original, so the publisher's product is worth less afterward — This is analog thinking. Copying a song to tape once cost quality, but digital duplication is lossless, which is precisely why the chapter says digital media made this problem so much larger.
- **C.** Because IP means creations of the mind with commercial value, and this uses one without permission or payment — **correct**. The license let you install and use the software while enrolled; using it afterward and handing it on goes past what was granted, and intellectual property is a creation of the mind with commercial value.
- **D.** Because the school owns the software, so passing it on takes school property rather than the publisher's — A license is permission to use, not ownership. The school holds an agreement with the vendor allowing enrolled students to install the software, and the vendor keeps the intellectual property throughout.

**4.** Besides its list of prohibitions, what does the Computer Ethics Institute's guidance actually recommend that people do?

- **A.** Publish an organizational code of conduct, so that every user of the system knows in advance which uses of it the organization will and will not stand behind — Writing guidelines is what businesses, universities, and public school systems do, and those guidelines tell users to act responsibly, ethically, and legally and to follow online etiquette and the law. That is an organization's job, not one of the two habits the institute recommends to an individual user.

- **B.** Secure your accounts and files so that nobody can reach them without authorization — Good practice, and close to the Integrity/Security fair information practice, but the institute states it from the other side, as a prohibition on using other people's computer resources without authorization. It is not one of the two recommendations.
- **C.** Think about the social consequences of programs you write and systems you design, and show consideration for others — **correct**. These are the two positive recommendations that follow the prohibitions, and they are the only items asking you to consider people you will never meet rather than simply to refrain from an act.
- **D.** Avoid using a computer to harm others or to snoop in other people's files — This is genuinely on the institute's list, which is what makes it tempting, but it sits among the prohibited behaviors. The question asked what the guidance recommends beyond the prohibitions, which is the two positive habits.

## Turning information systems into competitive strategy

Two grocery chains buy the same checkout software. One cuts labor cost with it; the other learns what households buy and makes offers the first cannot match. Same software, different **intent**.

**How this supplement relates to the chapter.** The chapter defines competitive advantage and cites Porter, but does not present Five Forces or the value chain as an objective. They extend Objectives 1.1–1.4 into practice.

### What “strategic” means here

When something your company does makes a customer choose you instead of a rival, and the rival cannot easily copy it, you have a **competitive advantage**. That is not the same as being good.

Two shops can both make good coffee, but one that knows when to alert you about your usual bean has an advantage a rival must build customer history to match. The systems at **Zoom** and **FedEx** were **strategic in their intent**: built and continuously updated to gain or sustain advantage.

There are three reasons to spend on a system, and only the third is strategic.

**Efficiency** — the same work with fewer people or less waste, so only the cost moves.

**Return on investment** — it pays back its cost, which is what a system must do, not the most it can do.

**Strategic use** — it changes the customer’s reason to choose you, being part of the offer rather than support for it.

Size does not decide which one you get: a corner boutique can beat its rivals with technology too.

The purchased software is never the advantage. Amazon and Zoom built theirs by combining commoditized technologies with proprietary systems and processes; for Google the customer data are the value.

### A clearly hypothetical practice situation

You advise a regional grocer. These assumptions exist only for this practice situation.

- Its shoppers are loyal out of habit, so nothing they would lose keeps them from leaving.

- Managers phone produce orders to a few suppliers, so nobody can say chain-wide what is paid.
- Inventory sits in a separate system per store, so headquarters cannot see stock.
- A warehouse club has entered local markets, and a meal-kit service advertises nearby.

## Porter's Five Forces: where the pressure comes from

Before naming the technology a company needs, name what squeezes it. Porter's argument is that profit in any industry is pressed by five forces, and the wrong name buys the wrong system.

**Competitive rivalry** — how hard the firms already in your industry fight over the same shoppers.

**Threat of new entrants** — how easily an outsider could compete tomorrow, given the **barriers to entry**: cost, scale, know-how, loyalty.

**Threat of substitutes** — a different product meeting the same need, so for a grocer the meal kit, not another grocer.

**Bargaining power of buyers** — your customers' ability to push price down or service up, because leaving costs them little.

**Bargaining power of suppliers** — the reverse, when few alternatives exist, as with Apple's assembly volumes.

Each force has a signal, and an initiative that relieves it.

### INTERACTIVE DIAGRAM

## The five forces, one at a time

Pick a force to redraw it: what the pressure looks like, the signal that it is the strong one, and the kind of IT initiative that relieves it.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Competitive rivalry** — Pressure from the companies already selling what you sell

The practice grocer (Several locations, familiar brands) → Two rival chains (Same aisles, same suppliers) → One shopper (Fills a single cart each week) → The usual weapon (Price cuts that thin both margins)

- **What it is** - how hard the firms already inside your industry fight one another for the same finite set of customers. Two gas stations across the street changing their signs on the same morning is rivalry you can watch happen.

- **Signal it is strong** - discounting is the only tool anyone reaches for, a promotion is matched within days, and unit volume stays flat while margin falls. Nobody is winning; everyone is paying.
- **What IT does about it** - moves the fight off price. One chain-wide inventory record so the advertised item is actually on the shelf, faster checkout so the line is shorter, and personalized offers built from purchase history a rival does not have.
- **The catch** - anything a rival can also buy becomes table stakes within a season. The chapter's point holds here: advantage sits in the proprietary systems and business processes you build on top of commoditized technology, not in the software licence.

#### **Threat of new entrants** — Pressure from companies not in your industry yet

Outsiders (Have capital, want your customers) → Barriers to entry (Cost, scale, know-how, loyalty) → A warehouse club (Enters some local markets) → The practice grocer (Suddenly shares its shoppers)

- **What it is** - how easily a company that does not compete with you today could start tomorrow. What holds outsiders back is called barriers to entry: the money, scale, expertise, or customer loyalty a newcomer would have to match first.
- **Signal it is strong** - newcomers keep appearing, starting up takes little capital, and nothing you own would take a competitor years to reproduce. If your only barrier is that customers are used to you, you have no barrier.
- **What IT does about it** - builds the barrier that money alone cannot buy: years of purchase history, saved shopping lists, subscription reorder, and a loyalty record that makes leaving cost the customer something real.
- **The catch** - a well-funded entrant can buy scale and buy price. The barrier that actually holds is the one that takes time to accumulate, which is why the data asset matters more than the store count.

#### **Threat of substitutes** — Pressure from a different answer to the same need

The real need (Feed a family on Tuesday) → Your answer (Groceries from an aisle) → A different answer (Meal kits, delivery, restaurants) → The result (The category shrinks, not just your share)

- **What it is** - a product or service from outside your industry that meets the same underlying need, so the customer never enters your industry at all. For a grocer the substitute is the meal-kit box and the restaurant, not the grocer across town.
- **Signal it is strong** - your price cuts stop working, because the customer you lost is not comparing you against another grocer. Whole categories decline at once, and the shoppers who leave do not come back for a coupon.
- **What IT does about it** - extends your offer into the substitute's territory: online ordering with reliable delivery windows, subscription staples that reorder themselves, and prepared-meal ordering that competes for the same Tuesday evening.

- **The catch** - substitutes are the force companies notice last, because management watches competitors with the same sign on the roof. Nothing in your sales data names the substitute; you have to go look for it.

### **Bargaining power of buyers** — Pressure from the people who hand you money

Shoppers (Phone in hand, price app open) → Switching cost (Near zero - a rival is nearby) → The practice grocer (Shelf price effectively set by the shopper)

- **What it is** - your customers' ability to force your price down or your service up, because walking away costs them almost nothing. Power grows when they can compare easily, when switching is trivial, or when a few large accounts are most of your revenue.
- **Signal it is strong** - shoppers compare your shelf price in seconds and leave over pennies, promotions buy visits but not repeat visits, and your busiest customers cannot be named because you have no record of who they are.
- **What IT does about it** - gives the customer something to lose by leaving: a loyalty record, a saved list, an order history, a personalized offer that only exists because the system knows this household.
- **The catch** - a loyalty program that is only a discount teaches shoppers to wait for the discount. The relief comes from the data and the switching cost, not from the coupon.

### **Bargaining power of suppliers** — Pressure from the firms you buy from

A small supplier pool (Supplies nearly all the produce) → The practice grocer (Learns of a price rise on the invoice) → Alternatives (Unqualified, unpriced, unknown) → The result (Terms set on the other side of the table)

- **What it is** - the ability of the firms you buy from to raise prices, cut quality, or dictate terms, because you have few alternatives and switching is slow. The chapter reports this pressure without labelling it a force: there are few suppliers worldwide who can, on relatively short notice, produce the numbers needed to meet demand for Apple's products, so shifting suppliers is not easy for Apple.
- **Signal it is strong** - a handful of suppliers cover most of your purchasing, price changes arrive as surprises rather than negotiations, and nobody can produce a list of qualified alternates.
- **What IT does about it** - a purchasing system that puts every quote side by side, tracks contract expiry before renewal, keeps a qualified alternate list current, and shares demand forecasts so the supplier plans with you instead of guessing.
- **The catch** - a long exclusive contract feels like control and is usually the opposite; it lowers this quarter's price and raises next year's dependence.

One lost sale can be told as a story about any of three forces.

**CHECK YOURSELF****Name the force**

For each situation, decide which of the five forces is actually at work before you decide what to build.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** The practice grocer's housewares aisle is losing sales. Shoppers say they now rent a pressure washer from a tool-rental app instead of buying one. Which force is this?

- **A.** Competitive rivalry, because another company is taking sales that used to belong to the grocer — Rivalry is pressure from firms selling the same product inside the same industry - another housewares retailer down the road. The rental app does not sell pressure washers at all, which is precisely what makes it something other than a rival.
- **B.** Threat of new entrants, because the rental app is new to the market — A new entrant is a company that starts selling what you sell. This would be the right answer if the app opened a store selling washers. Being young or newly launched is not the test; entering your industry is.
- **C.** Threat of substitutes, because renting meets the same need without buying the product at all — **correct**. A substitute satisfies the same underlying need - a clean driveway - by a different route, so demand leaves the whole product category rather than shifting between sellers inside it.
- **D.** Bargaining power of buyers, because shoppers are choosing where to spend their money — Buyer power is the pressure customers apply while still buying from your industry, by forcing your price down or your terms up. Here they are not negotiating with anyone in the industry; they left the category.

**2.** The practice grocer buys nearly all its produce from a small supplier pool. Suppliers raise prices, and store managers discover the change only when invoices arrive. Which force is strong?

- **A.** Bargaining power of buyers, because the grocer is the buyer in this relationship — Buyer power names the pressure a company's own customers exert on it, not the fact that the company happens to be doing the purchasing. Every force is named for whoever holds the leverage, and here the growers hold it.
- **B.** Bargaining power of suppliers, because alternatives are few and suppliers set terms the grocer discovers after the fact — **correct**. Supplier power is strong when suppliers are few, switching is slow or costly, and terms change unilaterally. All three appear here, and the invoice surprise also shows the grocer has no visibility into its own purchasing.
- **C.** Competitive rivalry, because other grocers are competing for the same produce — Rivalry would be the right label if other grocers were undercutting the practice grocer on the shelf. Competing for scarce supply is real, but the pressure described - price set by the supplier, learned late - sits on the supply side of the business.

- **D.** Threat of new entrants, because new growers could enter this market and bring prices back down again, so the grocer should wait for that correction rather than change anything now — This is a correct description of new entrants applied to the wrong side of the business, and it names a possible relief rather than the current pressure. Easy entry by new growers would weaken supplier power; nothing here says that is happening.

**3.** You have concluded that buyer power is the strongest pressure on the practice grocer. Which framework do you turn to next, and for what reason?

- **A.** The value chain, because it maps the activities inside the company and shows where a system could actually be built — **correct.** The five forces say what is squeezing the firm from outside; the value chain breaks the firm into nine internal activities, so an external finding becomes a specific place inside the business to build something.
- **B.** The five forces again, run location by location, because exposure to the warehouse club differs across the region — Re-running the forces per location does sharpen a diagnosis, and it is worth doing when the evidence is thin. It still cannot produce a project, because the five forces describe conditions outside the company and never name an activity inside it.
- **C.** The value chain, because it ranks the five forces from strongest to weakest — The value chain is the right framework named for the wrong reason. It contains no forces and performs no ranking - it divides the firm into five primary and four support activities.
- **D.** Neither - move straight to comparing vendors, because the diagnosis is finished and the frameworks have nothing left to contribute — Naming a force says what is wrong outside the firm, not what to change inside it, so a shortlist assembled now is a list of products in search of a problem. This would have been right only if the diagnosis had already identified the activity the system must strengthen.

## Porter's Value Chain: where inside the company to act

The five forces describe the weather outside; they never say which department to change. The **value chain** cuts a firm into nine activities, each adding value a customer will pay for or destroying it.

**Primary activities** — the five stages a product passes through, so a change to one reaches the buyer.

**Support activities** — the four touching no single product but making the others possible.

| ACTIVITY                            | WHAT IT COVERS                                                 |
|-------------------------------------|----------------------------------------------------------------|
| <b>Inbound logistics</b><br>primary | Receiving, storing, and moving goods before they are sellable. |
|                                     |                                                                |

| ACTIVITY                                    | WHAT IT COVERS                                                               |
|---------------------------------------------|------------------------------------------------------------------------------|
| <b>Operations</b><br>primary                | Turning inputs into what the customer buys: a priced shelf.                  |
| <b>Outbound logistics</b><br>primary        | Getting the finished product to the customer.                                |
| <b>Marketing and sales</b><br>primary       | Making customers aware and giving them a reason to choose.                   |
| <b>Service</b><br>primary                   | Keeping the customer whole after the sale: returns and repairs.              |
| <b>Firm infrastructure</b><br>support       | General management, finance, accounting, and planning.                       |
| <b>Human resource management</b><br>support | Recruiting, training, scheduling, and keeping people.                        |
| <b>Technology development</b><br>support    | Improving the product and the processes: research, design, in-house systems. |
| <b>Procurement</b><br>support               | Finding, qualifying, and contracting the firm's suppliers.                   |

**The distinction most often missed. Procurement** is deciding whom to buy from; **inbound logistics** is what happens to the goods after that. A quote portal is procurement, a dock scanner is inbound logistics.

**Sort**

### Which value chain activity does this system improve?

Place each proposed system for the hypothetical grocer in the value chain activity it actually strengthens - four of the nine activities appear here.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Inbound logistics** — primary: receiving, storing and moving goods that have arrived, before they are sellable

- A dock scanner that records every pallet the moment it is unloaded, so on-hand counts are right before goods reach the shelf — The goods were bought weeks ago; this system handles what happens to them between the truck and the shelf, which is inbound logistics.
- Temperature sensors in the refrigerated warehouse that raise an alert before a pallet of dairy spoils — The dairy has arrived but is not yet sellable, so protecting it in storage is part of receiving and handling incoming goods.
- A cross-dock routing system that decides which store should receive each incoming case — Directing arrived goods to their destination inside the company is internal handling of inputs, so it is inbound logistics rather than delivery to a customer.

**Operations** — primary: turning inputs into the thing the customer buys - for a grocer, the stocked and priced shelf

- A restock list that turns each register scan into a task telling a clerk exactly which shelf to refill — For a grocer, a stocked, priced, shoppable shelf is the product itself, so producing it is operations.
- A batch planner in the in-store bakery that says how many loaves of each kind to bake before opening — Producing the item the customer buys is operations, whether the input is flour or a pallet of packaged goods.

**Marketing and sales** — primary: making customers aware and giving them a reason to choose you

- A loyalty app that notices a household stopped buying diapers and sends that household an offer — It creates awareness and a reason to choose the grocer for a specific customer, which is marketing and sales.
- A campaign tool that measures which of two weekly circular headlines pulled more shoppers into stores — It measures and improves how customers are attracted, which is squarely marketing and sales.
- A basket report showing that shoppers who buy charcoal also buy paper plates, used to plan next month's promotions — The analysis exists to shape offers and promotions, so it strengthens marketing and sales even though the data comes from the register.

**Procurement** — support: finding, qualifying and contracting suppliers - the buying decision itself

- A bidding portal where approved growers quote a price on the same order of tomatoes — Choosing whom to buy from and on what terms is procurement - a support activity performed for the whole company, not the physical receiving that follows it.
- A contract database that flags supplier agreements approaching expiry — Renewal timing is a supplier and terms decision, so it lives in procurement even though no goods move.

## From a finding to a recommendation

A proposal an executive will fund has four parts, in this order.

- 1 **Name the force or activity** exactly: a recommendation that opens with technology has skipped the diagnosis.
- 2 **Give the evidence**, the observable fact behind it, so the meeting is not two opinions.
- 3 **State the initiative** concretely: not “improve digital capabilities” but “one chain-wide inventory record.”
- 4 **Commit to a measurable outcome**: the number that moves, and the date you check it.

**Say it in one sentence.** Because [force or activity] is [strong or weak], shown by [evidence], build [initiative], judged by [measure] at [date].

### **DECIDE** Advising a regional grocer

Work the four decisions in this hypothetical situation; after each one, compare every outcome before continuing.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice case. A regional grocer's sales are slipping, leadership wants to spend on technology, and nobody has yet said what problem the spending is meant to solve. You are the analyst in the room.

**Decision 1.** A members-only warehouse club enters some local markets. Sales fall only at the grocer's exposed locations. Leadership asks you to name the pressure before anyone names a product.

- Threat of new entrants - a company that did not sell here before now does — **the stronger call**. Right - a company that was not competing here is now selling the same goods to the same shoppers, and the hypothetical evidence is unusually clean: only locations exposed to the club declined, which ties the loss to the entrant rather than to a region-wide cause. Naming the force this precisely lets you reject any initiative that would not touch it.
- Threat of substitutes - shoppers have found a different way to meet the same need — A substitute meets the need without buying from your industry at all, such as a meal kit or a restaurant. The club sells the same groceries from the same suppliers, so it competes inside the industry, not outside it. Hold that thought - a real substitute is in this region too.
- Competitive rivalry - a competitor is taking share, which is what rivalry means — Understandable, and it is not far off, but rivalry is pressure from the firms already established in the industry. The distinguishing fact here is that this competitor was not in the market before, which is exactly what makes it an entrant. The difference matters: rivalry pushes you to defend, an entrant tells you your barriers to entry are low.

**Decision 2.** You have named the force. Now the response. The grocer knows little about who its shoppers are: no membership, no history, no way to distinguish a regular household from an occasional visitor.

- Match the club's advertised prices across every store — This starts a price war against an opponent with better buying power and lower overhead. Margin falls chain-wide, including at locations that were never threatened, and the club can outlast you. Price is the dimension where the entrant is strongest.
- Launch a loyalty program that records what each household buys, and use that history to make personalized offers — **the stronger call**. This raises the customer's switching cost and builds an asset the entrant cannot instantly buy: accumulated purchase history. It also fixes the blindness by helping the grocer recognize its regular customers.
- Expand the weekly circular's print run so more households see the grocer's specials — This creates more impressions of the same undifferentiated message. It costs money, reaches the club's members too, and produces no customer history, so the grocer still cannot say which households responded.

**Decision 3.** Separately, produce suppliers raise prices, and managers learn of it only from invoices. Purchasing happens through separate calls from individual stores.

- Sign a long-term exclusive contract with the cheapest supplier to lock in the price — This lowers near-term cost and raises supplier power later. Under exclusivity, the grocer has no alternate, no comparison price, and little leverage at renewal - the same trap the chapter describes when few suppliers can meet demand on short notice.
- Instruct store managers to cut produce orders until prices come down — Empty produce cases drive shoppers to the store that has tomatoes. The invoice falls only because the grocer reduced the reason to visit.
- Put purchasing on one system that shows every store's prices and volumes, and qualify additional growers — **the stronger call**. Now the grocer sees its own buying for the first time, can compare quotes side by side, and has credible alternates - which is what moves a negotiation. This is a procurement fix, not an inbound logistics fix, and saying so keeps the project in the right department.

**Decision 4.** The board will fund one initiative. It asks for a concise defense of the loyalty program.

- 'It will modernize us. Customers expect a digital experience from a grocer in this decade.' — No force, no evidence, no measure. Everything in that sentence would be equally true if the initiative were a new website, a mobile game, or nothing at all - which is why a board cannot tell whether it worked.

- 'Buyer power is high, so we will run a loyalty program and judge it on repeat visits.' — **the stronger call**. That is the full shape once you say the rest of it out loud: the force is buyer power, the evidence is that switching costs are near zero and the grocer cannot identify repeat customers, the initiative is a loyalty program that gives you purchase history for differentiated offers, and the measure is repeat-visit rate and share of basket at affected stores, read at the scheduled review. It can be argued with, which is the point - a proposal that cannot be argued with also cannot be evaluated. Add one caution out loud: any advantage from this is fleeting, because rivals can eventually do the same thing, so the plan must include what gets improved after launch.
- 'Two of our competitors already run loyalty programs, so we are behind and need to catch up.' — This justifies parity, never advantage, and it hands your technology roadmap to competitors. It also tells the board nothing about why the program would work here or how anyone would know if it did.

Two cautions limit what any system can do.

**The business model comes first** — firms with bad models fail whether or not they use technology, because a system amplifies an idea rather than replacing it.

**The advantage can be fleeting** — it fades because competitors eventually do the same thing, which is why FedEx keeps rebuilding.

Two questions on both of them.

#### CHECK YOURSELF

#### Right framework, right place

Three situations that test whether you can place a system in the value chain and defend what it will do.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** The practice grocer wants a system that tells every store manager what is on the next truck and when it will dock. Which value chain activity does it improve?

- **A.** Procurement, because the system is about goods the grocer purchased — Procurement is choosing suppliers and agreeing on price and terms - the purchase decision. Those goods were procured weeks ago; what happens to them from the truck onward is a separate activity with separate systems.
- **B.** Inbound logistics, because it covers goods on their way to the shelf — **correct**. Inbound logistics covers everything that happens to incoming goods before they are sellable, and knowing what is arriving and when is exactly that problem.

- **C.** Operations, because store staff will be using it during the working day — Operations is transforming inputs into what the customer buys - the stocked, priced shelf. Who touches a system and when they touch it does not determine its activity; the activity it improves does.
- **D.** Outbound logistics, because it moves product closer to the customer — Outbound logistics is getting the finished product out to the customer, such as batching pickup orders or routing deliveries. This system handles the trip from supplier in, which is the opposite direction.

2. An applicant-tracking and shift-scheduling system substantially reduces cashier turnover at the practice grocer. Where does it sit in the value chain?

- **A.** Operations, because cashiers do their work on the sales floor — Operations is producing what is sold. Where employees physically stand is not the test - recruiting, scheduling and retaining people is its own activity that supports operations rather than being part of it.
- **B.** Service, because cashiers are the staff who deal directly with customers — Service is what keeps the customer whole after the sale - returns, complaints, repairs. Cashiers do face customers, but a system that recruits and schedules staff acts on the workforce, not on the post-sale experience.
- **C.** Human resource management, a support activity that makes the primary activities possible — **correct.** Human resource management covers recruiting, hiring, training, scheduling and retention, and it is classed as support because every primary activity depends on the right people being there.
- **D.** Firm infrastructure, because hiring and payroll are run centrally out of headquarters — Firm infrastructure is general management, finance, accounting, legal and planning. Being run from headquarters is not the test: the value chain gives people decisions their own support activity, and this would be firm infrastructure only if the system managed the company's finances or planning.

3. The practice grocer's board is told a new loyalty system will give the chain a permanent advantage because no rival has one. Given this chapter, what is the most accurate response?

- **A.** The claim holds, because a system built to gain or sustain advantage keeps producing that advantage for as long as it runs — Sustaining is work, not a property the system acquires at launch: FedEx sustains its advantage by reengineering and improving performance on average twice a year. This would have been right if the chapter said advantage persists once built, and instead it says advantage can be fleeting.
- **B.** Advantage from information systems can be fleeting, because competitors can eventually do the same thing — **correct.** This is the chapter's own caution, and it explains the phrase gain or sustain: the advantage lives in continuing to improve the system after launch, not in the launch itself.
- **C.** The claim fails because information systems are now a commodity, so IT spending belongs on cost reduction and risk mitigation — This is the commodity argument the chapter states and then rejects, pointing to rapidly changing IT and to companies from Amazon to Zoom that built advantage by combining commoditized technologies with proprietary systems and business processes. It is a real position, but it is not this chapter's conclusion.

- **D.** The claim fails because loyalty software is sold off the shelf, so a rival can license the same package next quarter — The software is buyable, which is true and is why the launch by itself is not the advantage. It still misses the chapter's point: advantage comes from combining commoditized technology with proprietary systems, processes and data, so accumulated purchase history is the part a rival cannot license.

The five companies below are invented. Naming the force, placing a system, and catching a claim of permanent advantage are the same three judgments.

#### APPLY IT

### Five companies you have not seen before

Each question is a different, invented business. Judge the force, the placement, or the claim before you read why.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A two-screen independent movie theater has seen steady walk-in traffic for a decade. In the last two years, several regular customers have said they now just wait for the same films to reach a streaming service a few months later. Which competitive force is doing the most damage right now?

- **A.** Rivalry among nearby theaters showing the same films — Nothing here describes another theater competing for the same audience on price or seating; the customers are not choosing a rival theater, they are choosing not to go to one.
- **B.** Suppliers, since film studios set the terms for which theaters get a release — Studio terms would explain which films a theater can show, not why a customer skips the trip entirely once a film is available at home.
- **C.** The threat of substitute entertainment from outside the theater industry — **correct.** Streaming is not another theater; it is a different way to meet the same need, arriving from outside the industry the theater competes in.
- **D.** New entrants building additional theaters in the area — No new theater is described. The pressure is a way of watching the film that removes the need to visit any theater at all.

**2.** A dental lab makes crowns and dentures to order for local dentists. A new system lets a dentist's office upload a digital scan and see, in real time, whether the case has been received, is being milled, or has shipped. Which value-chain activity does this system improve?

- **A.** Outbound logistics, because it tracks a case on its way back to the dentist — **correct.** Outbound logistics covers getting the finished product to the customer, and a status feed that follows the case from milling through shipping is exactly that movement made visible.
- **B.** Service, because dentists are the customers being kept informed — Service is what happens after delivery — fixing a problem with a crown that does not fit. Being kept informed while the case is still in production is a different activity from after-sale support.

- **C.** Operations, because the lab is milling crowns while the case is open — Operations is producing the crown itself — the milling. The tracking system does not do the milling; it reports on where the case sits, which is closer to the movement of the product than its production.
- **D.** Firm infrastructure, because the system touches the lab's general management — Firm infrastructure is general management, finance, legal and planning. A case-tracking system for dentists has nothing to do with how the lab is financed or governed.

**3.** A boutique fitness studio's owner lists two problems in a planning memo: (1) class bookings are still taken by phone and are double-booked most weeks, and (2) the studio's Instagram account "does not feel cool enough" compared to a newer studio down the street. Which one is a genuine value-chain opportunity a system could address?

- **A.** Both, since either problem could be fixed with the right kind of app — Wanting a fix is not the test. One of these names something the studio does; the other names an opinion about a competitor's image.
- **B.** Neither, since a small fitness studio does not really run a full value chain — Every business runs a full value chain, including the scheduling and service work happening around the classes, not only the classes themselves.
- **C.** Only the Instagram one, since brand perception is what drives a customer's choice of studio — Brand perception can matter, but "does not feel cool enough" names no activity a system operates on — there is nothing here to build.
- **D.** Only the double-booking one, since it names an activity a system could change — **correct.** Booking and scheduling sit inside operations or service, are something the studio does today, and a system could plainly change how they are done. A rival's image is a competitive fact, not an activity in this studio's own chain.

**4.** A regional pest-control company installs the first online same-day booking tool in its market and tells investors this guarantees a permanent lead over every competitor. Given the chapter's own caution, what is the most accurate response?

- **A.** Advantage from an information system can be fleeting, since competitors can eventually copy it — **correct.** Being first does not make an advantage permanent; the chapter is explicit that a rival can eventually do the same thing, which is why sustaining it takes continued work rather than a single launch.
- **B.** The claim holds, since being first to market with a system locks in the advantage for good — This is the claim being tested, not the chapter's position. Nothing about launching first prevents a competitor from building the same tool next season.
- **C.** The claim fails only because booking software is inexpensive to build — Cost is beside the point. An advantage can fade even when a system was expensive to build, and a cheap one can still be worth having; the issue is durability, not price.
- **D.** The claim fails because customers do not care about booking speed for pest control specifically — Nothing in the scenario says customers are indifferent to booking speed, and the question is about whether an advantage lasts, not whether the feature matters at all.

5. A specialty coffee roaster sells almost entirely through a handful of long-standing wholesale accounts and has been losing accounts to a competitor with fresher relationships. The owner commissions an expensive mobile app for direct-to-consumer sales, expecting it to fix the account losses. What does the chapter's caution about business models say to expect?

- **A.** The app will likely succeed regardless of the account losses, since a genuinely good app overcomes almost any weakness underneath it — This reverses the chapter's point. A system amplifies whatever model is already there; it does not overcome a weakness the model has in an entirely different part of the business.
- **B.** The app targets a different problem than the one described; a system amplifies a business model, it does not repair one — **correct.** The business problem named is account relationships; the app is a consumer-sales tool. A system built for one activity does not repair a different one, however well it is built.
- **C.** The app is doomed to fail outright, since no app can be built while wholesale relationships remain unfixed — Nothing requires fixing wholesale first — the app can be built and can even succeed on its own terms. The caution is about what it can fix, not whether it can exist.
- **D.** The account losses stop mattering once the app launches, since consumer and wholesale sales run on entirely separate tracks — The two channels can certainly interact, but more importantly the question is about whether the app addresses the stated problem at all, which it does not.

## Using AI to improve business workflows

Two chains subscribe to the same AI assistant. One hands it the work and stops checking; the other decides, task by task, where it drafts and where a person signs. Same subscription, different **method**.

**How this supplement relates to the chapter.** The AI material is the chapter's own, spread through it rather than gathered into an objective. What this adds is a **method** for deciding where AI belongs.

### What generative AI actually is

**GenAI** is artificial intelligence that learns patterns, relationships, natural language, and problem-solving from existing datasets. Three parts explain behavior you meet later.

**Learned from existing datasets** — trained on free internet content, so what it knows came from outside your organization.

**Generative** — it creates new content from a prompt, so one question asked two ways returns two answers.

**Tuned system by system** — all summarize and classify well, but they differ by strength, so choosing one is a business decision.

ChatGPT, released in 2022, reached a million users in a month and 100 million by January 2023.

### Where AI sits among the systems already covered

AI is not a separate universe. In the catalogue of categories it is the **intelligent system**, which emulates or enhances human capabilities: loan analysis, self-driving cars, Siri.

**The systems around it report** — a transaction processing system records events, a management information system summarizes them, a decision support system models them.

**An intelligent system judges** — a larger claim, because it takes on part of the judgment, not just the numbers underneath.

It does not replace the systems around it: a loan application still has to be recorded somewhere first.

### CHECK YOURSELF

## What the technology is, and where it fits

Three questions on the chapter's own account of GenAI and on where it sits among the systems organizations already run.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. In the term generative AI, what does the word generative refer to?

- **A.** The system creates new content from what it learned in its training data, in response to a user prompt — **correct**. The generative aspect is the ability to create new content based on what the system learned from its training data in response to a prompt, rather than retrieving a stored answer.
- **B.** The system generates its own training data rather than depending on datasets collected by people — This inverts where the data comes from. Many of these systems were trained on freely available content from the internet, discussion boards, social media and other sources - material people created, not material the system produced for itself.
- **C.** The system generates a profit for the organization that deploys it, which is why it is worth buying — Return on investment is a reason an organization might buy a system, and it is a genuine test any system has to pass, but it is not what the word generative describes. The word names how the output is produced.
- **D.** The system generates its answer by retrieving the closest ready-made response stored during training — Retrieval is the opposite of what the word names. The chapter describes a system creating new content from what it learned in response to a prompt, rather than selecting a stored answer that was written out in advance.

2. In the chapter's table of the categories of information systems used in organizations, which category do ChatGPT, Gemini, Siri and self-driving cars appear under?

- **A.** Transaction processing system, because these systems handle enormous volumes of day-to-day events — A transaction processing system handles day-to-day business event data at the operational level, the grocery checkout register being the chapter's example. Volume is not the test; recording business events as they happen is.
- **B.** Decision support system, because they provide analysis tools that support quantitative decision making — A decision support system supplies analysis tools and access to databases for quantitative decision making, such as demand forecasting. That is close, and the difference matters: it hands a person numbers rather than emulating the judgment itself.

- **C.** Intelligent system, whose stated purpose is to emulate or enhance human capabilities — **correct.** The chapter lists the intelligent system as the category that emulates or enhances human capabilities, with an AI system for analyzing bank loan applications, self-driving cars, Siri, Alexa, ChatGPT and Gemini as its sample applications.
- **D.** Business intelligence system, because they analyze Big Data to understand aspects of a business — A business intelligence system analyzes Big Data to better understand aspects of a business, with analytical processing and data visualization as its examples. It reports on the business, whereas an intelligent system stands in for a human capability.

3. A manager says a routine report is now written entirely by the assistant, with nobody reading it before it circulates. On the chapter's reasoning, what is the specific problem?

- **A.** Circulating the report costs the organization more than it saves, so the return on investment is negative — Cost is a fair question to ask about any system, and it is not the chapter's objection here. The objection stands even if the arrangement is cheap, because the problem is that nobody can vouch for what goes out.
- **B.** Nobody can say whether the output is correct or even reasonable, so the manager has made the role irrelevant — **correct.** Asking the system to do the work for you leaves no credible way of knowing whether the output is correct or even reasonable, which is the chapter's reason for calling the arrangement a knowledge replacement.
- **C.** The assistant works faster than the manager, which means the role is genuinely no longer needed at all — Speed is the reason people reach for these tools, so it is not the flaw. The chapter's argument is that retaining control of the decision is what makes a person more valuable, not what makes them slower.
- **D.** The report should have been produced by a management information system, which is the category built for reporting — Naming the right system category does not rescue an unreviewed output, and a reporting system would face the same objection. The failure here is the missing human decision point, not the choice of software.

## Companion, not replacement

The chapter's principle is plain: GenAI should be a **knowledge companion and not a knowledge replacement**. The argument is about credibility, not effort.

Ask the system to do the work and you make yourself irrelevant: it produces an output you have no credible way of judging. Ask for assistance while keeping the decision and you make yourself more valuable. The workflow decides which, not the subscription.

## A method for deciding where AI belongs

"Use AI" is not a decision, and neither is "ban AI." Four steps, in order.

- 1 **Name the task and where it sits** — one task, not a department, and its value chain activity.

- 2 **Choose the mode** — augment, automate, or leave alone — and say which out loud.
- 3 **Place the decision point** where being wrong is expensive: the cost of an error decides who signs.
- 4 **Commit to a measure and a date:** the number that should move, and when you look.

The second step is the one people skip.

**Augment** — it drafts or summarizes and a person decides, where a starting point saves time but judgment matters.

**Automate** — it acts inside limits set in advance, for high-volume, low-stakes, reversible work like sorting messages.

**Leave it alone** — the task stays human because the stakes or sensitivity are too high, as with one named employee.

Now walk the method through one task.

#### INTERACTIVE DIAGRAM

### The four steps, one at a time

Pick a step to redraw it: what the step produces, how to tell you have done it properly, and the mistake that step exists to prevent.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Name the task** — Step one - one specific piece of work, placed in one value chain activity

Too broad to act on (Use AI in customer service) → The actual task (Draft the first reply to a routine complaint) → Where it sits (Service - after the sale, keeping the customer whole) → What this buys you (Something narrow enough to measure)

- **What the step produces** - a single task written as a verb and an object, plus the value chain activity it belongs to. Drafting a complaint reply is a task; improving customer service is a wish.
- **How to tell you did it properly** - you can say who does the task today, how often it happens, and how long one instance takes. If you cannot, you have named a department rather than a task.
- **The mistake it prevents** - buying a tool and then hunting for somewhere to put it. A proposal that opens with the technology has skipped the diagnosis, which is the same failure the previous supplement warned about for any IT recommendation.

- **Why the activity matters** - naming it keeps the project in the right department and tells you whose numbers should move. A drafting aid in service will not show up in procurement's results no matter how well it works.

### **Choose the mode** — Step two - augment, automate, or leave it alone

Augment (System drafts or ranks, a person decides) → Automate (System acts inside limits set in advance) → Leave it alone (Stakes or sensitivity too high to hand over) → The deciding question (What does one wrong answer cost?)

- **What the step produces** - one word, chosen deliberately and written down, because augment, automate and leave it alone produce three completely different designs from the same subscription.
- **Augment** fits work where a strong starting point saves real time but the judgment still matters. This is the chapter's principle in operational form: the system augments the human rather than replacing the human.
- **Automate** fits work that is high in volume, low in stakes, and reversible, such as sorting arriving messages by topic. The limits have to be written before launch, because an automated task nobody bounded will eventually meet a case its designer never imagined.
- **Leave it alone** is a real answer, not a failure of nerve. A decision about one named employee, or anything that would require sending information out of the company that should not leave it, stays with a person.

### **Place the human decision point** — Step three - where a person signs before anything becomes real

Cheap to be wrong (Reversible, small, easily corrected) → Expensive to be wrong (Money, safety, reputation, a person's record) → The checkpoint (Sits just before the irreversible act) → What the reviewer needs (Time to look, and a reason to bother)

- **What the step produces** - a named place in the workflow where a person reviews and approves, and a named person who does it. A checkpoint with no owner is a diagram, not a control.
- **How to place it** - sort the possible errors by what they cost rather than by how likely they are. Difficulty is not the test; a simple task with an expensive failure still needs a signature.
- **Make the review real** - give the reviewer enough time and enough information to disagree. A person clicking approve on forty drafts an hour is not a control, and everyone in the process knows it.
- **The mistake it prevents** - the drift the chapter describes, where people ask the system to do everything for them. Keeping the decision is exactly what keeps the person more valuable than the tool.

### **Set the measure and the date** — Step four - the number that should move, and when you will look

The number (Average time to first reply) → The counter-number (Drafts rewritten from scratch) → The date (Fixed before launch, not after) → The decision waiting (Keep it, change it, or stop it)

- **What the step produces** - one sentence naming a number and a date, so the initiative can be judged rather than merely defended. This is the same commitment the previous supplement required of any recommendation put to a board.
- **Always pair the numbers** - a speed measure alone rewards fast bad work. Pairing time to first reply with the share of drafts rewritten from scratch stops anyone from claiming a win that customers would not recognize.
- **Fix the date in advance** - a review date chosen after results arrive is not a review, because whoever proposed the initiative gets to pick the week that flatters it.
- **Name the possible outcomes now** - keep, change, or stop, with the third one genuinely available. An initiative that cannot be stopped was never really being tested.

Most workplace disagreements about AI are really disagreements about mode.

**SORT**

### Augment, automate, or keep it human?

Place each task from the hypothetical practice grocer in the mode it belongs to, judging by what one wrong answer would cost.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Augment** — the system drafts, ranks or summarizes and a person decides before anything is acted on

- Writing the first version of a reply to a routine emailed complaint, which an agent edits before sending — A blank screen is the expensive part and a wrong promise is the expensive risk, so the system supplies the start and the agent keeps the send.
- Condensing a long supplier contract into a one-page brief a buyer reads before negotiating — The summary saves hours of reading, but a missed clause changes what the buyer agrees to, so a person reads the passages that matter before the meeting.
- Turning a week of register data into a plain-language summary a store manager then interprets — Describing what moved is drafting work; deciding what to do about it is judgment, and the manager keeps that half of the task.
- Suggesting three headline options for the weekly circular, from which the marketing lead picks one — Options are cheap and choosing is the decision, so producing candidates for a named person to choose among is the textbook shape of augmentation.

**Automate** — the system acts on its own inside limits set in advance - high volume, low stakes, reversible

- Sorting arriving customer messages into topic folders so the right team opens them first — It is high volume, low stakes and reversible - a message filed under the wrong topic is refiled in seconds, which is exactly when acting without a reviewer is worth it.
- Sending an acknowledgement that a message arrived, with a reference number and an expected reply time — The content is fixed and factual, nothing is being promised about the outcome, and the cost of sending one to a duplicate message is close to nothing.
- Renaming and filing arriving supplier invoices by vendor and date into a shared folder — This is repetitive clerical work with an obvious right answer and an easy correction, so a defined rule can run it without anyone reviewing each file.

**Keep human-only** — the stakes or the sensitivity are too high, so the task stays entirely with a person

- Deciding which of two employees is promoted when their managers disagree — The stakes for the person are high, the decision is hard to reverse, and it rests on judgment about people that a manager has to be able to explain and own.
- Writing the letter that tells a long-standing produce supplier the chain is ending the relationship — It is a consequential business decision in a relationship built over years, and the person accountable for ending it should be the person who words it.
- Reviewing employees' medical leave notes to propose next week's staffing — The information is sensitive enough that sending it to an outside system is itself the harm, which makes this an information privacy question before it is an efficiency question.

## Output quality follows input quality

**Garbage in, garbage out** holds twice: quality depends on training data you do not control, and on the prompt you do. Compare "Write an email to a customer" with "Write a short, polite email to a customer who received the wrong item, offering a replacement and a discount."

Three things moved between them.

- **Specificity** — the wrong item, and the remedy.
- **Tone and format** — short and polite, not left to chance.
- **Audience** — a customer let down once already.

A prompt for a recurring task is worth writing once, reviewing, and reusing.

## What goes wrong

Four failures cause most of the damage, and each has a control belonging inside the workflow, not in a policy nobody opens.

**EXPLORE****Four ways this goes wrong**

Open each failure and read all four facets before moving on - the workplace illustrations belong to the hypothetical practice grocer, and the control matters as much as the risk.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**Hallucination** · Fluent, confident, and wrong

- **What the failure is:** GenAI can make mistakes called hallucinations. The chapter's example is asking a system about the only survivor of the sinking of the Titanic: early models would gladly describe that survivor, which is a hallucination, because there were more than 700 survivors.
- **What it looks like at work:** A drafted reply cites a return window the chain does not offer, or a contract summary describes a delivery term that appears nowhere in the contract. Nothing in the writing signals which sentences were invented, because the invented ones read exactly like the rest.
- **What it costs when nobody checks:** The wrong return window reaches a few hundred customers, and honoring it costs money the chain never agreed to spend. Worse, staff begin treating every output as checked, because it has been right often enough to earn trust it did not earn.
- **The control that belongs in the workflow:** Someone who already knows the answer verifies the checkable parts - prices, dates, policies, names - before anything leaves the building. The working rule is the chapter's: if you have no way to judge whether an output is reasonable, you are not in a position to use it.

**Inherited bias** · The training data shows through

- **What the failure is:** GenAI output can be biased if the datasets used to train the model are biased. The chapter is careful here: that does not mean a human would not be biased as well, but quality system outputs rely on quality system inputs, and the inputs came from outside your organization.
- **What it looks like at work:** Draft job descriptions consistently favor one kind of background, or suggested replies are noticeably warmer to some customers than others, because the patterns in the training material carried through into the output.
- **What it costs when nobody checks:** A slanted pattern gets applied at machine scale and with an appearance of neutrality, which makes it harder to spot and harder to argue with than one manager's slanted judgment would have been.
- **The control that belongs in the workflow:** Review outputs in batches rather than one at a time, because a pattern is invisible in a single sample and obvious across fifty. Keep the tasks where bias is most damaging, such as hiring and discipline, in the human-only mode.

**Sensitive data leaving** · An outside system is still outside

- **What the failure is:** Most of these systems are services run by another company. Pasting a customer list, an employee record, or an unreleased contract into a prompt hands that information to an outside party, which is an information privacy question of exactly the kind Objective 1.4 raises.
- **What it looks like at work:** An agent pastes an entire complaint thread, including the customer's address and card details, so the draft reply will have the full context. Nothing about the interface suggests that anything unusual just happened.
- **What it costs when nobody checks:** Information a customer revealed for one purpose is now held somewhere they never agreed to, which is the accessibility and property side of the same privacy problem, and no efficiency gain compensates for it.
- **The control that belongs in the workflow:** Write down what may go into a prompt and what must be removed first, then design the workflow so the redacted version is the easy path. A rule that depends on each person remembering it under time pressure is not a control.

**Accountability** · The decision is still yours

- **What the failure is:** Responsibility does not move to the vendor when the draft comes from a system. Whoever sends, approves, or acts on an output owns the consequence, and owes an explanation of it to a customer, a colleague, or a regulator.
- **What it looks like at work:** A refund is issued on terms nobody at the chain authorized, and when the finance manager asks who approved it, the answer offered is that the system suggested it and it looked right at the time.
- **What it costs when nobody checks:** The organization cannot say who decided, so it cannot correct the process that produced the decision. Errors repeat because there is nobody whose job it was to catch them.
- **The control that belongs in the workflow:** Every automated or augmented step names an accountable person, and that name is recorded with the output. This is what makes keeping the decision valuable rather than merely cautious - the person who can explain the decision is the person worth employing.

The third deserves a note: handing a customer list to an outside system is an information privacy question of the kind Objective 1.4 raises.

**Accountability does not transfer.** A person owns the decision, so “the system suggested it” is not an explanation a regulator accepts. If you cannot judge an output, you cannot defend what you built on it.

## The method on a hypothetical situation

Return to the practice grocer, a hypothetical chain facing a warehouse club. Four conditions exist only here.

- Customer service answers hundreds of emailed complaints a week, each a quarter hour from blank.
- Most fall into four familiar categories, so the same paragraphs are rewritten daily.
- A few report a food-safety concern and must reach a store manager the same day.
- Nobody has written down what a good reply looks like.

Now run the four steps.

- 1 **The task** is drafting a routine complaint reply, in the service activity.
- 2 **The mode is augment:** editing beats a blank screen, and a wrong refund promise costs a customer.
- 3 **The decision point** sits before sending, and a rule routes anything mentioning injury to a manager.
- 4 **The measure** is time to first reply plus the share of drafts fully rewritten, on a fixed date.

Now build the same workflow yourself.

#### DECIDE

### Redesigning one workflow at the practice grocer

Work the four decisions in this hypothetical situation; after each one, compare every outcome before continuing.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

This is a hypothetical practice case. The regional grocer's customer service team answers a few hundred emailed complaints a week, each reply starts from a blank screen, most messages fall into four familiar categories, and a small number report a food-safety concern or an injury. Leadership has bought a GenAI subscription and wants to know what to do with it. You are the analyst in the room.

**Decision 1.** First decision: what exactly is the task? Everyone in the room has a different sentence for it, and the sentence you choose determines everything downstream.

- Roll the assistant out to the whole customer service team and let each agent find their own uses for it — This names no task, so there is nothing to design and nothing to measure. Ten agents will invent ten different practices, the sensitive messages will be handled inconsistently, and in six months nobody will be able to say whether it helped.
- Drafting the first version of a reply to a routine complaint, a task in the service activity of the value chain — **the stronger call**. Right. It is one task, stated as a verb and an object, placed in a value chain activity. You can already say who does it, how often it happens, and how long one instance takes, which is exactly what the later steps need.

- Improving customer satisfaction, since that is the outcome leadership actually cares about — That is the outcome you hope for, not the task you would change. Nothing about it tells you what a person would stop doing, so it cannot be assigned a mode, a checkpoint, or an honest measure.

**Decision 2.** Second decision: the mode. Drafting a reply is repetitive, the four familiar categories cover most messages, and a wrong promise about a refund or a return costs a customer relationship.

- Augment it - the system drafts, the agent edits and sends — **the stronger call.** Correct, and for the stated reason: the drafting is the slow part and the promise is the risky part, so the system takes the first and the agent keeps the second. This is the chapter's principle made operational - the system augments the human rather than replacing the human.
- Automate it - the system writes the reply and sends it, since the categories are familiar and the volume is high — Volume and familiarity argue for speed, and the cost of an error argues louder. An unreviewed reply can promise a refund the chain never offered, and the customer has already read it before anyone at the chain has. This is the arrangement the chapter calls a knowledge replacement.
- Leave it alone - complaint replies touch customers, so no system should be near them — Leave it alone is a legitimate mode, and it is the wrong one here. It is reserved for stakes or sensitivity high enough that no review makes the risk acceptable. A routine complaint reply that a person reads before sending does not meet that bar, and refusing the help costs the team real hours for no reduction in risk.

**Decision 3.** Third decision: the human decision point. Most messages are routine, but a small number report a food-safety concern or an injury, and those have to reach a store manager the same day.

- One rule for everything: an agent reads and approves every draft before it is sent — The universal review is the right instinct and an incomplete design. It treats a spoiled-product report exactly like a late-delivery complaint, so the message that needed a manager today sits in the same queue as everything else and is answered politely rather than escalated.
- Approve in batches at the end of each day, so agents can review many drafts quickly — This turns the checkpoint into a formality. A reviewer moving through forty drafts to finish the day is not really deciding, and the urgent messages have now waited a full day on top of it - the review exists on the diagram but not in practice.
- An agent approves every draft, and illness or injury routes to a store manager — **the stronger call.** Right on both counts. The agent approves each draft before it is sent, so the checkpoint sits immediately before the irreversible act, and anything mentioning illness or injury reaches a store manager undrafted - the cases where being wrong is most expensive are pulled out of the automated path entirely rather than being handled a little more carefully inside it.

**Decision 4.** Fourth decision: leadership asks how the chain will know whether this worked, and by when.

- Ask the agents in a few months whether the assistant has made their work easier — Their experience is worth hearing and it is not a measure. It arrives at a date nobody set, it says nothing about what customers received, and a team that knows leadership funded the tool is being asked to judge the decision its managers already made.
- Track average time to first reply, and the share of drafts agents rewrote from scratch, both read at a review date fixed now — **the stronger call**. That is the full shape. One number should improve, the second guards against the first improving for the wrong reason, and the date is set before anyone has seen results. The honest possible outcomes - keep it, change it, stop it - are all still available at that review.
- Count how many drafts the system produced each week, which shows how much work it is doing — This measures activity rather than results. Output volume rises simply because the tool is switched on, and it would keep rising even if every draft were rewritten from scratch and every customer were annoyed by the reply that arrived.

## From knowledge workers to learning workers

The chapter ends forward-looking: efficiencies should let knowledge workers be more creative, and the line between knowledge and manual work blurs as every organization runs on systems.

Almost every worker is now a knowledge worker, but workers of the future must become **learning workers**, because not the knowledge itself but the knowledge of how to learn will matter most.

### CHECK YOURSELF

### Method, prompts, and failures

Three questions on deciding the mode, writing the prompt, and owning what the system produces.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** The practice grocer wants the system to file arriving supplier invoices by vendor and date into a shared folder, with no person reviewing each one. Which mode is that, and is it defensible?

- **A.** Augment, and defensible, because a person will eventually open the folder and notice anything filed wrongly — Augment means a person decides on each item before it is acted on, and nobody is deciding here. A later chance to notice a mistake is not the same as a review standing between the system and the action.
- **B.** Automate, and defensible, because the work is high in volume, low in stakes, and a misfiled invoice is corrected in seconds — **correct**. Automate is the right mode exactly when volume is high, stakes are low, and errors are cheap to reverse, and filing a document by vendor and date meets all three of those conditions.
- **C.** Automate, and indefensible, because a task without a human checkpoint has replaced the human rather than augmenting them — This applies the companion principle too widely. The chapter's concern is with handing over decisions you cannot vouch for, not with clerical steps whose only possible error is an easily corrected filing mistake.

- **D.** Keep human-only, and defensible, because invoices are financial records and financial records need a signature — Financial records do need controls, and the payment decision is where the signature belongs. Naming the file is not the payment decision, so treating the two the same wastes the reviewer's attention on the harmless half.

**2.** The chapter contrasts a weak prompt - Write an email to a customer - with a stronger one - Write a short, polite email to a customer who received the wrong item, offering a replacement and a discount. What changed between them?

- **A.** The prompt became longer, and longer prompts reliably produce better output than shorter ones — Length happens to increase here, and it is a side effect rather than the cause. Padding a vague request with more words leaves it just as vague, which is why the exercise names specific improvements instead of a word count.
- **B.** The prompt named the system it was written for, so the request was matched to that system's strengths — Choosing a suitable system is a real decision, since these systems are tuned differently, but naming a product inside the prompt is not one of the improvements the exercise describes. Nothing in the strong prompt mentions any product.
- **C.** The prompt became more specific about the task, clearer about tone and format, and better targeted — **correct.** Those are the three improvements the exercise names, and you can see each one in the strong prompt: the wrong item and the offer, short and polite, and a customer who has already been let down.
- **D.** The prompt corrected the training data, which is the underlying cause of a weak response — A prompt cannot reach the training data, which is the half of garbage in, garbage out that you do not control. That is precisely why the exercise concentrates on the half you do control.

**3.** An agent sends a drafted reply that promises a refund on terms the chain does not offer. Asked about it, the agent says the system produced the wording and it looked reasonable. What does the chapter's reasoning say about this?

- **A.** The vendor is answerable for the error, because the vendor's system produced the incorrect wording — Buying a service does not transfer responsibility for how you use it, and the customer received the promise from the chain rather than from the vendor. The chain has to honor or explain what it sent.
- **B.** The agent still owns the decision, because sending an output you cannot vouch for is the choice that caused the harm — **correct.** The person who sends an output owns the decision built on it, which is the chapter's reason for saying that retaining control of decision making is what makes a person more valuable.
- **C.** The error proves the task was in the wrong mode, so the reply should have been sent without any review — Removing the review would make this failure more frequent rather than less, since nothing would stand between an invented refund term and the customer. The mode was right; the checkpoint was not treated seriously.

- **D.** The error is unavoidable, because these systems make mistakes called hallucinations and no workflow can reduce them — Hallucinations are real and a workflow can absolutely limit their reach, which is the whole point of verifying checkable claims such as prices, dates and policies before anything is sent.

## The concepts as code

Everything so far has been described in words, because that is how the material is taught and examined. Words are also where these ideas go to become vague. This supplement writes four of them as functions instead: a function either returns the right answer for a case you did not think of, or it does not.

**How this supplement relates to the chapter.** Nothing here is new material. Each exercise is a definition you have already read, restated as a contract with test cases attached: the network effect, the difference between data and information, the categories of organizational information system, and Mason's questions about who may see what. The code is the check on whether the definition was actually understood, not an addition to it.

### Why a definition is worth writing as a function

Prose lets you agree with a definition without ever committing to what it means at the edges. Code does not offer that comfort, and the edges are where the understanding lives:

**A function has to handle the empty case** — a report over no rows, a network with one member — and deciding what those return forces you to say what the concept actually claims when there is nothing to claim about.

**A function has to resolve conflicts** — when a need looks like two categories at once, the order of your checks is your real definition, whatever the prose said.

**A function fails visibly** — a misunderstanding that survives an essay shows up here as a red test with the value you actually returned printed next to the one you should have.

If you can write it, you understood it. If you can only describe it, you may only have recognised it.

### How these run

Each exercise gives you a stub, a stated contract and a set of tests. Write the body, run the tests, and read the failures — the value your code returned is printed beside the value expected, which is usually enough to see the mistake without a debugger.

- Your code runs in this page, with no network and nothing installed.
- **Run the tests**, or press Ctrl-Enter while typing.

- Tab indents rather than moving focus, so the editor behaves.
- **Reset the stub** puts the starting code back.
- A loop that never ends is stopped after two seconds.
- **Show a solution** reveals one that passes, when you want it.

There is more than one correct implementation of every one of these. The tests judge behaviour, not style, so a version that reads better than the one shown is a better answer, not a different one.

Every record in the test data below is **hypothetical** — invented so the functions have something to run against. No real customer, company or network is described.

#### WRITE THE CODE

### Four definitions, four functions

Write each function so every test passes; a failing test prints what your code returned next to what was expected.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

**1.** A network's value comes from the connections it makes possible, not from the count of members. Implement **connections(n)**: how many distinct two-way links exist among n members?

**Signature:** `connections(n) -> number`

```
function connections(n) {
  if (n < 2) return 0;
  return (n * (n - 1)) / 2;
}
```

This is the network effect in one line. Members grow linearly and links grow with the square, which is why a platform that is slightly ahead on users can be far ahead on value, and why the second-place platform struggles to catch up.

**2.** Turn raw records into a management report. Implement **report(rows)**: given transaction rows, return an object mapping each region to its total revenue, including only completed sales.

**Signature:** `report(rows) -> { region: totalRevenue }`

```
function report(rows) {
  return rows
    .filter(function (r) { return r.status === 'complete'; })
    .reduce(function (acc, r) {
      acc[r.region] = (acc[r.region] || 0) + r.units * r.unitPrice;
      return acc;
    }, {});
}
```

You just wrote the difference between data and information. The rows are data: true, but unusable for a decision. The returned object is information, because it answers a question a manager actually asked. Nothing was added except structure and a rule about what counts.

**3.** Route a need to the right class of system. Implement **classify(need)** returning 'TPS', 'MIS', 'DSS' or 'ESS'.

**Signature:** `classify(need) -> 'TPS' | 'MIS' | 'DSS' | 'ESS'`

```
function classify(need) {
  if (need.recordsEvent) return 'TPS';
  if (need.comparesScenarios) return 'DSS';
  if (need.scope === 'organization') return 'ESS';
  return 'MIS';
}
```

The last test is the interesting one. A system that must record events is a transaction processing system no matter how senior the person asking, because everything else is built on the record it keeps. Getting the order of the checks right is the same as getting the reasoning right.

**4.** Mason's **property** and **privacy** questions become a function. Implement **redact(record, purpose)**: return a copy holding only the fields the stated purpose actually needs.

**Signature:** `redact(record, purpose) -> object`

```
function redact(record, purpose) {
  var allowed = {
    shipping: ['name', 'address'],
    analytics: ['region', 'totalSpend'],
    support: ['name', 'orderId']
  }[purpose] || [];
  var out = {};
  allowed.forEach(function (f) {
    if (f in record) out[f] = record[f];
  });
  return out;
}
```

The allow-list is the whole point. Written as a deny-list, this function is correct today and wrong the moment a new column lands in the table, because nobody remembers to add it. That is how most real privacy failures happen - not by decision, but by default.

## What the four have in common

Read back over what you wrote and notice that none of the four functions needed a framework, a database or a network. Each is a small pure function over plain data, and each encodes a decision that

a person would otherwise make inconsistently:

- 1 **connections** turns an intuition about popularity into arithmetic, and the arithmetic is what explains why platform markets tip toward a single winner rather than settling into a tie.
- 2 **report** is the whole data-to-information distinction: a filter that says what counts and an accumulation that says what question is being answered.
- 3 **classify** shows that a taxonomy is only as good as the order in which its tests are applied, which is why two people can agree on the definitions and still sort the same case differently.
- 4 **redact** makes a privacy principle enforceable, and shows that the enforceable version is an allow-list, because a deny-list is a promise that decays every time the schema changes.

#### THE TRANSFERABLE IDEA

Business rules are code whether or not anyone writes them down. When they stay in prose they are applied differently by different people on different days; when they are written as a function with tests they are applied the same way every time and can be argued about precisely. A great deal of what an information system does for an organization is exactly that conversion.

### Where this shows up outside a course

The four functions are small, but the shape is the one that recurs wherever software meets a business rule. Somebody states a policy in a sentence, somebody else has to make a machine apply it a million times, and the gap between those two is where most defects and most arguments live.

**The specification is rarely wrong; it is usually incomplete** — the sentence covers the cases everyone pictured, and the function has to answer the ones nobody did, which is why the empty and tied inputs above were the hardest part of each exercise.

**The tests are where the disagreement surfaces** — two people can nod at the same definition and encode different behaviour, and a failing case is a much shorter conversation than two competing paragraphs.

**The default matters more than the rule** — every one of these functions has a path taken when nothing matches, and what happens there decides how the system behaves on the day something unexpected arrives.

None of that is specific to information systems. It is simply what it looks like when a policy stops being a description and starts being enforced, and this chapter's subject is an organization full of exactly

that.

### CHECK YOURSELF

### What the code made explicit

These ask about the reasoning the exercises forced, not about JavaScript syntax.

**Lesson and answer summary** — the interactive version of this activity loads when scripting is available.

1. Your `classify` function checks `recordsEvent` before it checks anything else, so a need that both records events and compares scenarios comes back as 'TPS'. What does that ordering actually assert?

- **A.** That executives matter less than operational staff when systems are prioritised — Priority between people is not what the branch expresses. The same ordering applies whoever raised the need, which is precisely why it is a definition rather than a preference.
- **B.** That a system which must capture events is first of all a record of them, because everything reported or modelled later is built on that record — **correct.** Right. The order of the checks is the claim: if events must be captured, that obligation is foundational, because a summary or a model can only ever be as good as the record it reads from.
- **C.** That transaction processing systems are more common in organizations than any other category, so the safe assumption for a system nobody has classified yet is that it is one of them — Frequency is not what the function measures, and it never sees a population of systems. Even if transaction systems were rare the ordering would be unchanged.
- **D.** That comparing scenarios is not a real requirement unless it is the only requirement stated — The scenario requirement is real and is handled on the very next line. It is subordinate here, not dismissed, which is a different claim entirely.

2. In report, the filter for completed rows runs before the totals are accumulated. A colleague suggests totalling everything first and subtracting refunds afterwards. What is the risk?

- **A.** Nothing at all, because subtracting the refunded rows afterwards reaches exactly the same total by a slightly longer route and the two queries can be used interchangeably — The two routes agree only while completed and refunded are the only statuses that exist. As soon as a pending or disputed row appears, the filter leaves it out and the subtraction quietly keeps it in.
- **B.** The totals would be slower to compute on a large number of rows — Performance is not the objection, and on realistic volumes the difference would not be noticeable. The concern is which rows end up in the number.
- **C.** Any row whose status is neither complete nor refunded is silently included in the total, because subtracting handles only the case you remembered — **correct.** Right. Filtering states what belongs in the answer; subtracting states what to remove from it. The second only handles cases you anticipated, so a new status defaults into the total instead of out of it.

- **D.** Refunded rows would be counted twice rather than removed — Double counting is not what happens; each row is read once either way. The problem is the rows that are neither completed nor refunded.

3. `connections(100)` returns 4,950 while `connections(10)` returns 45. A competitor has 60 members to your 100. What does the arithmetic support saying?

- **A.** That your network is worth about 1.7 times theirs, in direct proportion to the number of members each of the two has signed up so far — That is the linear reading, and it is the one the arithmetic contradicts: value here is not proportional to members, which is the entire reason the function is worth writing.
- **B.** That your network offers roughly 2.8 times as many possible connections, so a lead in members is a larger lead in what the network can do — **correct.** Right. 4,950 against 1,770 is close to 2.8 times, so a 1.7-fold lead in members is a 2.8-fold lead in possible connections. That gap is what makes catching up hard.
- **C.** That their network cannot grow, because yours has already taken the available members — Nothing in the calculation says membership is fixed or exclusive, and both networks can grow at once. The function counts links, not market share.
- **D.** That connections are the only thing that determines what a network is worth — The function counts possible connections, which is one input to value rather than the whole of it. Whether those connections are used at all is a separate question.

4. `redact` builds its result from an allow-list. Six months later a `dateOfBirth` column is added to the customer table. What happens to the analytics output?

- **A.** It gains the new field, because analytics receives whatever the record contains — That is the deny-list behaviour, and it is exactly the failure the allow-list prevents. Under a deny-list the new column would flow through untouched.
- **B.** It is unchanged, because a field has to be named before it can be returned — **correct.** Right. Nothing is returned unless it appears in the allow-list, so a new column is excluded by default and stays excluded until somebody decides otherwise. Safe by default is the whole design.
- **C.** It throws an error, because the function encounters a field it does not recognise — The function never inspects fields it was not asked for, so an unfamiliar one cannot raise an error. It is simply not read.
- **D.** It gains the field only if the record was created after the column was added — When the record was created makes no difference; the function reads the same allow-list every time it runs, regardless of the record's age.

## Chapter vocabulary

Every term the chapter defines, in one place, written so it makes sense on its own. Search by word, or filter by the objective the term belongs to. Use this while you read rather than after — most of the confusion in this material comes from two terms that sound alike meaning genuinely different things.

### 3D printing

1.4

A way of making a solid physical object by adding thin layers of material one on top of another from a digital model, instead of cutting the shape out of a block. Because it adds material rather than carving it away, it wastes far less and can reproduce almost any design cheaply.

**For example:** Printing a replacement plastic clip for a broken pair of headphones instead of buying the whole headset again.

### APIs (application programming interfaces)

1.1

Standard connection points that let one company's software use another company's data or features over common web protocols, without needing intimate knowledge of how that software works inside. The chapter compares an API to a power socket: you plug in and get electricity without knowing how the utility produces or delivers it.

**For example:** Lyft shows you a live map inside its own app by plugging into Google Maps rather than building a mapping system from scratch.

### apps

1.1

Software programs written to perform one particular, well-defined function, usually on a phone or tablet. Companies now build them because customers reach for a phone far more often than a laptop.

**For example:** Opening Venmo for fifteen seconds just to send a friend your half of the pizza, then closing it.

### artificial intelligence

1.1

Using information technology to imitate the things human intelligence does, such as recognizing patterns, understanding language, and making judgments. It is what lets software make sense of enormous streams of data no person could read.

**For example:** Asking ChatGPT or Siri a question in plain English and getting an answer written the way a person would write it.

## Big Data

1.1

Datasets so enormous and so tangled that ordinary tools struggle with them. Three things make them hard: volume, meaning sheer size; variety, meaning many different types of data mixed together; and velocity, meaning the data arrive and must be analyzed faster and faster.

**For example:** Every skip, replay, and playlist add from hundreds of millions of Spotify listeners piling up every second of the day.

## BYOD

1.1

Short for bring your own device: the workplace trend of employees using their personal phones and laptops for work instead of equipment the company issues. Workers like it and it can raise productivity, but it makes security, compliance, and tech support much harder for the employer.

**For example:** Reading your work schedule and answering your manager's messages on the same phone you use for TikTok.

## climate change

1.1

Large-scale, long-term shifts in temperatures and weather patterns, both regionally and worldwide. Its effects — rising seas, more severe storms, changed weather — create problems for individuals, businesses, and whole societies.

**For example:** A coastal town rebuilding after a severe hurricane far more often than it did a generation ago.

## competitive advantage

1.3

Whatever lets a company beat the rival firms selling to the same customers — doing something better, faster, cheaper, or in a way nobody else can copy yet. Information systems can create it, but the chapter warns it is often fleeting because competitors eventually do the same thing.

**For example:** FedEx delivers a quarter of the packages it handles each day within one business day, helped by sorting hubs that decide in a few hundred milliseconds where to divert each scanned package.

## computer ethics

1.4

The standards of right and wrong conduct that apply specifically to how people use computers and information systems. It covers questions such as who may read whose files, who owns data about you, and what counts as stealing when nothing physical is taken.

**For example:** Deciding not to log in with a classmate's credentials to reach a research database your account does not cover.

## computer fluency

1.1

Being able to teach yourself new technologies as they appear and judge what they will do to your work and your life. It goes past knowing today's tools, because the tools keep changing.

**For example:** Picking up a new scheduling app your employer rolls out in one afternoon on your own, and noticing which parts of your shift it will change.

## computer literacy

1.1

Knowing how to operate a computer and use common applications. The chapter says it can be the difference between being employed and being unemployed, but that it is no longer enough on its own.

**For example:** Being able to build a spreadsheet, attach it to an email, and find the file again next week.

## consumerization of IT

1.1

The pattern where new technologies show up first in products sold to ordinary shoppers and only later get adopted inside companies. It forces businesses to keep watching the consumer market to see what will change how they operate.

**For example:** Tablets were built and sold as consumer gadgets first and only later became standard equipment in warehouses, showrooms, airplane cockpits, and hospitals.

## data

1.2

Raw symbols such as characters and numbers, with no meaning attached to them. They stay nearly worthless until something processes them and gives them a context.

**For example:** The string 465889727 sitting by itself tells you nothing at all — you cannot say what it is.

## data privacy statements

1.4

The posted notice on a website that says what data are gathered, what they are used for, who gets access, whether giving them is required or voluntary, and how confidentiality will be handled. It tells you what a company intends to do; it does not by itself protect you.

**For example:** The privacy policy link at the bottom of a checkout page that almost nobody opens before paying.

## data quality

1.2

How trustworthy data are for making a decision, judged on five things: completeness, accuracy, timeliness, validity, and consistency. Bad inputs produce bad decisions no matter how good the system is.

**For example:** A mailing list holding an old address and a duplicate record sends your acceptance letter to an apartment you moved out of last year.

## demographic changes

1.1

Shifts in the makeup of a population — how old people are, how many babies are born, and who moves where. Developed countries are aging fast while regions such as Africa are growing quickly, which changes where workers and customers will be.

**For example:** Your hometown high school closing a wing because far fewer children are being born there than in your parents' day.

### digital density

1.1

The amount of connected data produced by each unit of activity. As it climbs, ordinary actions throw off more data that can be linked together, which is what makes new services and new business models possible.

**For example:** A run used to produce nothing; now a single run produces pace, route, heart rate, elevation, and a sleep score the moment your watch syncs.

### digital divide

1.1

The gap between people who can get to information systems and people who cannot, where the first group holds a large advantage over the second. It falls hardest on rural communities, older people, people with disabilities, minorities, and developing countries that lack affordable internet access.

**For example:** A student who can only get online at the public library falls behind a classmate with home broadband and a laptop.

### globalization

1.1

The knitting together of economies all over the world, made possible by innovation and technological progress. You see it in the way goods, money, information, and workers now move across borders.

**For example:** Your phone was designed in California, assembled in China, and supported by a call center on a third continent.

### hardware

1.2

The physical equipment of an information system — the machines and parts you could actually drop. Today that includes not just computers and printers but sensors, cameras, and other input and output devices.

**For example:** The laptop in front of you, plus its screen and keyboard.

### healthcare IS

1.1

Information systems built to support the work of medicine, from diagnosing and treating patients to analyzing patient and disease data to running the business side of a clinic or hospital.

**For example:** The patient portal where your doctor's office posts your lab results and lets you book the follow-up visit.

### home automation

1.1

Monitoring and controlling a house's lighting, heating, and appliances from a distance instead of by hand at a switch or a dial. The chapter uses this as another name for smart home technology.

**For example:** Turning the porch light on from your phone while you are still on the bus home.

## Industrial Internet of Things (IIoT) 1.1

The use of internet-connected devices inside manufacturing, which brings the information technology side and the factory-floor operations side together. It can improve efficiency, product quality, and flexibility, letting companies mass-produce customized products and watch their supply chains closely.

**For example:** Sensors on a plant's machines reporting wear so a part gets swapped out before the whole line stops.

## information 1.2

Data that have been formatted, organized, or processed until they mean something. Information represents a piece of reality and answers questions about who, what, where, and when.

**For example:** The same digits shown as 465-88-9727 in a field labeled SSN inside John Doe's record — now the number tells you something.

## information privacy 1.4

The question of how much personal detail you should have to hand over to others — an employer, a store, a website — through your job or through transactions such as online shopping, and what they may do with it afterward. It gets harder as organizations combine scattered pieces into an increasingly complete picture of you.

**For example:** A shopping site greeting you by name and showing you the exact jacket you looked at on a different site yesterday.

## information system (IS) 1.2

The combination of people and information technology that creates, collects, processes, stores, and distributes useful data. The people are part of the definition, not an afterthought, because the technology exists to serve someone.

**For example:** Your school's registration system: the staff who run it, the software, the network, and the class-schedule data it hands back to you.

## information technology (IT) 1.2

The technology portion of an information system: hardware, software, and telecommunications networks. It leaves out the people, although the chapter notes the gap between the terms IS and IT has been shrinking and many people now use them interchangeably.

**For example:** The servers, the app, and the campus Wi-Fi that let you check a grade — but not the registrar who maintains the records.

## intellectual property (IP) 1.4

Creations of the mind that have commercial value, such as songs, software, writing, photographs, and product designs. Digital copying makes it easy to take and share these without permission, attribution, or payment to whoever made them.

**For example:** Copying a friend's entire music library with no loss in quality and passing it along to other friends.

## Internet of Things (IoT)

1.1

A network of everyday physical objects that can share data over the internet on their own. Anything that generates or uses data can be connected, monitored, or controlled remotely.

**For example:** A car tire with a pressure sensor that reports itself as low, and a home meter that reports energy use back to the utility.

## internetworking

1.2

Linking host computers and their separate networks together so they form one much larger network. The internet itself is the result of doing this on a global scale.

**For example:** Your dorm network joins the campus network, which joins your provider's network, which is how your message reaches a friend overseas.

## knowledge

1.2

The ability to understand information, form an opinion about it, and make a decision or prediction from it. It also covers the rules and guidelines you apply to organize or manipulate data for a particular task.

**For example:** Knowing that one Social Security number identifies exactly one person, so two records sharing one must be an error.

## knowledge society

1.1

A society in which what people know counts for as much as land, labor, or money once did, which makes education the thing that determines how far people can go. Peter Drucker predicted it would emerge as knowledge workers grew in number and importance.

**For example:** A graduate being considered for a job a classmate without a degree cannot even apply for, in an economy where median earnings for workers 25 and over run US\$67,256 with a bachelor's degree against US\$39,428 with only a high school diploma.

## knowledge worker

1.1

A professional, usually well educated, whose core job is creating, changing, or pulling together knowledge rather than making or moving physical things. Peter Drucker coined the term in 1959 and predicted these workers would be paid better and hold more bargaining power than earlier agricultural and industrial workers.

**For example:** A financial analyst who spends the day reading reports and turning them into one investment recommendation.

## management information systems

1.2

Systems that take the record of daily transactions and turn it into detailed reports that help someone manage a firm or one part of it. The same phrase is also used as a name for the academic field that studies information systems.

**For example:** The weekly report telling a store manager which sizes sold out so she knows what to reorder.

### micro-moments

1.1

The short bursts when a person reaches almost instinctively for a mobile device to get one thing done — to buy something, know something, do something, or go somewhere. Customer contact now happens in these bursts rather than in long sessions at a desk.

**For example:** Pulling out your phone in the middle of an argument to settle who won a game in 2019.

### network effect

1.1

The idea that a network, or a tool built on one, becomes more valuable as more people use it. It explains why the leading platform in a category is so hard to unseat.

**For example:** A messaging app is useless when none of your friends are on it and nearly impossible to quit once all of them are.

### office automation systems

1.2

Software that supports the routine day-to-day work of individuals and small groups, sometimes called personal productivity software. Word processors, spreadsheets, presentation tools, and email clients all belong here.

**For example:** Writing a paper, building a budget in a spreadsheet, and emailing both to someone, all from the same suite of programs.

### opt in

1.4

Signaling that you agree to have your data collected or used further, for example by checking a box. Nothing is supposed to happen until you actively say yes.

**For example:** Ticking the box that says yes, send me deal emails, while ordering a pair of sneakers.

### opt out

1.4

Signaling that your data may not be collected or used in some further way. Here the collection is the default and the burden is on you to say no.

**For example:** Clicking unsubscribe at the bottom of a marketing email, or switching off ad personalization in your account settings.

### outsourcing

1.1

Handing a business process or task — accounting, manufacturing, security, customer support — to another company or another country. Firms do it to cut costs, reach skills they lack, move faster, and stay focused on what they do best.

**For example:** A U.S. hospital sending X-rays overseas to be read by radiologists while doctors at home are asleep.

## quantified self

1.1

Logging every part of your own daily life — activity, performance, what you eat, mood, heart rate — in order to improve your health and performance. Activity trackers such as the Fitbit, worn and passively used on a regular basis, are what make that logging practical.

**For example:** Wearing a Fitbit all day and checking your steps, resting heart rate, and sleep score every morning.

## resource scarcity

1.1

The problem that fossil fuels and other natural resources are limited while population growth, global trade, and consumption keep pushing demand higher. Humans already consume beyond what the planet can supply.

**For example:** The lithium inside your phone battery gets harder and more expensive to source as every new device needs some.

## robotics

1.1

Putting programmable machines to work on manual, physical tasks that a person would otherwise do by hand. Continuous input from sensors, paired with machine learning and artificial intelligence to make sense of that input, is what has driven the fastest advances here.

**For example:** Autonomous Caterpillar mining trucks hauling loads with nobody in the cab.

## sensors

1.1

Devices that detect, record, and report changes in the physical world around them. Cheap sensors paired with wireless radios are the reason so many ordinary objects can now feed data to the internet.

**For example:** A strip in a road surface that reads the temperature and triggers a lower speed limit when ice becomes likely.

## shifts in economic power

1.1

Long-term changes in which countries hold purchasing power and control over natural resources. Established economies are losing their dominant positions, which creates political struggles that have to be worked out.

**For example:** Companies that once wrote off developing countries now sell into them, because a fast-growing middle class there holds purchasing power it did not have a generation ago.

## smart home technologies

1.1

Technology that lets someone monitor and control a home's lighting, heating, or appliances from a distance. The market for it is expected to top US\$250 billion by 2029.

**For example:** A Nest thermostat that learns your schedule and lets the house cool down while nobody is in it.

## software

1.2

A program, or a set of programs, that tells the computer hardware what tasks to carry out. Without it the hardware sits there doing nothing useful.

**For example:** The operating system running your laptop and the browser you open on top of it.

## strategic

1.3

Describes a choice made with the intent of helping a company gain or sustain an advantage over its rivals. The chapter contrasts this with using technology only to run existing work more efficiently or to earn a return on the investment.

**For example:** FedEx reengineering and improving its package-handling systems about twice a year so its delivery promise stays ahead of competitors.

## sustainable development

1.1

Development that meets the needs of people now without wrecking the ability of future generations to meet their own needs. It is the standard the chapter applies to growth in a world of limited resources.

**For example:** A company signing long-term wind and solar contracts so its data centers can keep growing without burning more fossil fuel.

## systems integration

1.2

Connecting separate, often modular information systems and their data together, frequently using APIs, so business processes run better and decisions get made on fuller information. It is why few real systems fit neatly into just one category anymore.

**For example:** A store's checkout feeding the inventory system automatically, so shelves get restocked without anyone retyping numbers.

## telecommunications networks

1.2

Two or more computer systems linked together by communications equipment so they can share data and services. They are what make global collaboration, communication, and commerce possible.

**For example:** The campus Wi-Fi that lets your laptop reach the library servers and then the wider internet.

## transaction processing systems (TPS)

1.2

Systems that handle the flood of routine day-to-day business events at the operational level of an organization, such as sales and registrations. They also generate huge amounts of data that other systems later analyze.

**For example:** The grocery checkout register that scans bar codes and prints coupons on the back of your receipt.

## urbanization

1.1

The movement of people out of rural areas and into cities. Roughly 50 percent of the world's population now lives in cities, and keeping those cities livable as they grow is a serious challenge.

**For example:** Someone leaving a farming town for a city apartment because that is where the jobs are.

## wearable technologies

1.1

Clothing or accessories with electronics built into them, such as a smartwatch. Their sensors record body data like movement and heart rate as well as surrounding conditions like light, orientation, and altitude.

**For example:** An Apple Watch tracking your heart rate during a workout and buzzing when a text lands on your phone.

### PUT IT TOGETHER

## Final challenge

Thirty situations rather than thirty definitions. Each one presents a documented chapter case or a clearly hypothetical organizational situation and asks what is actually going on, which is the form the material takes once you are the person being asked. Answer all of them, then score: the result breaks out by chapter objective and supplement, so if one area is weak you will see exactly which section to reread instead of a single number that hides it.

### FINAL CHALLENGE

## Final challenge

**Answer key** — the interactive, self-scoring version loads when scripting is available.

1. **1.1** A county buries temperature sensors in a stretch of highway. The readings travel over the network and automatically lower the posted speed limit when ice is likely. Road use that generated no connected record now generates a continuous stream that other systems act on. Which idea does this best illustrate?

- **A.** Big Data, because the sensors produce a high volume of readings at high velocity from many different points — Big Data names the datasets themselves: extremely large and complex, high in volume, variety, and velocity. These readings may well pile up into Big Data, and that would be the right answer if you were asked to classify the stored result rather than the change in the road.
- **B.** Increasing **digital density**, because the road now generates connected data per unit of activity — **correct**. Digital density is the amount of connected data per unit of activity. Traveling this road used to leave no connected reading; now it feeds a speed-limit decision, which is exactly the new value-added interaction that rising digital density enables.

- **C.** The network effect, because each sensor added to the road makes the system more valuable to every driver using it — The network effect is the notion that a network, or a tool or application based on one, increases in value with the number of other users. It would fit if drivers made the service better for each other; here the readings are collected whether one car passes or a thousand.
- **D.** The consumerization of IT, because sensor technology reached ordinary consumers before it reached government agencies — Consumerization of IT is the pattern of technologies appearing in the consumer marketplace first and being pulled into organizations later, as smartphones were. That would be the answer if the county had adopted a device its residents already had at home.

2. **1.1** A kitchen and bath retailer opens operations in three countries at once. In the first, supplier contracts are hard to enforce because commercial law works differently. In the second, warehouse broadband is unreliable and few local workers have logistics training. In the third, the product photography offends local norms. Which of these is a **goeconomic** challenge as the chapter uses the term?

- **A.** The contracts that are hard to enforce, because a country's laws and regulations are part of how its economy functions — That is a governmental challenge: differences in political systems, regulatory environments, laws, standards, or individual freedoms. It would be the right answer to a question asking which obstacle comes from how a country governs commerce.
- **B.** The photography that offends local norms, because what customers are willing to accept is economic behavior — That is a cultural challenge: differences in languages, beliefs, attitudes, religions, or life focus, and different viewpoints about intellectual property. Choose it when the obstacle is what people find acceptable rather than what a country can support or permit.
- **C.** None of the three, because goeconomic challenges are about currency exchange rates and tariffs on imported goods — Currency and tariffs are genuine trade obstacles, but the chapter defines goeconomic challenges more broadly as differences in infrastructure, demographics, welfare, or workers' expertise. Reading the term as a synonym for exchange rates is the usual misstep.
- **D.** The unreliable broadband and the shortage of trained logistics workers, because that is infrastructure and expertise — **correct.** Goeconomic challenges are differences in infrastructure, demographics, welfare, or workers' expertise. Unreliable broadband is infrastructure and a shortage of trained logistics staff is expertise, so both belong here rather than under law or culture.

3. **1.1** A rural clinic wins funding for a telehealth program, but enrollment stalls. Its own survey finds that the patients who would benefit most — older residents, people with disabilities, and households on unpaved roads with no broadband — are the least able to join. How does the chapter frame what the clinic has run into?

- **A. The digital divide**, because the groups named lag national averages for internet access and computer literacy — **correct**. The digital divide is the gap between those with access to information systems and those without, and the chapter names rural communities, the elderly, people with disabilities, and minorities as the groups still behind on internet access and computer literacy. It calls this an ethical challenge because computer literacy is tied to a person's ability to compete.
- **B. Resource scarcity**, because the clinic lacks the funding it would need to reach every household in its service area — Resource scarcity in this chapter means the limited availability of fossil fuels and other natural resources, a societal megatrend rather than one organization's budget. It would fit a question about pressure on the planet's finite supply.
- **C. The knowledge society**, because the program assumes patients have the education that knowledge work depends on — The knowledge society is Drucker's prediction that possessing knowledge would matter as much as land, labor, or capital, with education as its cornerstone. That explains why access matters so much; the thing physically blocking these patients is the access itself.
- **D. Demographic changes**, because the population this clinic serves is aging faster than the country as a whole — Demographic changes are shifts in the structure of populations related to age, birth rates, and migration. Age does appear in the scenario, but the barrier described is connectivity and skills, which the chapter names separately.

4. **1.1** A neighborhood tool-lending app is of little use while membership is small because few requested tools are listed nearby. As membership grows, requests are much easier to fill, and new members say they joined because so many others already had. What is happening?

- **A. Increasing digital density**, because every loan now generates connected data the app can act on — Digital density, the amount of connected data per unit of activity, is genuinely rising here as well. It explains what each loan leaves behind, not why the member who joined last is better off than the member who joined first.
- **B. A micro-moment**, because a member instinctively picks up a mobile device to accomplish a goal — A micro-moment is that instinctive reach for a mobile device to buy something, know something, do something, or go somewhere, and this app is used in exactly those moments. It describes one moment of use, not why the service as a whole got more useful as it grew.
- **C. The network effect**, because the value of the application increases with the number of other users — **correct**. The network effect is the notion that the value of a network, or of a tool or application based on a network, increases with the number of other users. That is precisely what the members describe, and it is the mechanism behind business models such as Uber and Airbnb.
- **D. Consumerization of IT**, because a tool built for consumers is now being used by contractors for paid work — Consumerization of IT means innovations are introduced in the consumer marketplace first and then adopted by organizations. It would be the answer if the story were about a company adopting the app because its staff already used it privately.

5. **1.1** An insurance firm never issued phones to its adjusters. Staff began checking work email on their own phones, and now they open customer relationship management records on them in the field. The IT director's memo lists security and compliance worries plus the cost of supporting many different devices, and in the same paragraph lists higher productivity, better retention of good adjusters, and happier customers. What is being described?

- **A. BYOD**, which the chapter calls a major concern of managers precisely because it carries real risks and real opportunities — **correct**. Bring your own device is the use of personal devices and applications for work-related purposes. The chapter frames it exactly as this memo does — worrying for security, compliance, and support, yet capable of raising productivity, retention, and customer satisfaction — and calls managing it a major concern of business and IT managers alike.
- **B. Outsourcing**, because work the firm used to equip and control is now being done on hardware the firm does not own — Outsourcing is the moving of business processes or tasks, such as accounting, manufacturing, or security, to another company or another country. These adjusters are still employees doing their own work; only the hardware changed hands.
- **C. The digital divide**, because adjusters with newer personal phones can do more of the job than adjusters with older ones — The digital divide is the gap between people with access to information systems and those without, which the chapter discusses at the level of communities and countries. Uneven phone models inside one staff is a support problem rather than that divide.
- **D. Systems integration**, because personal phones are being connected to the firm's customer relationship management system — Systems integration is connecting separate and often modular information systems and data, using technologies such as APIs, to improve business processes and decision making. Integration work may follow from this situation, but it does not name the trend the memo is about.

6. **1.2** A warehouse scanner emits an unlabeled identifier. A screen places it in a row labeled **SKU**, beside its aisle and a low on-hand quantity. The shift lead orders more because experience says that quantity will not meet weekend demand. Which part of that sequence is **knowledge**?

- **A. The unlabeled identifier**, because it is what the system actually captured and everything else is derived from it — That is data: raw symbols such as characters and numbers, with no meaning in and of themselves and little value until processed. Asked what the scanner produced, this would be the right answer.
- **B. The labeled row showing SKU, aisle, and quantity**, because a person can now read it and understand what it refers to — That is information: data formatted and organized so that it answers who, what, where, and when. The contextual cues, meaning the labels, are what turn the raw string into something readable, but reading it is not yet deciding anything.
- **C. The scanner and the network that carried the reading**, because without them none of the rest could happen — That is the technology component of the information system, the hardware and telecommunications networks. The chapter treats those as what carries and processes data, not as a form of understanding.

- **D.** The lead's rule that the displayed quantity will not meet expected demand, because it lets her decide — **correct**. Knowledge is the ability to understand information, form opinions, and make decisions or predictions from it — a body of governing procedures, guidelines, or rules used to organize data for a task. The reorder rule lives in the lead, and without it the row on the screen would just sit there.

7. **1.2** A distributor spends heavily on servers, a database, a network, and licences. After implementation nobody has been trained, no report reaches a decision maker, and the sales team still keeps its own spreadsheets. A consultant tells the owner: you bought information technology, but you do not yet have an information system. On the chapter's definition, what makes that a fair statement?

- **A.** The equipment is under-powered for the transaction volume the distributor runs, so it cannot process the data quickly enough — Capacity problems are real and would explain slow or failing reports. Nothing in the scenario says the machines cannot keep up; it says nobody uses what they produce.
- **B.** An information system is people plus technology producing **useful** data, and both the people and the usefulness are missing — **correct**. The definition has two halves. Information technology — hardware, software, and telecommunications networks — is one of them; the other is people, and the data have to be useful for someone, somewhere. Technology that produces no useful data for anyone has not become a system.
- **C.** The statement is unfair, because the chapter says many people use IS and IT synonymously, so buying one is buying the other — The chapter does note that the difference is shrinking and that many use the terms interchangeably. It also keeps the distinction that matters here: technologies do not exist for their own sake, they are created to serve a useful purpose for people.
- **D.** The distributor has hardware and software but no telecommunications network, the third component of information technology — A network is listed among the purchases, so this fails on the facts of the scenario. It would be the right objection if the pieces could not share data at all, since telecommunications networks are what let computers share data and services.

8. **1.2** A pharmacy chain wants every sale recorded at the register the moment an item is scanned, so the day's takings reconcile at close and the shelf count drops as the goods leave the store. Which class of system is doing that work?

- **A.** A management information system, because it produces the detailed information a manager needs to run the stockroom — A management information system produces detailed information to help manage a firm or part of a firm, and inventory management and planning is the chapter's own example of one. It normally runs on data the transaction system captured first, which is the step this question describes.
- **B.** A decision support system, because the chain will use the numbers to decide how much of each item to order next — A decision support system provides analysis tools and access to databases to support quantitative decision making, such as product demand forecasting. Deciding how much to reorder would use one; recording the scan that sold the item would not.

- **C. A transaction processing system**, because it processes day-to-day business event data at the operational level — **correct**. The chapter lists a grocery store checkout cash register connected to a network as a sample application of a transaction processing system. The defining trait is capturing day-to-day business events at the operational level as they occur.
- **D. A business intelligence system**, because it analyzes sales data to reveal patterns in what customers buy — A business intelligence system analyzes Big Data to better understand aspects of a business, using tools such as online analytical processing and data visualization. That is what gets built on top of years of sales records, not the record keeping itself.

9. **1.2** A regional manager wants to see what next quarter's demand would look like under three different pricing plans. The tool she uses pulls history out of several databases, lets her change the assumptions, and models each plan so she can compare outcomes side by side. Which category fits best?

- **A. A decision support system**, because it provides analysis tools and database access to support quantitative decision making — **correct**. This matches the chapter's definition line for line, and product demand forecasting is its own sample application. The giveaway is a person testing alternatives quantitatively rather than recording events or reading a standard report.
- **B. A transaction processing system**, because the historical sales it draws on were captured one transaction at a time — A transaction processing system processes day-to-day business event data at the operational level, and it is indeed the source of the history she is modeling. That is the chapter's point that transaction data feed a broad range of other systems, including this one.
- **C. An office automation system**, because she is working in a spreadsheet-style environment to produce her own analysis — Office automation systems — word processors, spreadsheets, presentation software, email clients — support a wide range of predefined day-to-day work activities of individuals and small groups. The interface may look similar, but the purpose here is modeling a decision, not producing a document.
- **D. An enterprise resource planning system**, because drawing on several databases means the system spans the whole organization — Enterprise resource planning supports and integrates all facets of the business, including planning, manufacturing, sales, and marketing. Reading from more than one database is not the same as integrating the business, which is a far larger claim.

10. **1.2** Two finalists for a systems analyst job. One has deeper hands-on expertise with the company's hardware and network than anyone currently on staff. The other knows enough about hardware, software, and networking to judge what each can do for the firm, understands how the order-to-cash process actually works, and has run projects to a deadline. Which candidate matches what the chapter says makes IS personnel valuable?

- **A. The first**, because technical competency is the hardest area to maintain given the pace of innovation, so depth there is scarcest — The chapter does say the technical area is perhaps the most difficult to maintain because of the rapid pace of technological innovation. It draws the opposite conclusion from that fact: the IS professional must know just enough about these areas to understand how they work and what they can do for an organization.

- **B.** Either one, because the chapter treats technical, business, and systems knowledge as interchangeable routes into the same career — The chapter treats the three domains as things one person integrates rather than as alternatives to choose among. It says directly that business and systems competency are what set the IS professional apart from someone with only technical knowledge and skills.
- **C.** The first, because business and project skills can be picked up on the job while technical depth has to be hired in — This inverts the chapter's argument. Business competency is the area it says may save a person's job in an era of increased outsourcing, and business skills are what propel IS professionals into project management and higher-paying management positions.
- **D.** The second, because good IS personnel integrate technical, business, and systems knowledge and direct the deeper specialists — **correct.** The core competency table covers technical, business, and systems domains, and the chapter states that the IS professional need not be a technical expert — typically that person manages or directs those who have deeper, more detailed technical knowledge.

**11. 1.2** A manufacturer wants one system in which accepting a customer order immediately reserves the inventory, schedules the production run, posts the entries to the general ledger, and releases a shipment, so that sales, manufacturing, finance, and planning all work from the same record. Which category is that?

- **A.** A supply chain management system, because the whole point is coordinating what suppliers deliver and when they deliver it — Supply chain management supports the coordination of suppliers, product or service production, and distribution, with procurement planning as the chapter's example. It covers part of this picture and would be right if the question stopped at inbound supply, but it does not reach the general ledger or the sales side.
- **B.** An **enterprise resource planning system**, because it supports and integrates all facets of the business, planning through sales — **correct.** Enterprise resource planning is defined by exactly this breadth: financial, operations, and human resource management in one system spanning planning through sales. The chapter groups it with customer relationship management and supply chain management as systems that deliberately cannot be filed under one older category.
- **C.** A collaboration system, because four departments will work from shared information instead of emailing each other — Collaboration systems enable people to communicate, collaborate, and coordinate with each other, with email plus an automated shared calendar as the example. They coordinate people; they do not execute the business transactions described here.
- **D.** A customer relationship management system, because everything in the chain is triggered by a customer order — Customer relationship management supports interaction between the firm and its customers, with sales force automation and lead generation as sample applications. A customer starts this chain, but reserving stock and posting ledger entries are not customer-facing interactions.

12. 1.3 A hospital moves scheduling, charting, and pharmacy onto a single cloud vendor. For a sustained period it books more procedures than competitors and reduces records work. Then the vendor has a serious weekday outage and the hospital cancels scheduled elective procedures. Which reading of this matches the chapter?

- **A.** The hospital's business model was unsound, and bad business models tend to fail whether or not the firm uses technology — The chapter does say that companies with bad business models tend to fail regardless of whether they use information technology. Nothing here suggests that situation: the hospital's sustained operational gains are evidence that the model worked.
- **B.** This is a security failure, since consolidating three clinical functions gives an attacker a single point to strike — Concentrating systems does raise security questions, but an outage is an availability failure rather than a breach. This would be the right frame if data had been stolen, or encrypted by an attacker demanding a ransom.
- **C.** This is the **dual nature** of information systems: the system that gave the advantage also created the dependence — **correct.** The chapter compares information technology to a sword: effective as a competitive weapon, and dangerous to those who live by it. Its Zoom case makes the same point in almost the same words, noting that relying on a single provider can quickly disrupt many activities in unforeseen ways.
- **D.** The earlier gains should be discounted, because a system that can go down has not produced a genuine advantage — This treats any exposure to risk as proof there was never a benefit. The FedEx case shows large, complicated, failure-prone systems producing sustained advantage, which is the chapter's point that the advantage and the risk arrive together rather than cancelling out.

13. 1.3 A distributor's IT committee has two proposals and money for one. **A:** retire aging servers and move the same workloads to a hosted provider, producing substantial operating savings with nothing a customer would notice. **B:** build reordering that reads each customer's own consumption history and proposes the next order before they run out, which no competitor offers. Which is **strategic** in the chapter's sense?

- **A.** A, because the savings are certain while B's revenue is not, and a system should return its investment before it chases growth — A is an efficiency case, and the chapter agrees that information systems must provide a return on investment. It presents that as the floor rather than the ceiling, adding that technology use can also be strategic and a powerful enabler of competitive advantage.
- **B.** Both equally, because an investment decision of this size shapes the firm's direction whichever proposal wins — Spending decisions always matter, but the chapter reserves the word strategic for systems developed and continuously updated to gain or sustain competitive advantage, as at Zoom and FedEx, rather than for any purchase that keeps operations running.
- **C.** Neither, because the chapter reserves strategic use of information systems for very large organizations such as FedEx — The chapter says the reverse in as many words: whether it is a small mom-and-pop boutique or a large government agency, every organization can find a way to use information technology to beat its rivals.

- **D. B**, because it aims to gain or sustain competitive advantage over rivals rather than lower the cost of work already being done — **correct**. Strategic here means developed, and continuously updated, to gain or sustain competitive advantage over rivals. B changes the customer's reason to buy from this distributor, while A changes only what the distributor pays to do what it already did.

**14. 1.3** A regional bank is first in its market with photo check deposit, and new account openings run well above trend. Later, every competing bank offers the same feature and the bank's growth returns to normal. What is the best explanation?

- **A. Competitive advantage from the use of information systems can be **fleeting****, because competitors can eventually do the same thing — **correct**. This is the caution the chapter attaches to its own argument: information systems can enable advantage, but that advantage can be fleeting because competitors can eventually do the same thing. It is also why FedEx reengineers and improves its systems on average twice a year even while leading.
- **B. The feature was not strategic to begin with, because it was built on standard technology rather than something invented at the bank** — The chapter rejects that test. Companies from Amazon to Zoom created competitive advantage by combining commoditized technologies with proprietary systems and business processes, so standing on standard technology does not disqualify an initiative from being strategic.
- **C. The bank stopped investing in the system after launch, so rivals caught up with a feature it had left unimproved** — Continued investment does matter — FedEx reengineers and improves its systems on average twice a year — but nothing in the scenario says this system was left alone, and the growth ended exactly when rivals matched the feature rather than when the bank stopped work.
- **D. The early growth came from the marketing around the launch, since a deposit feature by itself does not move account openings** — Launch marketing might explain a brief spike, not sustained account openings above trend that return to normal precisely as competitors match the feature. The timing points at the rivals, not at the advertising.

**15. 1.3** Two airlines licence the identical off-the-shelf revenue management package. The first runs it as delivered. The second loads years of its own booking, weather, and gate-turn history into it and rebuilds how crews are staged around what it learns, sustaining a margin lead. What does the chapter say accounts for the difference?

- **A. The second airline must have licensed a more capable edition, since identical software would otherwise produce identical results** — A more capable edition would be a difference in the purchased product, and both airlines licensed the identical package. The scenario holds the software constant on purpose, so the remaining difference has to be what each firm did with it.
- **B. Nothing durable accounts for it, since standardized software is a commodity that every competitor can licence just as easily** — That is the commodity argument the chapter states and then answers. It accepts that information systems have become standardized and ubiquitous, and it still points to Amazon, Zoom, Google, and Facebook building advantage on top of them.

- **C.** Advantage comes from combining commoditized technologies with **proprietary systems and business processes** on top — **correct**. The chapter's phrasing is that companies created competitive advantages by combining certain commoditized technologies with proprietary systems and business processes. Accumulated history and a rebuilt staging process are things a competitor cannot licence.
- **D.** The first airline's mistake was licensing at all, since a strategic system has to be developed in house to count as one — The chapter never requires in-house development, and its API discussion makes the opposite point: Uber built almost its entire app around functionality other companies provided. What matters is what gets built on top, not who wrote the base.

**16.** **1.4** A fitness app told users at sign-up that their workout data would be used to improve coaching inside the app. After a change of ownership, the company sells those data to an insurance broker. Which assessment best connects the chapter's ethics lesson to current privacy rules?

- **A.** Database control is the main issue, so agreement to the original collection is enough for the later sale — This is the ownership-only reading of the chapter's property question. It would fit only if the original notice, consent, promises, data type, and applicable law placed no limit on the new use; the stated inside-the-app purpose supplies such a limit.
- **B.** The changed purpose breaks the stated promise, creating an ethical problem and potentially a legal one under applicable rules — **correct**. The chapter's example treats a reversed purpose promise as an ethical failure. Current enforcement can also make deceptive privacy promises or particular data sales legally significant, though the exact rule depends on jurisdiction and facts.
- **C.** Jurisdiction is the main issue, so the original promise has no independent ethical weight — Jurisdiction determines legal rights and duties, but ethics and an organization's stated commitment remain relevant even when a specific statute does not decide the case. This would fit a question limited solely to which statute applies, not this ethical assessment.
- **D.** Removing names resolves the concern, because the broker receives records without direct identifiers — De-identification can reduce some privacy risk, but it does not erase the changed purpose or automatically remove a record from every privacy rule. This would fit only if the promise and applicable standard expressly permitted that form of sharing.

**17.** **1.4** A retailer combines three things it collected separately: loyalty-card purchase records, answers to a mall survey shoppers filled out for a free coffee, and transaction data from its store credit card. Merged, the file predicts household events accurately enough to time mailings to them. Which concern does the chapter raise about exactly this?

- **A.** This is a data breach, since data gathered for one purpose has ended up inside another system — A breach is unauthorized access, disclosure, or destruction of data, which is what the integrity and security practice exists to prevent. Everything here was done deliberately by the company that already holds the data.

- **B.** This is an intellectual property violation, because the survey answers are creations of the people who wrote them — Intellectual property is creations of the mind that have commercial value, such as music, software, articles, and product designs. Marketing survey answers are personal data about the respondent, which the chapter handles under privacy rather than property.
- **C.** There is no concern to raise, because each of the three datasets was provided by the shopper voluntarily — This is the argument companies rely on, and it is why the chapter says they are often walking a fine line. Voluntary provision at three separate moments is precisely the consent the chapter says was never given for the merged picture.
- **D.** Providing data at different points is not agreement that they be combined, which is what **aggregation** does — **correct.** The chapter says that just because people provide data at different points does not mean they agree for the data to be combined to create a holistic picture, and it warns that predictive models built this way allow manipulating people and reduce human beings to objects that can be quantified.

**18.** **1.4** An online retailer's privacy page states plainly what it collects, why, who will see it, and how confidentiality is maintained, and it offers a checkbox for marketing email. A customer who has moved twice writes in asking to see and correct the address on file, and is told there is no way to do that. Which fair information practice is the retailer missing?

- **A.** Notice/Awareness, because customers must be told what data are gathered and what the data are used for — Notice and awareness covers disclosing what is gathered, what it is used for, who will have access, whether provision is required or voluntary, and how confidentiality will be ensured. The privacy page in the scenario already does all of that.
- **B.** Choice/Consent, because customers must be given options about what will be done with their data — Choice and consent is about giving options on what will be done with the data, typically an opt-in or opt-out decision. The marketing checkbox is that practice, and it is present.
- **C.** Access/Participation, because customers must be able to see what is held about them and correct inaccuracies — **correct.** Access and participation is providing customers with means to access data collected about them, check for accuracy, and request correction of inaccuracies. A customer who cannot see or fix a stale address is being denied exactly that.
- **D.** Enforcement/Redress, because there must be a means to enforce these practices and to provide customers with remedies — Enforcement and redress is providing the means to enforce these practices, and remedies for customers, through self-regulation or appropriate laws and regulations. That is what someone would invoke after being refused; the practice being refused here is access.

**19.** **1.4** A hobbyist downloads a model file for a discontinued appliance part, prints copies on a home 3D printer with the manufacturer's logo raised on the housing, and sells them at a weekend market as genuine spares. Why does the chapter treat this as an **intellectual property** problem and not just a manufacturing story?

- **A.** Because affordable 3D printing lets anyone manufacture a copy, which causes tremendous losses in intellectual property — **correct**. The chapter puts 3D printing beside lossless digital duplication: as printers become better, faster, and more affordable, they open new avenues for quickly and inexpensively producing counterfeit goods, causing problems for consumers expecting the original and tremendous losses in intellectual property.
- **B.** Because copying a physical object requires a licence, while copying a digital file is a contract matter rather than an IP one — This has both halves backwards. Copying digital files is the chapter's central intellectual property example, and many physical objects can be reproduced freely; what matters is whose creation of the mind is being taken.
- **C.** Because the hobbyist never bought the original part, and intellectual property protects revenue on every unit sold — Lost revenue per unit is the argument the chapter shows people disputing, since buyers often claim they would have chosen a free alternative or bought nothing at all. The chapter still treats the behavior as a problem, on the ground that someone else's work is used without permission, attribution, or compensation.
- **D.** Because the printed part carries the maker's logo, and intellectual property is about the brand rather than the design — Trademark is one kind of intellectual property, and the fake logo does compound the harm. The chapter's point about 3D printing is narrower and harder: the design itself can now be copied by anyone, so an unbranded print sold as a genuine spare raises the same problem.

**20.** **1.4** Someone edits a photograph of a coworker so that it appears to show him somewhere he never was, then posts it to a group the whole department follows. Under the Computer Ethics Institute guidelines quoted in the chapter, which prohibition does this most directly break?

- **A.** Snooping in other people's files — Snooping is reading files that belong to someone else without authorization. It would be the answer if the photograph had been taken out of the coworker's own account; what is at issue here is what was published, not how it was obtained.
- **B.** Using a computer to bear false witness — **correct**. Bearing false witness is using a computer to present something untrue as fact about another person, which is what an altered photograph passed off as real does. The chapter raises this case directly when it asks whether the practice of rearranging and otherwise changing photographs is ethical.
- **C.** Appropriating other people's intellectual output — Appropriating other people's intellectual output means taking someone's creative or intellectual work and using it as your own. That would be the answer if the wrong were publishing a photographer's image without credit rather than falsifying what it shows.
- **D.** Interfering with other people's computer work — Interfering with other people's computer work means disrupting what someone else is doing with their machine or their systems. Nothing here stops the coworker from using a computer; the harm is to his reputation.

**21.** **1.5** A furniture maker's largest accounts supply most of its revenue. One demands longer payment terms and a lower price, and mentions that rival suppliers have quoted the same work. Which of the five forces is strongest here?

- **A. Bargaining power of suppliers**, because the builders are dictating the terms of the relationship to the maker — Supplier power is the pressure the firms you buy from apply to you, by raising prices or setting terms when you have few alternatives. The word supplier appears here because the furniture maker is the supplier, so choosing this means reading the situation from the wrong side of the table.
- **B. Competitive rivalry**, because two other suppliers have quoted the same work and are competing for it directly — Rivalry is the fight among firms already selling the same product to the same customers. Rivals do exist in this story and their quotes are what give the builder leverage, but the pressure is being applied by the customer and it is the customer who decides.
- **C. Bargaining power of buyers**, because a big share of revenue plus easy switching lets a customer force price and terms — **correct**. Buyer power grows when a few accounts make up most of your revenue, when comparison is easy, and when switching costs the customer little, and all three are present. That combination is the signature of this force, and it is why loyalty and switching costs are the usual response to it.
- **D. Threat of new entrants**, because rival suppliers are moving into business this maker used to hold on its own — New entrants are firms not competing in your industry today that could start tomorrow, held back by barriers to entry such as cost, scale, know-how, or loyalty. The two rival suppliers quoting are already in this business, so no barrier is being tested.

**22.** **1.5** A cinema chain's attendance falls for several periods, including in towns where it is the only theater and no new theater has opened. Over the same span, household spending on streaming subscriptions rises sharply. Which force explains the decline?

- **A. Competitive rivalry**, because the chain is losing to other companies competing for the same entertainment spending — Rivalry is pressure from firms inside your own industry selling nearly the same thing to the same customers. The scenario removes that possibility deliberately: in the towns described there is no other theater and none opened.
- **B. Bargaining power of buyers**, because moviegoers are refusing to pay the ticket prices the chain sets for them — Buyer power is customers forcing your price down or your service up while still buying from your industry, which happens when switching is cheap and comparison is easy. These households are not negotiating over ticket prices; they stopped buying tickets at all.
- **C. Threat of new entrants**, because streaming services entered a market the chain used to have to itself — A new entrant is a company that starts selling what you sell. A streaming service operates no theaters and sells no tickets to them, which is what separates an entrant from a different kind of product answering the same need.
- **D. Threat of substitutes**, because a product from outside the industry meets the same need, so demand leaves the category — **correct**. A substitute meets the same underlying need, an evening's entertainment, by a different route, so the customer never enters your industry. The tell is in the facts: the decline continues where there is no competitor at all, so nothing inside the industry can account for it.

**23.** **1.5** A grocery chain plans a portal where every approved grower submits a sealed quote against one consolidated weekly order, alternate growers stay qualified and visible, and contract expiry dates appear on the buying team's calendar before renewal comes due. Which value chain activity does this initiative improve?

- **A. Procurement**, because it is about finding, qualifying, and contracting the suppliers — **correct**. Procurement is deciding whom to buy from and on what terms. Quotes, qualification of alternates, and contract renewal are all parts of that decision, which is the activity an initiative like this belongs to.
- **B. Inbound logistics**, because it concerns goods arriving from suppliers before they are sellable — Inbound logistics is receiving, storing, and moving goods that have already arrived, such as a dock scanner counting a pallet as it comes off the truck. It begins after the procurement decision, and mixing the two is the most common way to file a recommendation in the wrong department.
- **C. Operations**, because better buying is what makes a stocked, priced shelf possible — Operations turns inputs into what the customer buys, such as register scans becoming restock tasks so the milk case refills before it empties. Almost anything upstream eventually helps operations, but naming the activity an initiative actually changes is what keeps the analysis usable.
- **D. Firm infrastructure**, because contracts and renewal calendars are matters of general management and planning — Firm infrastructure covers general management, finance, accounting, legal, and planning, such as a margin report by store by day. The buying team here is doing supplier selection, which the value chain names as its own activity.

**24.** **1.5** A retailer gives every returns desk a screen showing the customer's complete purchase history, so a refund is resolved quickly and no receipt is needed. Complaints fall, and repeat visits from customers who returned something rise. Where does this initiative sit in the value chain?

- **A. Marketing and sales**, because a smooth return is what persuades customers to choose this retailer next time — Marketing and sales is making customers aware and giving them a reason to choose you before they buy, such as a loyalty offer aimed at an item a household stopped buying. The downstream effect described is real, but the activity being improved happens after the sale.
- **B. Service**, because it keeps the customer whole after the sale — returns, complaints, and repairs — **correct**. Service is the primary activity that keeps the customer whole once the sale is done, and a returns screen showing purchase history so a refund is resolved quickly is an example of a system that adds value there.
- **C. Outbound logistics**, because it manages goods moving between the customer and the store — Outbound logistics is getting the finished product out to the customer, such as pickup and delivery batched by neighborhood into slots a driver can meet. A return travels the other way and is handled as service, not as a delivery problem.

- **D.** Technology development, because the retailer had to build a new screen and connect it to purchase records — Technology development is the support activity of improving the product and the processes themselves, such as building your own app rather than renting one every rival can rent. Building this screen involved that work, but the activity where the value reaches the customer is service.

**25.** **1.5** Four analysts each write one closing sentence at the end of the same study of a hypothetical regional group of vehicle service depots. Which one is a recommendation an executive can fund and later judge, rather than a preference?

- **A.** Our biggest exposure is disruption, so we should invest in digital transformation and modernize the customer experience across every depot we operate — This names a worry and a category of spending. Disruption is not one of the five forces, digital transformation is not something anyone can be assigned to build, and no measure is named, so a year later nobody could say whether it worked.
- **B.** We should adopt artificial intelligence this year, because the larger garage groups in our region already have and our labour costs per job are higher than theirs — This opens with a technology and reasons from what other firms are doing, which skips the diagnosis. The cost gap is a real observation, but nothing here names the force or the value chain activity the technology would relieve, or what would count as success.
- **C.** Buyer power is strongest — booking records show fleet customers splitting work across three garages — so build one shared service history, judged by return bookings — **correct.** It names the force, gives the observable evidence behind that claim, states a concrete initiative rather than a category of spending, and commits to a measure — the four parts that separate a fundable proposal from a preference.
- **D.** Turnaround time in the workshop varies widely from one depot to the next, which is costing us repeat bookings from fleet customers and ought to be dealt with soon — This is a finding, and a legitimate one that even names a consequence. It stops short of a recommendation because it proposes no initiative anyone could be assigned to build and no measure by which success could later be judged.

**26.** **1.6** A hypothetical regional grocery chain of twelve stores, under pressure from a members-only warehouse club, lists three possible uses for a GenAI system. **A:** draft the weekly produce order and transmit it to growers with no one reading it. **B:** draft each store's weekly staff schedule, which the store manager edits and approves before it is posted. **C:** decide which of two long-standing growers to drop from the supplier list. Which use matches the chapter's position on how GenAI should sit in a workflow?

- **A.** A, because ordering produce is repetitive clerical work, and repetitive work is what a system exists to take off a person's desk — Taking repetitive work off a desk is a fair reason to reach for the tool, and the produce order is genuinely that kind of work. This version transmits the order with nobody reading it, and the chapter's objection is exact: ask the system to do the work for you and you have no credible way of knowing whether the output is correct or even reasonable. A buyer's review before the order was transmitted would put a person back at the decision, which is what use B does.

- **B. C**, because dropping a supplier is a hard judgment call, and hard judgment is where the analysis a system can do is worth the most — Analysis does help on hard calls, and the chapter presents these systems as giving users access to advanced knowledge and problem-solving capabilities. Handing over the decision itself inverts what the chapter asks for, because its condition for making yourself more valuable is that you retain control of decision making. Having the system compare the two growers on delivery reliability while a buyer chooses would fit.
- **C**. None of the three, because a system capable of producing a hallucination has no business anywhere in a workflow that commits money — Hallucinations are real and the chapter names one — an early model describing the sole survivor of the Titanic when more than 700 people survived. The chapter's response is to keep a person between the output and the decision, not to keep the tool out of the building; it calls GenAI a knowledge companion, which is a role inside the work rather than outside it.
- **D. B**, because the system produces a draft while the manager keeps control of the decision, which is what makes the person more valuable — **correct**. The chapter's rule is that GenAI should serve as a knowledge companion and not a knowledge replacement: ask it for assistance while retaining control of the decision and you have made yourself more valuable. A draft schedule the manager edits and approves puts the person at the decision and the system in front of it, which is what augmenting the human rather than replacing the human looks like in one workflow.

**27. 1.6** A distributor's service team drafts customer replies with a GenAI assistant. One reply, sent without review, told a customer the return window is sixty days and cited a policy clause by number. The distributor's stated return window is thirty days, and no clause carries that number. The passage read fluently and hedged nothing. In the chapter's terms, what went wrong?

- **A**. The assistant produced a **hallucination**, a confident output that is simply not true, which is what the review step exists to catch — **correct**. The chapter calls a GenAI mistake of this kind a hallucination, and its example is a model that would gladly tell you about the only survivor of the Titanic when there were more than 700 survivors. Invented detail delivered with no hedging is the same failure, and it is precisely why the chapter asks that a person retain control rather than pass the output straight through.
- **B**. The training data were biased, because the assistant learned return policies from firms whose windows are longer than this distributor's — Bias is a different fault with a different cause: the chapter says output can be biased if the datasets used to train the model are biased, which shows as a systematic slant across many responses. One fabricated clause number is a single false statement, so this would fit if the assistant were consistently harsher with one group of customers than another.
- **C**. The prompt was too vague, and the chapter's own prompting exercise shows that a vague request produces a weaker response — Prompt quality matters, and the chapter's exercise does show a specific request beating a vague one — asking for a short, polite email to a customer who received the wrong item rather than just an email to a customer. A sharper prompt would not have brought a nonexistent clause number into existence, so the defect is in the truth of the output rather than the clarity of the request.

- **D.** The assistant was not integrated with the distributor's policy documents, so it had no access to the text it was describing — Connecting the assistant to the firm's own policy text is a sensible remedy, and the chapter defines systems integration as connecting separate systems and data to improve business processes and decision making. Naming a remedy is not naming the failure, and this failure already has a name in the chapter.

**28.** **1.6** A staffing firm builds an AI screening system on ten years of its own hiring decisions and asks it to rank new applicants. Reviewers find the rankings faithfully reproduce the firm's past preferences, including patterns the firm has since concluded were wrong. Nothing is malfunctioning: the system is doing what it was built to do. Which chapter idea explains the result?

- **A.** The dual nature of information systems, because the system that saves the recruiters time is the same system now exposing the firm to harm — The dual nature is the chapter's sword image: information technology is effective as a competitive weapon and dangerous to those who live by it. That describes the shape of the exposure rather than its source, and this scenario tells you the source outright — the records the model learned from.
- **B.** A hallucination, because the rankings the system reports do not correspond to anything true about the applicants being ranked — A hallucination is a confident false statement, like the chapter's account of a single Titanic survivor. These rankings invent nothing; they are a faithful reflection of what the system was shown, which is a different fault and needs a different fix, namely attention to the training data rather than fact-checking of an answer.
- **C. Bias** carried in from the training data, because output can be biased when the datasets used to train the model are biased — **correct.** The chapter states that GenAI output can be biased if the datasets used to train the model are biased, and adds that quality system outputs rely on quality system inputs. Ten years of decisions the firm itself now judges to have been wrong is exactly such a dataset, and careful prompting cannot undo what the training already encoded.
- **D.** The digital divide, because applicants without reliable internet access are underrepresented in ten years of application records — The digital divide is the gap between those with access to information systems and those without, which the chapter discusses for rural communities, the elderly, people with disabilities, and minorities. It would be the right frame if some applicants could not reach the firm at all; here everyone applied, and the unfairness is in how the applications were ranked.

**29.** **1.6** A benefits analyst wants a summary of a spreadsheet holding employee names, salaries, and health plan selections. She plans to paste it into a public GenAI assistant the firm holds no contract with, whose provider states that submitted content may be retained and used to improve the service. What should her manager raise first?

- **A.** That the summary could contain a hallucination, so no figure in it can be relied on until it is checked against the spreadsheet — A summary can certainly contain a hallucination, and the chapter's insistence that the person retain control of the decision applies to every figure the assistant reports back. That is a reason to check what comes out, and it would be the first concern if the file held nothing sensitive; it says nothing about what happens to what goes in.

- **B.** That pasting the file moves **confidential employee records outside the firm**, to a provider that says it may keep them — **correct**. The moment the file is submitted it has left the firm's control, and the provider states it may retain the content. The chapter's privacy discussion turns on what people were told their data would be used for, and an employee who disclosed a health plan selection to an employer was not agreeing to have it kept by an outside service. The chapter also warns that combining data collected at different points builds a picture nobody consented to.
- **C.** That many systems of this kind were trained on freely available content from the internet and social media, so they handle benefits records poorly — The chapter does say many GenAI systems were trained on freely available content from the internet, discussion boards, social media, and other sources. How well the assistant reads an unfamiliar document affects the quality of the summary, which is a separate question from whether the document should be sent at all.
- **D.** That summarizing a spreadsheet is the analyst's own work, so handing it over makes her less valuable to the firm — The companion-not-replacement argument is about who holds the decision, not about who does the arithmetic, and a summary the analyst then reads and interprets is assistance of exactly the kind the chapter endorses. What is wrong here is what the assistant is being handed, not that it is being asked for help.

**30.** **1.6** Four managers at a wholesale distributor each bring one GenAI proposal to a planning meeting. Which one could be built, run, and later judged?

- **A.** Adopt GenAI across the company this year, because competitors are already using it and the firm cannot afford to fall behind — This names a technology and a fear rather than a piece of work. It does not say which task changes, who reviews the output, or what number would move, so nobody could be assigned to build it and nobody could say a year later whether it worked.
- **B.** Put a GenAI assistant on customer email end to end, replying without review, so the service team can be cut by half next quarter — This one is concrete and headcount is genuinely measurable, which is what makes it tempting. It removes the reviewer the chapter insists on — ask the system to do the work for you and there is no credible way of knowing whether the output is correct — so a wrong answer reaches the customer unread. Keeping a representative on the reply would rescue it.
- **C.** Use GenAI wherever it saves time, and let each department work out for itself where that turns out to be — Letting departments choose sounds practical, and the people doing the work do know where the time goes. As written it names no task, no reviewer, and no measure, so it produces scattered experiments rather than one change anyone can fund, staff, or evaluate.
- **D.** Draft the first reply to routine order-status emails for a representative to edit and send, judged by reply time and complaint volume after one quarter — **correct**. It names one specific task, keeps the representative at the decision so the system augments the person rather than replacing them, and commits to measures and a date on which they will be read. Those parts are what separate a proposal a manager can fund and later judge from a preference.

31. 1.7 A team writes a rule as prose in a specification: 'exclude refunded orders from the regional totals.' Six months later a 'disputed' status is added. What most likely happens to the totals?

- **A.** Nothing, because the rule already says which orders to exclude — The rule names one exclusion. It says nothing about a status that did not exist when it was written, so it cannot govern that status.
- **B.** Disputed orders are silently included, because a rule written as an exclusion only removes the cases somebody anticipated — **correct.** Right. An exclusion list handles the cases it names and lets everything else through. This is why a rule stated as what counts behaves differently from one stated as what does not.
- **C.** The reports fail loudly, because the code meets a status it does not recognise — Nothing errors. The row has a valid status and valid numbers; it is simply not the row anyone meant to count.
- **D.** Disputed orders are excluded, because anything not explicitly completed is left out by default — That would be true had the rule been written as an allow-list. Written as an exclusion, the default runs the other way.

32. 1.7 Two analysts agree on the definitions of a transaction processing system and a decision support system, yet classify the same request differently. What most likely differs?

- **A.** One of them has misunderstood one of the two definitions — Both can hold the definitions correctly and still disagree, which is exactly what makes this worth noticing. A misunderstanding is possible but not the likeliest explanation.
- **B.** The order in which they apply the two tests, because a request fitting both categories is settled by whichever is checked first — **correct.** Right. When a case satisfies more than one definition, the sequence of checks is the real decision rule, and two people who never wrote that sequence down will not share it.
- **C.** The seniority of the person who made the request to each of them — Who asked does not change what the system must do, and neither definition mentions the requester's rank.
- **D.** Whether their organization actually operates both kinds of system, since a test written for a category the firm does not have will never fire — Both systems existing or not affects what can be built, not which category the request belongs to.

33. 1.7 A network grows from 40 members to 80. A colleague says the network is now twice as valuable. What does the connection arithmetic say?

- **A.** It is roughly twice as valuable, since value tracks the number of members — That is the linear intuition the arithmetic contradicts. Members doubled, but the links among them did considerably more than double.
- **B.** The possible connections rise from 780 to 3,160, so value grows faster than membership — **correct.** Right. Forty members allow 780 links and eighty allow 3,160, close to a fourfold rise. This gap is why an early lead in members is so hard for a rival to close.

- **C.** The value cannot change until the new members actually connect to each other, because a possible connection nobody makes adds nothing anyone can use — Possible connections are what the count measures, and they are what makes the network worth joining. Whether each is used is a separate and later question.
- **D.** Doubling the members halves the value of each individual connection — Nothing in the arithmetic values individual connections, and no such halving occurs. The count of possible links simply rises faster than the membership.

**34.** **1.7** A function that strips a customer record down to the fields a purpose needs is written by deleting the sensitive fields it knows about. What is the design flaw?

- **A.** Deleting fields is slower than copying the ones that are wanted — Performance is not the issue here, and on a single record neither approach is measurably faster than the other.
- **B.** It removes fields from the original record rather than from a copy — That is a real bug in some implementations, but it is not the design flaw being described, and it is fixable without changing the approach.
- **C.** Every field added to the record in future is exposed by default, since the function removes only what it was named — **correct.** Right. A deny-list is correct only for the schema it was written against. The safe design states what may be included, so anything new is withheld until somebody decides otherwise.
- **D.** It cannot handle a record that is missing one of the sensitive fields, so the function fails the first time it meets a customer who never supplied one of them — A missing field is easy to handle either way, and it is not what makes the approach unsafe.

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**What to do next.** Two things make this material stick. First, take one organization you already deal with — an employer, a store you shop at, a service you pay for — and name its information systems out loud: what data goes in, what information comes out, who acts on it. Second, practise the strategy method on that same organization until naming a force and an initiative feels routine. Everything after this chapter assumes both habits.